

Time-Consistent Agents, Time-Inconsistent Firms: Investment in Heterogeneous-Agent Economies*

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Abstract

This paper studies firms' investment decisions in heterogeneous-agent economies with trading frictions in financial markets. Shareholders have time-consistent preferences but disagree on the timing of dividends. As a consequence, the firm's problem is endogenously time-inconsistent: the firm behaves as if it faces quasi-hyperbolic discounting, opening a corporate discount-rate wedge between the rate at which it discounts investment and its financial cost of capital. In a calibrated economy, larger trading frictions imply lower capital but only when firms cannot commit to an investment policy. In contrast, if firms can commit, trading frictions affect asset prices but have little effect on aggregate capital. Consistent with the mechanism, firms whose equity is less liquid apply larger wedges in the data. Moreover, firms' ability to commit materially changes aggregate-shock transmission: without commitment, the wedge varies over time, so the risk-free rate alone is not a sufficient statistic for investment incentives.

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1 Introduction

A central question in macroeconomics concerns the determinants of investment. Since [Tobin \(1969\)](#), a guiding principle has been that investment should compare the marginal value of installed capital with its replacement cost. This result emerges naturally in models where all owners agree that the firm should maximize its cum-dividend market value, such as in the neo-classical model with a representative agent or complete markets. In such environments, shareholders unanimously support the policy that maximizes firm value discounted at the market interest rate, and the firm’s problem is time-consistent: the optimal plan chosen today remains optimal tomorrow. But unanimity and time consistency rely on owners agreeing about how to discount future cash flows. In the heterogeneous-agent models now central to macroeconomics, including the HANK framework that has become a workhorse for monetary and fiscal policy analysis ([Kaplan et al., 2018](#)), shareholders may have different marginal rates of substitution and, consequently, different views on how the firm should trade off dividends today against investment for tomorrow. The firm must then aggregate heterogeneous valuations. What are the consequences for capital accumulation?

This paper shows that when shareholders disagree on how to discount future cash flows, the aggregation of their preferences leads to a problem for the firm that is time-inconsistent: the optimal investment plan at any point in time is suboptimal from the perspective of future dates. Moreover, we show that, under plausible assumptions, the firm’s problem is equivalent to one in which it discounts cash flows with *quasi-hyperbolic preferences* ([Laibson, 1997](#)). Notably, this result emerges from the aggregation of rational shareholders with standard exponential discounting, rather than from behavioral assumptions about managers or owners. The direction of the inconsistency depends on who discounts the firm’s continuation value more heavily: when current owners discount it more than the marginal buyer does, the firm exhibits present bias; when they discount it less, the firm exhibits future bias. We characterize the conditions under which each case arises and show that present bias is the empirically relevant case. Importantly, present bias generates a *corporate discount-rate wedge* between the rate at which the firm discounts investment and its financial cost of capital, the object documented by [Gormsen and Huber \(2025\)](#).

We develop the analysis in a production economy where the disagreement arises from trading frictions in asset markets. This focus is motivated by two considerations. First, asset liquidity matters for a wide range of macroeconomic outcomes, including asset prices (e.g., [Aiyagari and Gertler, 1991](#); [Amihud et al., 2005](#)), firm investment (e.g., [Amihud and Levi, 2023](#); [Cui and Radde, 2020](#)), household consumption responses to fiscal stimulus (e.g., [Kaplan et al., 2014](#); [Kaplan and Violante, 2014](#)), and the transmission of monetary policy (e.g., [Kaplan et al., 2018](#); [Jeenas and Lagos, 2024](#)). Second, liquid and illiquid asset positions, and more broadly household balance-sheet heterogeneity, are central features of the new generation of heterogeneous-agent models that have become workhorses for policy analysis (e.g., [Kaplan](#)

and Violante, 2014; Kaplan et al., 2018; Auclert et al., 2021). By studying how trading frictions generate shareholder disagreement, we provide a novel channel through which liquidity affects investment decisions, one that operates through the firm’s objective function rather than through asset prices or financing constraints.

We build a model with two standard blocks: incomplete markets and neoclassical firms, as in Aiyagari (1994). Households face idiosyncratic income risk and have access to two financial assets: shares in a representative firm and risk-free one-period bonds. In the spirit of Aiyagari and Gertler (1991) and Kaplan et al. (2014), we assume that trading equity shares is subject to transaction costs while the bond market is frictionless, although we impose a borrowing limit.¹ Unlike Aiyagari and Gertler (1991) and Kaplan et al. (2014), we consider a production economy, which allows us to study the implications of financial market frictions for aggregate investment. Crucially, we assume that firms make the investment decisions and, therefore, face a dynamic problem. In practice, firms rather than households make investment decisions. In contrast to irrelevance results that hold in frictionless environments (e.g., Carceles-Poveda and Coen-Pirani, 2010), in our setting trading frictions break the equivalence. However, the presence of transaction costs poses a challenge for specifying the firm’s objective, as households’ expected rates of return for holding the firm’s stock do not equalize. As a consequence, the firm’s owners disagree on the appropriate discount factor the firm should use and the implied investment plan. There is a long tradition in microeconomics that studies ways to specify the firm’s objective in this scenario (see DeMarzo, 1993; Dreze, 1974; Grossman and Hart, 1979; Makowski, 1983). In this paper, we assume that the firm maximizes an ownership-weighted valuation, where the weights are given by the initial stock holdings. In the spirit of Grossman and Hart (1979), we provide a microfoundation for an ownership-weighted objective of this form.²

Disagreement about the discount factor leads to the paper’s main result: the firm’s problem is *time-inconsistent*. To understand why, consider the tradeoffs involved in the design of the firm’s investment plan. Investment is an intertemporal decision: agents give up consumption in period t (through the distribution of dividends) for a higher level of production in $t + 1$. Owners include both holders and sellers of shares, and since selling is costly, their discount factors do not equalize. As a consequence, they disagree on the optimal investment plan for the firm: sellers prefer current dividend payouts to capital gains, which they can realize only by selling shares and paying a transaction cost, and therefore favor a lower level of investment. However, this disagreement alone does not generate time-inconsistency: a uniformly lower discount factor would simply reduce investment, leaving the plan time-consistent. The key force is that the valuation governing the firm’s trade-off differs across horizons. One period out, the split between the dividend and the ex-dividend price is settled by the ownership-weighted

¹Our quantitative analysis also includes a wedge between the borrowing and saving rate, as in Kaplan et al. (2018). All our theoretical results go through in this case.

²We show that our qualitative results are similar under alternative weights such as in Dreze (1974), or majority voting as in DeMarzo (1993).

valuation of current owners, sellers included. Everything beyond that is embedded in the ex-dividend price, which reflects the marginal buyer's valuation: the risk-free rate plus a *liquidity premium*. If buyers expect to keep holding the stock (which they do in our calibrated economy), they face little prospect of paying the transaction cost, so the marginal buyer discounts the continuation value less than current owners do on average. They are therefore willing to pay more today for a firm that commits to high investment. Thus time-inconsistency follows: today, the firm would like to promise high investment from $t + 1$ onward. But when $t + 1$ arrives, ownership has turned over, the new set of owners applies its own valuation to the new near horizon, and the firm is once again tempted to cut investment and promise high investment starting in $t + 2$. Notably, transaction costs are crucial for these results. In the absence of transaction costs (e.g., as in Aiyagari, 1994), every owner values the dividend and the ex-dividend price in the same way, so the problem of the firm is time-consistent.

More generally, the mechanism rests on two ingredients: a friction that drives a wedge between current owners' valuation of the ex-dividend price and the marginal buyer's, and a firm objective that aggregates owners' valuations into a single decision rule. Transaction costs are one such friction, which we interpret broadly as the effective cost a marginal seller bears to convert equity into liquid resources, including explicit trading costs, portfolio rebalancing costs, and institutional restrictions on selling. Capital gains taxes are a second such friction, and we show in an extension that they produce qualitatively similar results.³

In a calibrated economy, we find that present bias is the empirically relevant case. When there are no trading frictions, the model is isomorphic to the one-asset economy in Aiyagari (1994). In this case, the steady-state level of capital is higher than in a complete markets economy due to precautionary savings. In contrast, the transmission of trading frictions to the real economy is highly sensitive to the firm's ability to commit. When firms can commit, trading frictions affect asset prices but have minor consequences for aggregate capital. On the one hand, trading frictions raise the liquidity premium, which depresses asset prices, implying a lower steady-state level of capital. On the other hand, there is an increase in the precautionary savings motive, implying a higher steady-state level of capital. Quantitatively, these effects roughly cancel, leaving aggregate capital nearly unchanged. When firms cannot commit, trading frictions have a third effect: they increase the present bias, which directly reduces investment. Quantitatively, this effect dominates. We find that a 10-basis-point increase in the corporate discount-rate wedge reduces steady-state capital by approximately 1.9 percent in the no-commitment economy. In our calibration, steady-state capital is about 20 percent lower than under complete markets. This overturns the standard result from Aiyagari (1994) that incomplete markets lead to over-accumulation of capital.

³A proportional tax on dividends does not create the same distortion. A dividend payment lowers the ex-dividend share price one-for-one, so a dividend tax scales down both today's payout and the market value of future payouts, leaving the firm's intertemporal trade-off unchanged. A capital gains tax, by contrast, applies only when returns are realized through sale, generating exactly the kind of wedge emphasized here. We formally establish this distinction in the appendix.

Notably, only the model without commitment generates a corporate discount-rate wedge, which we read as evidence for lack of commitment. We also provide empirical support for the model’s association between stock liquidity, corporate discount rates and discount-rate wedges. We do this by combining the firm-quarter discount rates and costs of capital of [Gormsen and Huber \(2025\)](#) with relative bid-ask spreads from CRSP. Firms with higher relative spreads use higher discount rates, which is to be expected, as these firms have a higher cost of capital due to a higher liquidity premium. Importantly, more illiquid firms also have larger wedges, consistent with our model’s prediction that trading frictions generate present bias.

We extend the main framework in several dimensions. First, we allow the firm to borrow in illiquid corporate bonds markets. We find that the problem of the firm is still time-inconsistent. Corporate bonds can affect the firm’s financial structure but leave its investment problem unchanged, as that problem depends on shareholders’ valuations. Even if corporate bonds are more illiquid than stocks, a firm without commitment may issue bonds due to present bias. This result provides a reason for corporate borrowing that does not rely on the tax advantage of debt.

In a second extension, we study how the demand and supply of liquid assets affect aggregate capital. On the demand side, we consider an increase in idiosyncratic uncertainty. More uncertainty implies that the illiquid asset is traded more, which widens the wedge and offsets most of the increase in capital that the precautionary motive would otherwise generate. On the supply side, we compare economies with different amounts of liquid assets, which are supplied as liquid government bonds. When firms can commit, an increase in the supply of government bonds leads to a lower level of capital because the interest rate increases. When firms cannot commit, a higher supply of government bonds leads to the opposite effect, a higher level of capital, because the illiquid asset is traded less. Thus, under no commitment, a higher supply of government debt *crowds-in* capital investment.

Finally, we study transition dynamics following unanticipated aggregate shocks. These exercises show that the risk-free rate alone is not a sufficient statistic for investment: lack of commitment introduces a time-varying corporate discount-rate wedge into the firm’s effective discount rate. Because the wedge and the financial cost of capital are distinct objects, they can move in opposite directions, insulating firms’ discount rates from movements in their cost of capital, consistent with the low short-run pass-through documented by [Gormsen and Huber \(2025\)](#). Most strikingly, an increase in income uncertainty raises investment under commitment but reduces it without commitment. Heterogeneous-agent business-cycle models that specify the firm’s discount factor in reduced form may therefore miss not only the magnitude of the investment response but also its sign.

Related Literature. Our paper contributes to several strands of the literature.

First, we contribute to the recent heterogeneous-agent literature that incorporates asset liquidity frictions. For example, [Kaplan and Violante \(2014\)](#) and [Kaplan et al. \(2018\)](#) show that

the distinction between liquid and illiquid household portfolios shapes marginal propensities to consume and the transmission of fiscal and monetary policy. Cui and Radde (2020) and Jeenas and Lagos (2024) study the effects of liquidity on firms' investment decisions.⁴ However, this literature does not study the implications of owner disagreement. For example, in Kaplan et al. (2018) firms rent capital, and arbitrage between capital and equity pins down the firm's valuation, in Cui and Radde (2020) households own the capital, so the firm's problem is static, and in Jeenas and Lagos (2024) a single entrepreneur makes the investment decision. As a consequence, owner disagreement does not affect the firms' problem, and liquidity frictions affect investment decisions only through their impact on asset prices. In our paper, liquidity frictions affect asset prices but also generate disagreement among owners, which leads to a problem for the firms that is time-inconsistent.

Second, our model provides a rationalization for the corporate discount-rate wedge documented by Gormsen and Huber (2025). They show that firms move their discount rates with the cost of capital far less than one for one, so that the wedge has grown as the cost of capital has fallen. Among the potential drivers they consider are market power, firm-level risk, and financial constraints, which they document are associated with larger wedges in the cross section. Fukui et al. (2024) propose a macroeconomic account of the sluggish comovement, in which nominal discount rates are sticky with respect to expected inflation, so that the real wedge moves with inflation expectations and monetary policy is non-neutral even absent other nominal rigidities. Barry et al. (2026) show that an elevated hurdle rate can serve as a commitment device that strengthens a firm's bargaining position with counterparties. A large corporate-finance literature points to agency conflicts (e.g., Jensen, 1986; Albuquerque and Wang, 2008; Stein, 2003), limits on managerial capacity (Jagannathan et al., 2016), catering to stock-market mispricing (Polk and Sapienza, 2009), and short-termism induced by career or analyst pressures (Narayanan, 1985; Graham et al., 2005; Terry, 2023), a pattern documented for listed firms by Asker et al. (2015). We offer a different explanation, in which the wedge emerges from trading frictions with fully rational firms and households, providing a rational account of corporate short-termism. Our mechanism also makes a cross-sectional prediction the others do not share: the wedge is larger for firms whose equity is less liquid, which we test in Section 6.3. Angeletos (2007) and Meh and Quadrini (2006) obtain a premium over the risk-free rate from uninsurable idiosyncratic investment risk; we obtain a wedge without risky investment, from disagreement among shareholders. To the best of our knowledge, ours is the first paper to derive the corporate discount-rate wedge from primitives in a quantitative general-equilibrium model.

Third, the paper is also related to the literature that studies the problem of the firm in economies with incomplete markets (see Diamond, 1967; Dreze, 1974; Grossman and Hart, 1979; DeMarzo, 1993; Makowski, 1983).⁵ These papers study the problem of defining the

⁴Relatedly, see Jermann (1998) and Cochrane (2008) for production-based asset pricing models.

⁵See Magill and Quinzii (2002) for a review of the literature.

firm's objective when shareholders disagree, typically in settings with aggregate risk and a finite number of periods. We consider instead an infinitely lived economy with idiosyncratic risk and trading frictions but no aggregate risk, and show that, under plausible assumptions, the natural aggregation of shareholder preferences yields quasi-hyperbolic discounting. Notably, we do not assume quasi-hyperbolic discounting as a behavioral phenomenon (as in [Maxted et al., 2025](#); [Grenadier and Wang, 2007](#); [Kang and Ye, 2019](#)), but rather derive it endogenously from trading frictions in asset markets (see also [Amador, 2012](#); [Azzimonti, 2011](#)). Such an approach disciplines the firm's β and δ , which are determined by asset-pricing moments, income heterogeneity, and transaction costs. [Favilukis \(2013\)](#) and [Favilukis et al. \(2017\)](#) also confront the firm's discount factor in incomplete-market economies, using an ownership-weighted average of the agents' intertemporal marginal rates of substitution, but they do not consider the potential time-inconsistency problem.⁶ The connection between preference aggregation and time inconsistency also appears in [Jackson and Yariv \(2015\)](#), who show in a different setting that aggregating heterogeneous time preferences generically leads to it, with present bias arising because discounting of the distant future is governed by the most patient individual. In our economy all households share the same rate of time preference, and present bias arises instead from the gap between the valuations of current owners and those of the marginal buyer. [Hellwig \(2026\)](#) also obtains time inconsistency at the firm level, but from the market's failure to align incentives across periods rather than from disagreement within a period.

Closest to our paper is [Carceles-Poveda \(2009\)](#), who studies how the aggregation of heterogeneous shareholders' preferences shapes investment, comparing several firm objectives in an infinite-horizon economy with aggregate risk and portfolio restrictions. In their economy, the objective that weights shareholders by their beginning-of-period holdings, which is the one we study, delivers allocations almost identical to those of a standard economy in which firms rent capital. An owner's valuation departs from the market's only when a portfolio constraint binds, and owners are rarely constrained. Importantly, the paper does not examine whether the resulting firm problem is time-consistent. In contrast, in our economy transaction costs make owners' valuations differ whenever they expect to sell, which requires no constraint to bind and occurs far more frequently, so the firm's problem is time-inconsistent and, absent commitment, investment departs substantially from the benchmark. [Carceles-Poveda and Coen-Pirani \(2010\)](#) study the complementary question of whether an objective exists under which household and firm ownership of capital are equivalent, and show that value maximization at present-value prices admitting no arbitrage delivers it. Absent transaction costs, every owner in our economy values the firm at its cum-dividend market value, so the ownership-weighted objective is value maximization at the risk-free rate.⁷ We focus instead on the time-consistency properties of the

⁶Relatedly, [Espino et al. \(2018\)](#) and [Bisin et al. \(2025\)](#) study the insurance properties of firm ownership, and [Demarzo and Urošević \(2006\)](#) studies how a large shareholder shapes ownership dynamics and asset prices.

⁷Relatedly, [Carceles-Poveda and Coen-Pirani \(2009\)](#) show that in incomplete-market economies shareholders unanimously agree on the firm's investment plan when the production function has constant returns to scale and short-selling constraints do not bind. When the short-selling constraint binds for some agent, unanimity fails only

firm's problem.

Finally, our results relate to the literature on the effects of transaction costs on financial markets (e.g., Constantinides, 1986; Aiyagari and Gertler, 1991; Heaton and Lucas, 1996; Vayanos, 1998; Gârleanu and Pedersen, 2013; Abel et al., 2013).⁸ This literature considers either exogenous dividend streams or endowment economies and focuses on the asset-pricing implications, abstracting from how trading frictions affect the firm's problem and investment.

The paper is organized as follows. Section 2 presents the household and firm problems. Section 3 considers a simple three-period example to understand the sources of time inconsistency. Section 4 presents the firm's problem and defines the equilibrium in the infinite-horizon economy, and Section 5 describes the solution to the firm's problem with beta-delta discounting. Section 6 presents the quantitative evaluation, and Section 7 studies transition dynamics following unanticipated aggregate shocks. Finally, Section 8 concludes.

2 The Model

We study an Aiyagari economy augmented to incorporate transaction costs on financial assets. Time is discrete and denoted by $t = 0, 1, 2, \dots$. Households face labor income risk, and markets are incomplete. In particular, households have access to only two financial markets: a market for the stock of the representative firm and a market for a risk-free bond. There is no aggregate risk.

Households. The economy is populated by a measure one of households indexed by $j \in [0, 1]$. Households are subject to idiosyncratic labor shocks that determine the number of hours they can sell in the labor market, which we label as their *employment status* and denote by h . We assume that h is drawn from a finite set $\mathbb{H} = \{h_1, h_2, \dots, h_S\}$, where $h_1 < h_2 < \dots < h_S$, with associated transition density $dF(h_{s'}|h_s)$. Households' preferences can be represented by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (1)$$

where $\mathbb{E}_t$ represents the expectation operator conditional on the information set at period t , β is the households' subjective discount factor, c_t denotes consumption in period t , and $u(\cdot)$ is a continuously differentiable and strictly concave function that satisfies Inada conditions.

Markets in this economy are incomplete. Households can trade two classes of financial assets: the stock of the representative firm and a one-period risk-free bond.⁹ Trade in the stock

when their positions are strictly negative. In our economy unanimity obtains in the absence of transaction costs because all short sales are ruled out.

⁸See Vayanos et al. (2012) for a survey of this literature.

⁹All households have identical preferences and access to both assets, so portfolio heterogeneity is endogenous. This distinguishes our environment from explanations based on preference heterogeneity or limited asset market participation (Vissing-Jørgensen, 2002; Guvenen, 2009b).

market entails a transaction cost, which we model as a wasteful use of resources. Let q_t denote the price of a share, and Δ_t^- and Δ_t^+ denote the shares sold and bought by a household. We assume that the transaction cost incurred in a sale is $\frac{\phi}{2}(\Delta_t^-)^2 q_t$, with $\phi > 0$. Quadratic transaction costs simplify the solution of the agents' problem, as it avoids the existence of inaction regions.¹⁰ Short-selling is not allowed in this market. In contrast, trading bonds is frictionless but households face a borrowing limit $\bar{b}$.¹¹ Thus, the constraints faced by a household in period t are

$$c_t + q_t \Delta_t^+ + b_{t+1} \leq w_t h_t + d_t \theta_t + \left(\Delta_t^- - \frac{\phi}{2} (\Delta_t^-)^2 \right) q_t + b_t (1 + r_{t-1}) - \mathcal{T}_t, \quad (2)$$

$$\theta_{t+1} = \theta_t + \Delta_t^+ - \Delta_t^-, \quad (3)$$

$$\Delta_t^+ \geq 0, \quad \theta_t \geq \Delta_t^- \geq 0, \quad (4)$$

$$b_{t+1} \geq \bar{b}, \quad (5)$$

where b_t denotes the holdings of one-period risk-free bonds, r_t is the real interest rate, w_t is the real wage, θ_t denotes the household's holdings of stock at the beginning of period t , d_t denotes the dividends distributed by the representative firm, and $\mathcal{T}_t$ denotes a lump-sum tax on households to finance government debt. Equation (2) is a standard budget constraint. The household receives labor income, $w_t h_t$, dividend income, $d_t \theta_t$, the proceeds from the sale of stock net of transaction costs, $\left(\Delta_t^- - \frac{\phi}{2} (\Delta_t^-)^2 \right) q_t$, and maturing bonds, $b_t (1 + r_{t-1})$, net of taxes, $\mathcal{T}_t$. They use their income to consume, c_t , buy stock, $q_t \Delta_t^+$, and buy bonds, b_{t+1} . Equation (3) represents the law of motion of stock holdings. Condition (4) imposes natural constraints on stock trades, ensuring that purchases and sales are non-negative and that the household does not sell more shares than it owns. Finally, equation (5) represents the borrowing constraint. We assume that $\bar{b} \leq 0$ and that, for every $t \geq 0$, $\bar{b} > -\sum_{\tau=t+1}^{\infty} \left(\prod_{s=\tau+1}^t \frac{1}{1+r_{s-1}} \right) (w_\tau h_1 - \mathcal{T}_\tau)$, so that the borrowing limit is strictly tighter than the natural borrowing limit at every date.

Thus, the problem of a household is to choose processes $\{c_t, b_{t+1}, \theta_{t+1}, \Delta_t^+, \Delta_t^-\}_{t=0}^{\infty}$ in order to maximize (1) subject to (2), (3), (4), and (5) for every $t \geq 0$, and given an initial portfolio (θ_0, b_0) and prices, dividends, and taxes $\{w_t, q_t, r_t, d_t, \mathcal{T}_t\}_{t=0}^{\infty}$.

The government. The government issues one-period risk-free bonds and levies lump-sum taxes on households. Letting $B_t^G \geq 0$ denote the stock of debt outstanding at the beginning of period t , its budget constraint is

$$B_{t+1}^G = (1 + r_{t-1}) B_t^G - \mathcal{T}_t. \quad (6)$$

¹⁰See [Heaton and Lucas \(1996\)](#). Our theoretical results go through with any continuously differentiable, strictly convex transaction cost function $\phi \mathcal{G}(\Delta_t^-) q_t$ where $\mathcal{G}(0) = \mathcal{G}'(0) = 0$.

¹¹Appendix E.1 extends our baseline setting to one with a wedge between the borrowing and saving rate like in [Kaplan et al. \(2018\)](#), and shows that all of our theoretical results go through. Furthermore, our main quantitative exercise assumes a positive wedge.

We restrict attention to government policies for which B_t^G is bounded.

Firms. There is a representative firm that operates a non-increasing returns to scale technology that combines labor and capital to produce the final consumption good, given by $y_t = (l_t^\gamma k_t^{1-\gamma})^\psi$, where l_t denotes the amount of labor hired, k_t denotes the amount of capital operated, $\gamma \in (0, 1)$ and $\psi \leq 1$.¹² Moreover, the firm operates the economy's investment technology, $k_{t+1} = (1 - \delta)k_t + i_t$, where i_t denotes the level of investment in period t and $\delta \in (0, 1)$ is the depreciation rate.

As is standard, we assume that the firm acts in the best interest of its shareholders. Because the choice of labor is an intratemporal decision, optimality implies that $l_t = \gamma \psi \frac{y_t}{w_t}$. Let $\pi_t = y_t - w_t l_t$ denote the firm's per-period profits. Then, $\pi_t = (1 - \gamma \psi) y_t$, or $\pi_t = z_t k_t^\alpha$, where $z_t = (1 - \gamma \psi) \left(\frac{\gamma \psi}{w_t} \right)^{\frac{\gamma \psi}{1 - \gamma \psi}}$ and $\alpha = \frac{(1 - \gamma) \psi}{1 - \gamma \psi}$. Dividends are then given by $d_t = \pi_t - i_t$, or

$$d_t = F_t(k_t, k_{t+1}) \equiv z_t k_t^\alpha + (1 - \delta)k_t - k_{t+1}.$$

Investment is an intertemporal decision that requires knowledge of shareholders' intertemporal preferences. Because of the assumed financial frictions, shareholders may have conflicting preferences about the firm's investment plans. We proceed in two steps. First, we compute the shareholder's valuations. Then, we specify the firm's objective.

Shareholders' valuations. Let $\tilde{q}_t(\theta, b, h)$ denote the shareholder's valuation of the firm in units of the consumption good, which is given by

$$\tilde{q}_t(\theta, b, h) \equiv \frac{V_{\theta_t}(\theta, b, h)}{\lambda_t(\theta, b, h)} = d_t + \frac{\mu_t(\theta, b, h) + \eta_t(\theta, b, h)}{\lambda_t(\theta, b, h)},$$

where $V_t(\theta, b, h)$ denotes the value function in period t of a household with stock holdings θ , bond holdings b , and employment status h , $V_{\theta_t}(\theta, b, h)$ denotes the envelope condition of a shareholder (i.e., a household with $\theta > 0$) with respect to θ , $\lambda_t(\theta, b, h)$ denotes the marginal utility of wealth (measured as the Lagrange multiplier associated to the budget constraint (2)), $\mu_t(\theta, b, h)$ denotes the marginal utility of an extra unit of the stock (measured by the Lagrange multiplier of constraint (3)), and $\eta_t(\theta, b, h)$ denotes the Lagrange multiplier associated to the short-selling constraint in (4).

The shareholder's valuation has two components. First, shareholders value the dividend they receive, d_t . Since the dividend is in units of the consumption good, all shareholders agree on its valuation. The second term denotes the *ex-dividend* value of the stock. Since the stock is

¹²In the background, we are assuming that there is a measure one of identical firms with non-increasing returns to scale. Since all firms make the same choices in equilibrium, we have $k_i = k$ and $l_i = l$ for all i . Then, there exists a representative firm with the same technology that operates all the capital and labor, i.e., $\int_0^1 k_i di = k$ and $\int_0^1 l_i di = l$. Finally, we assume that the entry costs are such that there is no firm entry in equilibrium.

a long-lived asset, agents value holding shares of the firm above and beyond the dividend they receive in the current period. This term may differ across agents.

There are two types of shareholders: those who keep all their holdings (and potentially buy more) and those who sell at least part of their portfolio. We call them buyers and sellers, respectively. Absent transaction costs, this distinction would be inconsequential: at the margin, the valuation of buyers and sellers would always coincide. However, transaction costs introduce a wedge in the agents' valuation, generating disagreement. The following lemma characterizes the shareholders' valuation. All proofs are in Appendix A.

Lemma 1. *The shareholders' valuation of the firm is given by*

$$\tilde{q}_t(\theta, b, h) = d_t + (1 - \phi \Delta_t^-(\theta, b, h))q_t. \quad (7)$$

Lemma 1 characterizes the valuation of all agents that start the period with $\theta > 0$. Note that since buyers choose $\Delta_t^- = 0$, their valuation simplifies to $\tilde{q}_t(\theta, b, h) = d_t + q_t$, which is independent of agent-specific variables. In contrast, the seller's valuation is decreasing in the amount of stock they sell. The more the agent sells, the higher the transaction cost and the lower the benefits of holding the stock.

Next, we use Lemma 1 to specify the firm's objective.

The firm's problem. When shareholders disagree, assuming that the firm acts in their best interest is not enough to pin down its objective, since one must still specify *whose* interest prevails.¹³ In this paper, we assume that the firm maximizes an *ownership-weighted valuation*, which can be thought of as giving each shareholder votes in proportion to their holdings, in the spirit of Grossman and Hart (1979) (see also Favilukis, 2013). Appendix B.3 shows, using calculations analogous to those in Grossman and Hart (1979), that an ownership-weighted objective of this form arises from a constrained-efficiency criterion: there is no alternative investment plan, paired with zero-sum transfers of current income across shareholders, that is unanimously preferred to the plan that maximizes an ownership-weighted valuation of this form.¹⁴

Assumption 1. *The firm maximizes an ownership-weighted valuation given by:*

$$\int_{\theta, b, h} \theta [d_t + (1 - \phi \Delta_t^-(\theta, b, h))q_t] d\Gamma_t(\theta, b, h) = d_t + (1 - \bar{\Phi}_t)q_t, \quad (8)$$

where $\Gamma_t(\theta, b, h)$ denotes the cross-section distribution over the portfolio holdings and employ-

¹³See, e.g., Diamond (1967), Dreze (1974), Grossman and Hart (1979).

¹⁴In Appendices B.1 and B.2 we generalize the weights to encompass weighting by future ownership, as in Dreze (1974), and majority voting, as in DeMarzo (1993), and show that our results are robust to these alternative specifications.

ment status, $d_t = F_t(k_t, k_{t+1})$, and

$$\bar{\Phi}_t \equiv \phi \int_{\theta, b, h} \theta \Delta_t^-(\theta, b, h) d\Gamma_t(\theta, b, h). \quad (9)$$

Maximizing the firm's value involves choosing the path of capital that trades off the effects on the dividend distributed in period t , d_t , and the continuation market value, q_t . For example, higher investment today implies a lower current dividend but potentially higher dividends in the future. Crucially, the continuation value q_t is discounted by $\bar{\Phi}_t$, which represents the weighted-average marginal cost faced by *current owners*. Consider the valuations of owners who sell some of their holdings relative to owners who keep everything. Since the firm can distribute dividends costlessly, those who sell tend to favor the distribution of dividends at the expense of a higher continuation value. The firm aggregates these differences by internalizing a weighted-average transaction cost in its problem.

An important aspect of the firm problem is *timing*: shareholders vote on firm policy *before* they trade. At the moment of decision, sellers still own their full shareholdings, so their valuations appropriately receive weight proportional to their ownership stakes. The fact that they plan to sell creates a wedge in how they value the firm, but their influence on firm decisions correctly reflects their current ownership.

Even with Assumption 1, the firm's problem is not fully specified since it still needs to understand the sensitivity of the stock price, q_t , to its investment plan. Naturally, the firm does not directly control q_t , which is determined in equilibrium. However, the firm understands that the equilibrium value of q_t depends on its investment decisions through their consequences on the dividend payout plan. To tackle this problem, we first solve a simplified 3-period version of the model in Section 3. We turn back to the infinite-horizon setting in Section 4.

3 Motivating Example: A 3-Period Model

Before analyzing the infinite-horizon economy, we study a simplified three-period version that isolates the source of the firm's time-inconsistency in a transparent way. With only three time periods, $t = 0, 1, 2$, we have that $c_t = 0$ for $t \geq 3$, $q_2 = 0$, and equations (2) to (5) only hold for $t = 0, 1, 2$.

Besides assuming a shorter time horizon, we make two additional simplifying assumptions relative to the model described in Section 2. First, we assume that households face no income risk. Instead, we follow Woodford (1990) and assume that households' employment status oscillates in a deterministic fashion between a low value h_{low} and a high value h_{high} . We assume that half of the households receive the low labor endowment in period $t = 0$, and the other half receive the high labor endowment. We refer to these two groups as the low and high groups $j \in \{l, h\}$, respectively. Second, we assume that households cannot borrow, $\bar{b} = 0$, and that the government debt is zero, so $\mathcal{T}_t = 0$. Thus, there effectively is a single financial asset in the

economy, the illiquid stock. All other aspects of the example are as described in the previous section.

Our main object of interest is the firm's intertemporal problem. We first characterize the solution for a firm that commits in period 0 to an investment path for capital in periods 1 and 2. Then, we show that if the firm is allowed to reoptimize in period 1, it chooses a different allocation for capital in period 2. Hence, the problem is not time-consistent.

The firm maximizes shareholder value, which, by Assumption 1, is given by:

$$\sum_{j \in \{l, h\}} \frac{\theta_t^j}{2} \left[d_t + (1 - \phi \Delta_t^{j-}) q_t \right], \quad t \in \{0, 1\}. \quad (10)$$

As discussed in Section 2, while q_t is determined in equilibrium, the firm understands that its investment decisions (and, hence, its dividends) will impact its value. Here, we assume that the firm knows that the stock price satisfies the following households' Euler equations

$$(1 - \phi \Delta_t^{j-}) q_t = \beta \frac{u'(c_{t+1}^j)}{u'(c_t^j)} d_{t+1} + \beta \frac{u'(c_{t+1}^j)}{u'(c_t^j)} (1 - \phi \Delta_{t+1}^{j-}) q_{t+1}, \quad t \in \{0, 1\}. \quad (11)$$

Note that we focus on equilibria where the short-selling constraint does not bind. In the numerical illustration below, we choose parameter values that satisfy this requirement. Then, introducing (11) into (10), we get that the firm's value in period 0 is given by

$$V_0^F(k_0) = \max_{k_1, k_2 \geq 0} \sum_{j \in \{l, h\}} \frac{\theta_0^j}{2} \left[d_0 + \beta \frac{u'(c_1^j)}{u'(c_0^j)} d_1 + \beta^2 \frac{u'(c_2^j)}{u'(c_0^j)} d_2 \right], \quad (12)$$

where $d_t = F_t(k_t, k_{t+1})$ and the firm takes the households' consumption plans as given.¹⁵ The optimal investment plan from period 0's perspective is the solution to problem (12).

The discount factor of the firm in problem (12) corresponds to a weighted average of the shareholders' IMRS implied by the household's Euler equations (11), where the weights correspond to the shareholder's proportional ownership in the firm. This discount factor corresponds to the assumption in other papers such as Favilukis (2013) and Favilukis et al. (2017). Our focus is on the time-inconsistency that this discounting can generate.

Time Inconsistency. Consider now the problem of the firm in period 1. The firm starts the period with capital stock k_1 , which was chosen in period 0. Suppose we give the firm the option to reoptimize its investment plan, i.e., the choice of k_2 . If the plan in period 0 is time-consistent, the firm would not have an incentive to change its choice. However, we show next that the firm may choose to change its plan; that is, its problem may be time-inconsistent. To see this, note

¹⁵This is an application of the standard "big D little d" argument. The firm takes aggregate consumption, which depends on dividends, as given, but optimizes on its own dividend payments.

that the firm's problem in period 1 along the equilibrium path is

$$\begin{aligned} V_1^F(k_1) &= \max_{k_2 \geq 0} \sum_{j \in \{l, h\}} \frac{\theta_1^j}{2} \left[d_1 + (1 - \phi \Delta_1^{j-}) q_1 \right] \\ &= \max_{k_2 \geq 0} \sum_{j \in \{l, h\}} \frac{\theta_1^j}{2} \left[d_1 + \beta \frac{u'(c_2^j)}{u'(c_1^j)} d_2 \right] \end{aligned} \quad (13)$$

where $\theta_1^j = \theta_0^j + \Delta_0^{j+} - \Delta_0^{j-}$. Then, the firm's problem is time-consistent if and only if the discounting between $t = 1$ and $t = 2$ in problem (12) and problem (13) coincide, or, equivalently, if

$$\frac{\sum_{j \in \{l, h\}} \frac{\theta_0^j}{2} \beta^2 \frac{u'(c_2^j)}{u'(c_0^j)}}{\sum_{j \in \{l, h\}} \frac{\theta_0^j}{2} \beta \frac{u'(c_1^j)}{u'(c_0^j)}} = \sum_{j \in \{l, h\}} \frac{\theta_1^j}{2} \beta \frac{u'(c_2^j)}{u'(c_1^j)}. \quad (14)$$

First consider the case with $\phi = 0$. The Euler equation (11) becomes

$$\frac{q_t}{d_{t+1} + q_{t+1}} = \beta \frac{u'(c_{t+1}^j)}{u'(c_t^j)}.$$

In this case, the IMRS are equalized across agents. Then, the firm's problem is time-consistent:

$$\frac{\sum_{j \in \{l, h\}} \frac{\theta_0^j}{2} \beta^2 \frac{u'(c_2^j)}{u'(c_0^j)}}{\sum_{j \in \{l, h\}} \frac{\theta_0^j}{2} \beta \frac{u'(c_1^j)}{u'(c_0^j)}} = \frac{\frac{q_0}{d_1 + q_1} \frac{q_1}{d_2 + q_2}}{\frac{q_0}{d_1 + q_1}} = \frac{q_1}{d_2 + q_2} = \sum_{j \in \{l, h\}} \frac{\theta_1^j}{2} \beta \frac{u'(c_2^j)}{u'(c_1^j)}.$$

When financial markets are frictionless, agents' valuations of consumption across periods coincide. As a consequence, they will agree on the optimal investment plan for the firm across periods, and the firm's problem is time-consistent.

In contrast, when $\phi > 0$, the agents' IMRS might not equalize if there is positive trade (i.e., if $\Delta_t^{j-} > 0$ for some households). In that case, there is no guarantee that condition (14) will hold. In particular, as agents' valuation and ownership change, the optimal plan for the firm may also change. We illustrate this with a numerical example.

Suppose $\beta = 0.95$, $u = \ln(c)$, $\phi = 0.2$, $h_{low} = 1$, $h_{high} = 3$, $\gamma = 0.7$, $\psi = 1$, $k_0 = 1$, and $\delta = 0.1$. Initial asset holdings are $\theta_0^j = 1$ for $j \in \{l, h\}$. To assess whether the firm's problem is time-consistent, we compute equilibrium assuming that the firm solves its problem in period 0 and does not re-optimize in period 1. We later check if, given this equilibrium, allowing the firm to re-optimize in period 1 changes its investment decisions. The household equilibrium is summarized in Table 1, while asset prices are given by $q_0 = 1.1661$ and $q_1 = 0.9664$, dividends are given by $d_0 = 0.2602$, $d_1 = 0.6085$, and $d_2 = 1.2940$, capital is $k_1 = 1.1272$ and $k_2 = 0.9112$, and wages are given by $w_0 = 0.5686$, $w_1 = 0.5894$, and $w_2 = 0.5529$.

Table 1: Equilibrium in the 3-Period Model

Household type	c_0	c_1	c_2	Δ_0^-	Δ_1^-	θ_1	θ_2
<i>low</i>	1.2050	1.6566	2.1072	0.3337	0.0000	0.6663	1.2012
<i>high</i>	1.5767	1.8902	2.6924	0.0000	0.5349	1.3337	0.7988

Let us compare the firm's problem in $t = 0$ and $t = 1$ along the equilibrium path. Plugging in the equilibrium values into (12), the problem of the firm in period 0 is given by

$$V_0^F(k_0) = \max_{k_1, k_2 \geq 0} d_0 + 0.7417d_1 + 0.5223d_2,$$

and plugging into (13), the problem of the firm in period 1 is given by

$$V_1^F(k_1) = \max_{k_2 \geq 0} d_1 + 0.6935d_2.$$

From its period 0 perspective, the firm's discount factor between $t = 1$ and $t = 2$ is 0.7042, which is the ratio between 0.5223 and 0.7417. When period 1 arrives, the firm's discount factor between $t = 1$ and $t = 2$ is 0.6935. Hence, the problem of the firm is time-inconsistent: the time preference between $t = 1$ and $t = 2$ is different in $t = 0$ than in $t = 1$. Furthermore, in this example, the direction of inconsistency is towards *present bias*. That is, when period 1 arrives, the firm discounts period 2 more than it did in period 0.

Extensions. We consider several extensions of the three-period model, each of which leaves our conclusions intact. First, since the 3-period model ends in a period where the stock price is zero, a stationary equilibrium does not exist, in contrast to the infinite-horizon model we study in Section 5. In Appendix C.1, we present a natural extension of the three-period model that generates a stationary equilibrium and show that all of our results go through. Second, in Appendix C.2 we let the transaction cost apply symmetrically to purchases and sales, again without altering our conclusions. Third, in Appendix C.3 we ask whether there are equilibria in which firms have different compositions of shareholders, and show that there are none: any such deviation would create a cross-firm arbitrage opportunity, so the representative-firm assumption is without loss. Finally, a natural question is whether transaction costs are necessary for our findings, or whether other frictions can also make the firm's problem time-inconsistent. In Appendix C.4 we replace the transaction cost with a capital gains tax and obtain a firm problem comparable to (8), in which buyers again prefer a higher continuation value through investment and sellers a higher current dividend, so the firm's problem remains time-inconsistent. While capital gains taxes are quantitatively large, at rates as high as 20% or even 37% depending on the seller's holding period, we make the point in the three-period setting because in an infinite

horizon they introduce unrealized gains as an additional state variable.¹⁶ These results suggest that what the mechanism requires is not a transaction cost as such, but a friction that drives a wedge between the valuations of buyers and sellers, and we conjecture that other such frictions would act similarly.

From 3 periods to infinite horizon. The purpose of this example was to show, in a simple and transparent model, that time-inconsistency is a natural outcome in the absence of IMRS equalization across agents. Intratemporal disagreement among current shareholders, as well as intertemporal disagreement among current and future shareholders, can make the optimal plan from period 0's perspective suboptimal in period 1.

The key step that allowed us to characterize the solution was to replace the price of the stock, q_t , from the firm's objective by substituting the households' Euler equation in (13), which holds true when short-selling constraints do not bind. Note, however, that this substitution requires a great deal of information on the part of the firm. The firm needs to know not only the current preferences of its owners but also their *future* preferences. Crucially, the firm cannot recover these valuations from asset prices. While the firm's stock market price is observable, the *sensitivity* of the price to changes in the investment plan is unobservable.

Unfortunately, these problems get exacerbated in an infinite-horizon setting. The firm would need information on its shareholders' valuation from the present to the infinite future. Moreover, realistic settings will allow for the set of shareholders to change over time (rather than the same shareholders changing the amount they own), and some shareholders might be constrained by the short-selling constraint. These issues substantially complicate the firm's problem and the computation of equilibrium.¹⁷ To make progress, Section 4 discusses an additional assumption that renders the infinite-horizon problem tractable.

4 The Firm in Infinite-Horizon

In this section, we set up the problem of the representative firm in the infinite horizon setting. A simplifying assumption will allow us to sidestep the aforementioned complications; however, two outcomes from the previous section remain: time-inconsistency and present bias. Moreover, under this additional assumption, we show that the firm behaves as if it faced quasi-hyperbolic (β - δ) discounting.

¹⁶Interestingly, the type of tax matters. Because a proportional dividend tax scales down both the cash flow paid out today and the market value of future payouts, it leaves the firm's intertemporal trade-off unchanged. A capital gains tax, by contrast, applies only when returns are realized through sale, so it makes future price appreciation less attractive for sellers relative to current dividends and creates the disagreement emphasized here.

¹⁷See Moll (2023) on why this (i) makes it very hard/impossible to solve, and (ii) it is an unrealistic assumption.

4.1 The Stock Price Elasticity

A crucial element of the firm's problem is its understanding of how its stock price, q_t , changes in response to its future dividend choices, $\{d_{t+s}\}_{s \geq 1}$. Like in the 3-period model of Section 3, the firm understands that the answer to this question lies in the households' Euler equation. Let $\mathcal{C}_t(\theta, b, h)$, $\Theta_{t+1}(\theta, b, h)$, $\Delta_t^+(\theta, b, h)$, and $\Delta_t^-(\theta, b, h)$, denote the policy functions for consumption, stock holdings, stock purchases, and stock sales, given (θ, b, h) . To avoid excessive clutter, we simplify the notation by removing the dependence on (θ, b, h) and, instead, including the household's index j as a superscript.

The Euler equation of household j is given by

$$(1 - \phi \Delta_t^{j-}) q_t = \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} (d_{t+1} + (1 - \phi \Delta_{t+1}^{j-}) q_{t+1}) \right] + \eta_t^j,$$

where η_t^j is the Lagrange multiplier on the households' short-selling constraint for stocks in (4), and the expectation is taken with respect to h_{t+1} . Let

$$\Phi_t^j \equiv \mathbb{E}_t [\phi \Delta_{t+1}^{j-}] + \phi \frac{\text{cov}_t(u'(\mathcal{C}_{t+1}^j), \Delta_{t+1}^{j-})}{\mathbb{E}_t [u'(\mathcal{C}_{t+1}^j)]}, \quad (15)$$

Then, we can rewrite the households' Euler equation as

$$(1 - \phi \Delta_t^{j-}) q_t = \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} \right] [d_{t+1} + (1 - \Phi_t^j) q_{t+1}] + \eta_t^j. \quad (16)$$

The household uses its expected IMRS between t and $t+1$ to discount future cash flows. Moreover, the stock valuation has an extra component, Φ_t^j , which reflects the household's expected marginal transaction cost. A higher transaction cost (indexed by ϕ) reduces the household's valuation, as it lowers the stock return in a sale. This effect is amplified by the fact that households will sell (i.e. $\Delta_{t+1}^{j-} > 0$) when their marginal utility of consumption, $u'(\mathcal{C}_{t+1}^j)$, is relatively high. Hence, a positive covariance between marginal utility and quantity sold further depresses asset prices.

Equation (16) is at the core of the firm's investment decisions. In the 3-period model of Section 3, we used (16) to iteratively replace $(1 - \phi \Delta_t^{j-}) q_t$ from the firm's problem (10). This strategy is not helpful in the infinite-horizon version of the model for two reasons. First, recall that in the simple model, it was important that there were no agents against their short-selling constraints, so that $\eta_t^j = 0$ for all agents and periods. However, we cannot rule this out in the infinite-horizon model with income risk. Households that suffer long spells of low employment may deplete their stock holdings entirely and become constrained. The Lagrange multiplier

associated with the short-selling constraint would then be positive for those agents. Second, after introducing the Euler equations into the firm's problem, we obtained a characterization of the firm's objective that depended on the *whole distribution of marginal utilities at all horizons*. This would require the firm to obtain information on agents' *individual* marginal utilities over time, information that is not readily available from aggregate variables or the price system.

Our strategy will use equation (16), but it will leverage the fact that if some stock buyer is not constrained by the borrowing limit for bonds, it is possible to recover the sensitivity of the stock price to the firm's investment plan from two financial prices: the risk-free interest rate and a *liquidity premium*.

4.2 The Liquidity Premium

For buyers, condition (16) reduces to

$$q_t = \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} \right] \left[d_{t+1} + (1 - \Phi_t^j) q_{t+1} \right],$$

because $\Delta_t^{j-} = \eta_t^j = 0$ for them. In principle, buyers may disagree on how much they value the dividend d_{t+1} relative to the *ex-dividend* market price q_{t+1} .¹⁸ That is, $\mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} \right]$ and Φ_t^j could vary depending on the buyer's initial portfolio and realization of the employment shock. However, their valuations coincide if they are not constrained by the borrowing limit for bonds. To see this, note that the agents' optimality condition for bonds is given by

$$\frac{1}{1+r_t} = \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} \right] + v_t^j, \quad (17)$$

where v_t^j is the Lagrange multiplier associated with the borrowing constraint (5). For agents who are not constrained by the borrowing limit, $v_t^j = 0$, and hence

$$\mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} \right] = \frac{1}{1+r_t} \implies q_t = \frac{d_{t+1}}{1+r_t} + \frac{(1 - \Phi_t^b) q_{t+1}}{1+r_t}, \quad (18)$$

so Φ_t^b must be the same for all buyers who are not constrained by the borrowing limit, who we denote with the superscript b . In what follows, we focus on economies where there exists a buyer who is not constrained by the borrowing limit and verify the existence of that individual in the numerical exercise. The next lemma characterizes this claim analytically.

Lemma 2. *A stock buyer who is not constrained by the borrowing limit for bonds exists if and*

¹⁸All buyers agree on the stock's valuation; if they did not, they would not be willing to buy at the common price q_t .

only if

$$\{j : \Delta_t^{j+} > 0\} \cap \operatorname{argmax}_{j \in [0,1]} \frac{q_t(1 - \phi \Delta_t^{j-})}{d_{t+1} + (1 - \Phi_t^j)q_{t+1}} \neq \emptyset. \quad (19)$$

Note that the fraction in (19) equals $\mathbb{E}_t \left[\beta u'(\mathcal{C}_{t+1}^j) / u'(\mathcal{C}_t^j) \right]$ for households that are not short-selling constrained. Then, Lemma 2 says that the highest expected IMRS cross-sectionally belongs to a buyer. This connects directly to (17): maximizing the expected IMRS minimizes $v_t^j \geq 0$, as in Alvarez and Jermann (2000). Since β is common across households, a higher expected IMRS reflects a stronger saving motive rather than greater patience, so the condition requires that some buyer's saving motive is (weakly) the strongest in the economy. The same expression is the reciprocal of $R_t^{\theta,j} = \left[d_{t+1} + (1 - \Phi_t^j)q_{t+1} \right] / \left[(1 - \phi \Delta_t^{j-})q_t \right]$, the gross return a household earns by retaining a marginal unit rather than selling it. The condition therefore says that a buyer earns the lowest such return, which for an unconstrained buyer is exactly the gross bond return, $R_t^{\theta,b} = 1 + r_t$.

Rearranging (19) connects the condition to our transaction cost and, in particular, to the persistence in trading: some buyer b , that is, some b with $\Delta_t^{b+} > 0$, must satisfy

$$\Phi_t^j - \phi \Delta_t^{j-} \left(1 + \frac{d_{t+1}}{q_{t+1}} - \Phi_t^b \right) \leq \Phi_t^b, \quad \forall j \in [0, 1].$$

Such a buyer is not constrained by the borrowing limit, so Φ_t^b is the common value in (18). In a stationary equilibrium the condition reduces to $\Phi^j - (1 + r)\phi \Delta^{j-} \leq \Phi^b$. The left-hand side is the net cost of holding a marginal unit: the cost the household expects to pay when it sells in the future, less the cost it saves today by retaining the unit rather than selling it. The buyer b sells nothing today, so its net cost is simply Φ_t^b , and the condition requires that no seller face a higher one. The condition fails when a seller expects to sell sufficiently more next period, so that Φ_t^j is large, than it sells today, so that $\phi \Delta_t^{j-}$ is small. Such a seller would then earn the lowest return on the unit it retains, and (18) would no longer price the stock. We verify that the condition holds in our main calibration in Section 6 and across the parameterizations of Appendix E.3.

Under the conditions of Lemma 2, the buyers' expected marginal transaction cost has a familiar reading. Let the yield of the stock be $r_t^\theta \equiv \frac{d_{t+1} + q_{t+1}}{q_t} - 1$. Define the liquidity premium as the excess return of the stock over the return of the bond, i.e. $r_t^\theta - r_t = \Phi_t^b \frac{q_{t+1}}{q_t}$, which reduces to Φ^b in steady-state. Then, the pricing equation of the firm's stock depends on two widely used financial prices: the risk-free rate, r_t , and the liquidity premium, Φ_t^b . With this, we are ready to set up the firm's intertemporal problem.

4.3 The Firm's Problem

Recall that Lemma 1 together with Assumption 1 state that the objective of the firm is to maximize a weighted valuation given by

$$d_t + (1 - \bar{\Phi}_t)q_t. \quad (20)$$

The discussion in Section 4.2 implies that the stock price can be expressed as

$$q_t = \frac{d_{t+1}}{1 + r_t} + \frac{(1 - \Phi_t^b)q_{t+1}}{1 + r_t}. \quad (21)$$

The following assumption provides the final piece necessary to solve the firm's problem.

Assumption 2. *The firm takes $\{\bar{\Phi}_t, \Phi_t^b\}_{t \geq 0}$ as given.*

Assumption 2 states that the firm takes the average transaction cost $\bar{\Phi}_t$ and the liquidity premium Φ_t^b as given. Both are determined in equilibrium by households' trading patterns, summarized by the cross-sectional distribution of Δ_t^- (see their definitions in equations (9) and (15)). The content of the assumption is that the firm does not anticipate how its own investment decisions would alter these trading patterns. The firm always has the correct expectation with respect to the *levels* of these variables but ignores the potential *changes* that arise from the effect of its investment decisions on the households' trading strategies.¹⁹

The resulting pricing equation (21) is intuitive. Since dividends are distributed without frictions and the economy features no aggregate risk, the firm's dividend is discounted by the same discount factor as the bonds, i.e., $\frac{1}{1+r_t}$. Note that if $\phi = 0$, the pricing equation simplifies to $q_t = \frac{d_{t+1} + q_{t+1}}{1+r_t}$ and the problem of the firm is to maximize $d_t + q_t$. In contrast, when $\phi > 0$, the resale price q_{t+1} is further discounted by the liquidity premium Φ_t^b . Thus, the firm's problem will be to maximize (20) subject to (21).

4.4 Equilibrium

An equilibrium for this economy consists of household allocations

$\left\{ \left(c_{j,t}, b_{j,t+1}, \theta_{j,t+1}, \Delta_{j,t}^+, \Delta_{j,t}^- \right)_{\forall j \in [0,1]} \right\}_{t=0}^{\infty}$, firm allocations $\{l_t, k_{t+1}, d_t\}_{t=0}^{\infty}$ and aggregates $\{w_t, r_t, q_t, \mathcal{T}_t, B_{t+1}^G, \bar{\Phi}_t, \Phi_t^b, \Gamma_t\}_{t=0}^{\infty}$, such that, given $(\theta_{j,0}, b_{j,0})_{\forall j \in [0,1]}$, k_0 , B_0^G , and r_{-1} :

1. Given $\{d_t, w_t, r_{t-1}, q_t, \mathcal{T}_t\}_{t=0}^{\infty}$, households optimize;
2. Given $\{w_t, r_{t-1}, \bar{\Phi}_t, \Phi_t^b\}_{t=0}^{\infty}$, firms optimize²⁰;

¹⁹Grossman and Hart (1979) argue that since such an assumption involves an elasticity, the firm's beliefs are neither verified nor falsified in equilibrium.

²⁰The optimization of the firm will depend on whether we consider the solution with or without commitment. This will be made explicit in the next section.

3. $\{\bar{\Phi}_t, \Phi_t^b\}_{t=0}^\infty$ are consistent with the cross-sectional distribution $\{\Gamma_t\}_{t=0}^\infty$ according to (9) and (15);
4. The government budget constraint (6) holds for all t ;
5. Markets clear: $\int_{j \in [0,1]} h_{j,t} dj = I_t$, $\int_{j \in [0,1]} \theta_{j,t} dj = 1$, $\int_{j \in [0,1]} b_{j,t} dj = B_t^G$, for all $t \geq 0$. By Walras Law, the goods market also clears.

A steady-state equilibrium is an equilibrium in which firm allocations and aggregates are constant over time. Our analysis will focus on steady-state equilibria.

5 Equilibrium Characterization

In a steady-state equilibrium, $\bar{\Phi}_t = \bar{\Phi}$ and $\Phi_t^b = \Phi^b$ for all t . Iterating the price (21) forward and replacing it in the objective function, the problem of the firm simplifies to

$$V^F(k_t) = \max_{\{k_{t+s}\}_{s \geq 1}} F(k_t, k_{t+1}) + \frac{1 - \bar{\Phi}}{1 - \Phi^b} \sum_{s=1}^{\infty} \left(\frac{1 - \Phi^b}{1 + r} \right)^s F(k_{t+s}, k_{t+s+1}). \quad (\text{FP})$$

5.1 Quasi-Hyperbolic discounting

The program (FP) features quasi-hyperbolic discounting. To see this, let $\tilde{\delta} = \frac{1 - \Phi^b}{1 + r}$ and $\tilde{\beta} = \frac{1 - \bar{\Phi}}{1 - \Phi^b}$. Then, the value of the firm can be written as

$$V^F(k_t) = \max_{\{k_{t+s}\}_{s \geq 1}} F(k_t, k_{t+1}) + \tilde{\beta} \sum_{s=1}^{\infty} \tilde{\delta}^s F(k_{t+s}, k_{t+s+1}).$$

This is exactly the β - δ form of quasi-hyperbolic discounting (Laibson, 1997), with $\tilde{\beta}$ in the role of the present-bias parameter and $\tilde{\delta}$ in the role of the long-run discount factor. On the one hand, $\tilde{\delta}$ represents the “natural” discount factor obtained from the pricing equation (21): it combines the risk-free rate, r , and the liquidity premium, Φ^b . That is, $\tilde{\delta}$ reflects the discounting implied in asset prices. On the other hand, $\tilde{\beta}$ summarizes the disagreement between current and future owners. If $\bar{\Phi} = \Phi^b$, then current and future owners agree on how to value the firm, so $\tilde{\beta} = 1$ and the problem simplifies to a standard one with exponential discounting. However, if $\bar{\Phi} > \Phi^b$ ($\bar{\Phi} < \Phi^b$), then current owners are more impatient (patient) than the market, which implies that $\tilde{\beta} < 1$ ($\tilde{\beta} > 1$) and the firm exhibits present (future) bias. In the next proposition, we decompose the difference $\bar{\Phi} - \Phi^b$ to identify the factors contributing to time-inconsistency.

Proposition 1. *The difference $\bar{\Phi} - \Phi^b$ is the sum of a persistence effect, a concentration effect,*

and a risk premium:

$$\begin{aligned} \bar{\Phi} - \Phi^b = & \underbrace{\phi \left(\tilde{\mathbb{E}}_t \left[\Delta_t^{j-} \right] - \tilde{\mathbb{E}}_t \left[\mathbb{E}_t \left[\Delta_{t+1}^{j-} \right] \middle| \text{buyer} \right] \right)}_{\text{persistence effect}} + \underbrace{\phi \widetilde{\text{cov}}_t \left(\theta^j, \Delta_t^{j-} \right)}_{\text{concentration effect}} - \\ & \underbrace{\phi \tilde{\mathbb{E}}_t \left[\frac{\text{cov}_t \left(u'(\mathcal{C}_{t+1}^j), \Delta_{t+1}^{j-} \right)}{\mathbb{E}_t \left[u'(\mathcal{C}_{t+1}^j) \right]} \middle| \text{buyer} \right]}_{\text{risk premium}}, \end{aligned}$$

where $\tilde{\mathbb{E}}_t$ and $\widetilde{\text{cov}}_t$ are the equal-weighted cross-sectional mean and covariance across agents j under $d\Gamma$, $\mathbb{E}_t, \text{cov}_t$ are taken over an agent's own next shock $dF(h'|h)$, and the conditionings are on being a buyer who is not borrowing constrained today, that is, on belonging to the set of buyers that prices the stock.

Proposition 1 presents two results. First, it shows that time-inconsistency can only emerge in an economy with transaction costs, i.e., $\phi > 0$. Setting $\phi = 0$ gives $\Phi^b = \bar{\Phi}$ and $\tilde{\beta} = 1$, recovering the standard exponential discounting problem. Second, it shows that the degree and direction of time-inconsistency (i.e., present or future bias) depend on the interaction of three economic forces.

The persistence effect captures the difference in transaction costs for different agents. On the one hand, we have the average transaction cost across all agents. On the other hand, we have the expected transaction cost next period for the buyers who are not constrained in the bond market (who price the stock). If there is *persistence* in trades, so that buyers in t expect to also be buyers in $t + 1$, then the expected transaction cost of buyers is smaller than the average across all agents, i.e., $\tilde{\mathbb{E}}_t \left[\Delta_t^{j-} \right] > \tilde{\mathbb{E}}_t \left[\mathbb{E}_t \left[\Delta_{t+1}^{j-} \right] \middle| \text{buyer} \right]$, which represents a force towards present bias. This is the case we expect to obtain in our model, as purchasing the illiquid asset is profitable only if there is a relatively high chance that the agent will *not* sell the asset immediately.

The concentration effect arises because the firm internalizes the ownership-weighted transaction cost rather than the simple average across all agents. Because shares integrate to one, the two differ exactly by the cross-sectional covariance between ownership and sales: $\bar{\Phi}/\phi = \tilde{\mathbb{E}}_t \left[\theta^j \Delta_t^{j-} \right] = \tilde{\mathbb{E}}_t \left[\Delta_t^{j-} \right] + \widetilde{\text{cov}}_t \left(\theta^j, \Delta_t^{j-} \right)$. The covariance can be written as $\widetilde{\text{corr}}_t(\theta^j, \Delta_t^{j-}) \sigma_\theta \sigma_{\Delta^-}$, where σ_θ measures how concentrated ownership is, σ_{Δ^-} measures how dispersed sales are, and the correlation sets the sign. If the correlation is positive, so that large owners tend to sell more, their high costs receive extra weight. Holding fixed the correlation and the dispersion of sales, a more concentrated ownership structure then raises the concentration effect.²¹ The effect vanishes, however concentrated ownership is, if the large owners do not trade.

²¹The concentration effect is not special to the ownership-weighted objective of Assumption 1. In Appendix B.1, we generalize the weights used in the firm's problem and obtain the same term, with θ replaced by the aggregation weight w .

The third term in the expression is the risk premium, which captures that agents tend to sell the most precisely when they need the resources the most, that is, in high marginal-utility states. When this covariance is positive, the stock is a risky investment, and buyers price that risk, which represents a force towards future bias.

The net effect of the three forces depends on parameter values. In our calibrated economy, the persistence and concentration effects are positive and outweigh the risk premium, so that $\bar{\Phi} > \Phi^b$ and the firm exhibits present bias.

Next, we solve the problem of the firm analytically. First, we consider the problem of a firm that can commit to future policies. This firm chooses an investment plan in the initial period and never reoptimizes. Then, we study the problem of a firm that cannot commit to an investment plan but fully anticipates its future incentives and understands how present decisions can affect its future self.

5.2 Firm with commitment

A firm that can commit to a future investment policy chooses a sequence of capital $\{k_{t+s}\}_{s=1}^{\infty}$ in period t to maximize its value:

$$V^F(k_t) = \max_{\{k_{t+s}\}_{s \geq 1}} F(k_t, k_{t+1}) + \tilde{\beta} \sum_{s=1}^{\infty} \tilde{\delta}^s F(k_{t+s}, k_{t+s+1}).$$

The first-order condition with respect to k_{t+1} is

$$F_2(k_t, k_{t+1}) + \tilde{\beta} \tilde{\delta} F_1(k_{t+1}, k_{t+2}) = 0$$

and the first-order condition with respect to k_{t+s+1} for $s \geq 1$ is

$$F_2(k_{t+s}, k_{t+s+1}) + \tilde{\delta} F_1(k_{t+s+1}, k_{t+s+2}) = 0$$

where $F_1(k_t, k_{t+1}) \equiv z\alpha k_t^{\alpha-1} + (1 - \delta)$ and $F_2(k_t, k_{t+1}) \equiv -1$. Note that when $\tilde{\beta} \neq 1$, the choice of initial investment involves a different trade-off than future investment.

Focusing on steady-state equilibria and using that $z = (1 - \gamma\psi) \left(\frac{\gamma\psi}{w}\right)^{\frac{\gamma\psi}{1-\gamma\psi}}$, $\alpha = \frac{(1-\gamma)\psi}{1-\gamma\psi}$, and $w = \gamma\psi k^{(1-\gamma)\psi} H^{\gamma\psi-1}$ (where H is the unconditional expectation of the employment process), we get that the level of capital when the firm has commitment is

$$k^C = \left(\frac{(1-\gamma)\psi\tilde{\delta}}{1-\tilde{\delta}(1-\delta)} H^{\gamma\psi} \right)^{\frac{1}{1-(1-\gamma)\psi}}. \quad (22)$$

5.3 No commitment

We now turn to the problem of a firm that cannot commit to an entire sequence of investments but optimizes period-by-period. We assume that the firm fully understands its future-self incentives and takes them into account when making its choice. This is known as a *sophisticated solution* in the literature on time-inconsistent preferences.

We solve for a Markov perfect equilibrium (MPE). Given an initial level of capital k , the firm takes the future policy function, $k' = g(k)$, as given. Thus, we can write the firm's problem as

$$V^F(k) = \max_{k'} F(k, k') + \tilde{\beta} \tilde{\delta} W(k')$$

subject to

$$W(k') = F(k', g(k')) + \tilde{\delta} W(g(k')).$$

Let $k' = \zeta(k)$ denote the solution of this equation for a given level of initial capital k . A MPE requires $\zeta(k) = g(k)$. The function $W(\cdot)$ takes into account the preference disagreement between the current and the future firm. If $\tilde{\beta} = 1$, the problem simplifies to a standard Bellman equation, with $V^F(k) = W(k)$. However, when $\tilde{\beta} \neq 1$, there is disagreement between the current and future firm. In particular, the current firm uses the rate $\tilde{\delta}$ to discount the future two periods ahead yet, the future firm intends to discount the immediate future at the rate $\tilde{\beta} \tilde{\delta}$ (because $\zeta(k) = g(k)$).

The following proposition shows that there is a unique differentiable solution that is the limit of the finite-horizon version of this problem. Moreover, it shows how the firm's capital depends on the degree of present bias, coinciding with the commitment level when $\tilde{\beta} = 1$.

Proposition 2. *The firm's problem with no commitment can have multiple, differentiable, Markovian solutions, but only one is the limit of the unique finite-horizon equilibrium. This solution is given by*

$$k^N = \left(\frac{(1-\gamma) \psi \tilde{\beta} \tilde{\delta}}{1 - \tilde{\beta} \tilde{\delta} (1-\delta)} H^{\gamma \psi} \right)^{\frac{1}{1-(1-\gamma)\psi}}, \quad (23)$$

assuming that $\tilde{\beta} \tilde{\delta} (1-\delta) < 1$. Holding $\tilde{\delta}$ fixed, k^N is strictly increasing in $\tilde{\beta}$, with $k^N = k^C$ when $\tilde{\beta} = 1$.

Equation (23) coincides with the commitment expression (22) once $\tilde{\delta}$ is replaced by $\tilde{\beta} \tilde{\delta}$. A firm that cannot commit thus discounts as if it were more impatient, increasingly so the lower is $\tilde{\beta}$. The reason is that each self raises current dividends at the expense of future capital, a best response to its successors, who share the same incentive.

Commitment and the capital gap. Proposition 2 varies $\tilde{\beta}$ holding $\tilde{\delta}$ fixed. This isolates the role of $\tilde{\beta}$, but a committing and a non-committing firm need not face the same $\tilde{\delta}$, so the proposition does not by itself answer whether present bias reduces the capital stock of a firm.

To make progress on this question, we consider the following counterfactual: starting from an economy with only non-committing firms, we append a mass zero of firms, with the same technology and trading frictions, that can commit.²² The following lemma helps make the comparison precise.

Lemma 3. *Consider the stationary equilibrium of an economy with only non-committing firms. Appending a mass-zero set of committing firms to this economy leaves all aggregate quantities, the interest rate r , and the wealth distribution Γ unchanged. The liquidity premium Φ^b , and hence $\tilde{\delta}$, are common to committing and non-committing firms, and their steady-state price-dividend ratios coincide:*

$$\frac{q^C}{d^C} = \frac{q^N}{d^N} = \frac{1}{(1+r)(1-\tilde{\delta})} = \frac{1}{r+\Phi^b}.$$

Consequently, present bias depresses investment: $k^N < k^C$ if and only if $\tilde{\beta} < 1$.

Given this result, we characterize how the *capital gap*, i.e., the log difference between the capital of non-committing and committing firms, varies with the model's structural parameters.

Proposition 3. *In the economy of Lemma 3, let $\mathcal{P}$ be a parameter that affects the capital gap only through $\tilde{\beta}$ and $\tilde{\delta}$.²³ Then*

$$\frac{\partial}{\partial \mathcal{P}} \left[\log \left(\frac{k^N}{k^C} \right) \right] \propto \underbrace{\frac{1}{r^N + \delta} \frac{1}{\tilde{\delta} \tilde{\beta}^2}}_{>0} \frac{\partial \tilde{\beta}}{\partial \mathcal{P}} + \underbrace{\left(\frac{1}{r^N + \delta} \frac{1}{\tilde{\beta} \tilde{\delta}^2} - \frac{1}{r^C + \delta} \frac{1}{\tilde{\delta}^2} \right)}_{<0 \text{ iff } \tilde{\beta} < 1} \frac{\partial \tilde{\delta}}{\partial \mathcal{P}}, \quad (24)$$

where $r^N = \frac{1}{\tilde{\beta} \tilde{\delta}} - 1$ and $r^C = \frac{1}{\tilde{\delta}} - 1$.

The capital gap moves through two channels. The first operates through $\tilde{\beta}$, the disagreement between current and future owners. Its coefficient is positive, so any parameter change that lowers $\tilde{\beta}$ widens the gap; this is the *present-bias channel*, governed by the persistence, concentration, and risk-premium components of $\bar{\Phi} - \Phi^b$ in Proposition 1. The second operates through $\tilde{\delta}$, the asset-price discount common to both firms. When $\tilde{\beta} < 1$ its coefficient is negative, so parameter changes that raise $\tilde{\delta}$ also widen the gap. The reason is that, when $\tilde{\beta} < 1$, the non-committing firm's capital is less sensitive to $\tilde{\delta}$ than the committing firm's, so a common increase in $\tilde{\delta}$ raises k^C by more than k^N . Whether a change in a given parameter increases or

²²We require decreasing returns for this exercise to ensure interior solutions to both firms' problems. Since households now hold shares of two firms, the transaction cost is specified as a function of the *fraction* of a firm's outstanding shares that is traded: a household selling a fraction x of a firm with ε shares outstanding pays $\frac{\phi}{2} x^2 q \varepsilon$, which coincides with the cost in (2) when a single firm's shares are normalized to one, and which makes the friction independent of the number of shares issued.

²³This covers every parameter whose comparative statics we study below: β , ϕ , σ , ρ_h , σ_h , $\bar{b}$ and B^G . It excludes δ , γ and ψ . As the proof shows, $\log(k^N/k^C) = \frac{1}{1-\alpha} [\log(r^C + \delta) - \log(r^N + \delta)]$, so δ also moves the bracket directly and γ and ψ also move the factor $1/(1-\alpha)$.

reduces $\tilde{\beta}$ and $\tilde{\delta}$ depends on its general-equilibrium effects on asset prices and trading strategies, which lack closed-form solutions. Thus, we leave the analysis of the signs of $\partial \tilde{\beta} / \partial \mathcal{P}$ and $\partial \tilde{\delta} / \partial \mathcal{P}$, parameter by parameter, to the numerical section.

It is worth noting that the analysis so far has rested on Assumptions 1 and 2, but our main results are robust to relaxing them. Without Assumption 2 (the firm takes $\bar{\Phi}_t$ and Φ_t^b as given), the three-period example of Section 3 shows that time-inconsistency and present bias survive. The firm's discounting then need not collapse to the two-parameter β - δ form of program (FP): without Assumption 2, the firm's problem no longer reduces to iterating the marginal buyer's pricing equation, and nothing guarantees that its discount factors decline geometrically beyond the first period. Without Assumption 1 (the ownership-weighted objective), Appendix B shows that our main qualitative results survive with generalized weights $w(\theta, b, h)$. The qualitative mechanism is therefore robust to both assumptions; Assumption 2 is what sharpens the firm's discounting into the two-parameter β - δ form.

5.4 Commitment and the Corporate Discount Rate

The previous results provide a rationale to the findings in Gormsen and Huber (2025), which document the existence of a wedge between the discount rate used by firms in investment decisions and their cost of capital. To see this in our model, let Λ denote the (log of the) firm's discount factor. When the firm cannot commit to an investment plan, Λ is given by

$$\Lambda \equiv -\log \tilde{\beta} \tilde{\delta} \approx r + \bar{\Phi}.$$

Gormsen and Huber (2025) decompose the firm's discount factor into a *financial cost of capital*, r^{fin} , and a *discount rate wedge*, κ , such that

$$\Lambda = r^{fin} + \kappa.$$

In the model without commitment, we have

$$r^{fin} \equiv \log \left(\frac{1}{\tilde{\delta}} \right) \approx r + \Phi^b, \quad \text{and} \quad \kappa \equiv \log \left(\frac{1}{\tilde{\beta}} \right) \approx \bar{\Phi} - \Phi^b.$$

Thus, when firms cannot commit, present bias can generate a positive corporate discount-rate wedge.²⁴ In contrast, a firm that can commit to an investment plan discounts cash flows at $\tilde{\delta}$, so the wedge is zero. The wedge is therefore the same object that governs the present-bias channel in Proposition 3.

²⁴Gormsen and Huber (2025) define the financial cost of capital as the weighted average of the costs of debt and equity. Because our baseline firm issues only equity, this equals its cost of equity. The buyer's pricing equation implies $r_t^\theta \equiv (d_{t+1} + q_{t+1})/q_t - 1 = r_t + \Phi_t^b q_{t+1}/q_t$, which reduces to $r^\theta = r + \Phi^b$ in steady state.

5.5 Corporate Bonds

Up to now, we have assumed that the firm funds itself from retained earnings. We now explore the consequences of allowing the firm to also issue illiquid corporate bonds. Suppose the firm can borrow up to an exogenous limit at the gross interest rate $1 + r^{cb} = \frac{1+r}{1-\tilde{\phi}}$. The parameter $\tilde{\phi}$ captures a premium that corporate bonds pay relative to the risk-free rate in reduced form. In contrast, we assume that the rate of return from savings is equal to r . Because $\frac{1}{1+r} > \tilde{\beta}\tilde{\delta}$ and $\frac{1}{1+r} > \tilde{\delta}$, the firm never finds it optimal to save.

The firm's borrowing decision depends on how $\frac{1}{1+r^{cb}}$ compares to $\tilde{\beta}\tilde{\delta}$ and $\tilde{\delta}$. Note that the discount rate the firm uses for this comparison depends on its ability to commit to an investment plan. It might seem natural to conclude that, if the premium on corporate bonds is higher than the premium on stocks ($\tilde{\phi} > \Phi^b$), the firm does not issue corporate debt. However, this logic is correct only if the firm has commitment. If it does not, the firm uses $\tilde{\beta}\tilde{\delta} = \frac{1-\bar{\Phi}}{1+r}$ to discount future cash flows and, hence, the firm may borrow even if the corporate rate is high relative to the stock market discount. This result provides a rationalization for corporate bond issuance that does not rely on the canonical tax advantage of debt.

Important for our results is that incorporating corporate bonds does not affect our conclusions: the firm still discounts futures cash flows using $\tilde{\beta}\tilde{\delta}^s$.²⁵ The following proposition formalizes this discussion.²⁶

Proposition 4. *Suppose the firm suffers from present bias and it has access to the bond market with an interest rate of $1 + r^{cb} = \frac{1+r}{1-\tilde{\phi}}$. If $\tilde{\phi} < \Phi^b$ the firm always borrows to the limit independently of its degree of commitment. If $\Phi^b < \tilde{\phi} < \bar{\Phi}$ the firm without commitment borrows up to the limit, while the firm with commitment does not borrow. If $\tilde{\phi} > \bar{\Phi}$ neither firm borrows. Furthermore, optimal levels of capital are determined according to (22) and (23) with and without commitment, respectively.*

6 Quantitative Evaluation

We now evaluate the model quantitatively, focusing on how transaction costs affect the corporate discount-rate wedge and aggregate capital, and on how commitment governs the link between them. Section 5.4 showed that present bias generates a positive wedge only when firms cannot commit to future investment policies; with commitment, the wedge is zero. We therefore take the no-commitment economy as our benchmark and use the commitment economy as a counterfactual in which the wedge is absent.

²⁵Naturally, whether the firm can borrow or not can affect the general equilibrium determination of $\tilde{\beta}$ and $\tilde{\delta}$. However, its qualitative properties do not change.

²⁶Note that the firm's borrowing problem also suffers from time inconsistency. We focus on the same refinement of MPE as in Proposition 2.

Table 2: Parameter Values

Parameter	Value	Source/Target
Returns to scale ψ	1.000	Constant returns to scale
Production weight on labor γ	0.660	Labor share
Depreciation δ	0.013	5% annually
Risk aversion σ	2.000	Standard CRRA value
Labor persistence ρ_h	0.966	Quarterly income process Auclert et al. (2021)
Innovation s.d. of log labor income σ_h	0.238	Quarterly income process Auclert et al. (2021)
Borrowing limit $\bar{b}$	-0.660	One labor income, Kaplan et al. (2018)
Government debt B^G	1.040	26% of annual GDP, Kaplan et al. (2018)
Maximum borrowing wedge $\bar{\xi}_b$	0.0147	6% annually, Kaplan et al. (2018)
Borrowing-wedge scale s_b	0.000792	Smooth at zero
Discount factor β	0.9883	Capital/annual GDP = 3
Transaction-cost scale ϕ	2.460	Wedge = 4.3 p.p. Gormsen and Huber (2025)

Note: The model period is one quarter. Depreciation, discounting, the income process, and the borrowing wedge are parameterized quarterly. Stock-to-GDP ratios use annual GDP.

6.1 Calibration

We calibrate the model at a quarterly frequency, taking standard parameters from the heterogeneous-agent literature, following [Kaplan et al. \(2018\)](#) and [Auclert et al. \(2021\)](#) whenever possible. We internally calibrate the household discount factor and the transaction-cost scale jointly to match the capital-to-GDP ratio and the corporate discount-rate wedge from [Gormsen and Huber \(2025\)](#). Table 2 summarizes the parameter values.

We assume that the production function exhibits constant returns to scale, $\psi = 1$. The labor share is set to $\gamma = 0.66$ and the depreciation rate to $\delta = 0.013$, which corresponds to 5 percent annually. For the household, we set the coefficient of relative risk aversion to $\sigma = 2$. We also set the borrowing limit to $\bar{b} = -\gamma = -0.66$, equal in absolute value to average quarterly labor income, as in [Kaplan et al. \(2018\)](#).²⁷ In Appendix E.2, we investigate the alternative case of no private borrowing, $\bar{b} = 0$. Following [Kaplan et al. \(2018\)](#), we introduce government debt as the net supply of liquid assets. We calibrate it to match the total quantity of net liquid assets of around 26 percent of annual GDP. Given our normalization, this implies $B^G = 0.26 \times 4 = 1.04$.

Households also face a premium on borrowing: debt carries an interest rate $r_t + \xi_b$, with ξ_b bounded above by 6 percent annually, as in [Kaplan et al. \(2018\)](#). To avoid an inaction region at zero, we assume that the wedge is a smooth increasing function of the amount borrowed, $b^- \equiv \max\{-b, 0\}$ (see Appendix E.1 for the details):

$$\xi_b(b^-) = \bar{\xi}_b \left[1 - \frac{s_b}{b^-} \left(1 - \exp \left\{ -\frac{b^-}{s_b} \right\} \right) \right].$$

We set $\bar{\xi}_b = (1.06)^{1/4} - 1 = 0.0147$, so that the upper bound equals the 6 percent annual wedge in [Kaplan et al. \(2018\)](#). The scale parameter s_b controls how rapidly the borrowing wedge

²⁷We normalize quarterly GDP to 1, so average quarterly labor income is $\gamma = 0.66$.

Table 3: Model and Data Moments

	Model	Data
<i>Target</i>		
Corporate discount-rate wedge, annualized percent	4.3	4.3
Capital to annual GDP	3.0	3.0
<i>Non-target</i>		
Liquidity premium, annualized bps	17	35
Transaction costs to annual GDP, percent	0.14	0.53
Variance log consumption / variance log income	0.32	0.30

Note: The corporate discount-rate wedge is from [Gormsen and Huber \(2025\)](#). The capital-to-GDP ratio is constructed using BEA fixed-assets and annual GDP data. The liquidity premium is from [van Binsbergen et al. \(2022\)](#). The transaction-cost is from [Greenwood and Scharfstein \(2013\)](#). The ratio of the cross-sectional variance of log consumption to the cross-sectional variance of log income is from [Krueger and Perri \(2006\)](#).

approaches its upper bound. We set $s_b = 0.000792$ so that the wedge reaches approximately 98 percent of its upper bound when debt equals 6 percent of average quarterly labor income. Thus, our specification eliminates the kink at zero while rapidly converging to that constant 6 percent wedge.

For log labor income, we follow the two-asset calibration of [Auclert et al. \(2021\)](#), setting persistence to $\rho_h = 0.966$ and the unconditional cross-sectional standard deviation to 0.92. The implied quarterly innovation standard deviation is $\sigma_h = 0.238$.

This leaves the two parameters we calibrate internally. We choose $(\beta, \phi) = (0.9883, 2.46)$ jointly to match an annual capital-to-GDP ratio of 3.0 and an average annualized corporate discount-rate wedge of 4.3 percentage points.²⁸

Model and Data Moments. Although the model is stylized, it matches the targeted moments and generates non-targeted moments of the same order of magnitude as in the data, as shown in Table 3. The first two rows report the targeted moments discussed above: the corporate discount-rate wedge and the capital-to-GDP ratio.²⁹

The next three rows report non-targeted moments. First, we compare the model-implied liquidity premium with the Treasury convenience yield estimated by [van Binsbergen et al. \(2022\)](#). Using SPX option prices, they infer a convenience-yield-free riskless rate and measure the Treasury convenience yield as the difference between this rate and the Treasury yield. At the six-month maturity, this spread is 35 basis points. This provides a useful benchmark for two reasons. Because the convenience-yield-free rate comes from S&P 500 index options, the

²⁸The wedge target is the average of the within-firm U.S. series reported in Figure 6 of [Gormsen and Huber \(2025\)](#). This series controls for time-invariant firm heterogeneity and changes in sample composition and adjusts firms' reported discount rates for overhead costs.

²⁹We measure capital using the Current-Cost Net Stock of Fixed Assets (FRED series K1TTOTL1ES000) and output using annual nominal GDP (FRED series GDPA). The average capital-to-GDP ratio over 2000–2019 is 3.0.

spread compares a riskless rate implied by equity-market prices with the Treasury yield, which is closely related to our premium of equity returns over bond returns. In addition, both rates price riskless payoffs, so the spread does not reflect compensation for aggregate risk. The model generates a liquidity premium of 17 basis points. The two moments are not identical, however: our model premium reflects the liquidity services provided by bonds, whereas the empirical convenience yield may also incorporate collateral value, regulatory benefits, and other money-like services. We therefore interpret the empirical estimate as a broad benchmark rather than an exact empirical counterpart.

Our model also generates transaction costs of an empirically relevant magnitude. As a broad empirical benchmark, we use [Greenwood and Scharfstein \(2013\)](#)'s estimate that fees and commissions from equity trading amount to 0.53 percent of GDP. The model-implied transaction costs equal 0.14 percent of GDP, approximately one-quarter of the empirical benchmark. We interpret this comparison as showing that the model generates trading costs of a realistic magnitude rather than directly matching the empirical measure.³⁰

Finally, the ratio of the cross-sectional variance of log consumption to the cross-sectional variance of log income is 0.32 in the model and 0.30 in the data ([Krueger and Perri, 2006](#)). This non-targeted moment summarizes the extent to which household consumption is smoothed relative to income dispersion. Its close fit indicates that the model generates an empirically plausible degree of consumption smoothing.

Quantitative Implications of the Theory. Table 4 connects the calibrated economy more directly to the theoretical results. Our theoretical analysis relied on the existence of an unconstrained buyer to derive the pricing equation used by the firm (see Section 4). With the borrowing wedge, the requirement becomes the existence of a buyer who does not borrow, as shown in Appendix E.1. We verify the condition in our benchmark economy: 98.4 percent of equity buyers hold nonnegative bond positions (see Panel A of Table 4), and Appendix E.3 reports that this share remains close to 98 percent across the perturbations considered there.

Panel B quantifies the decomposition in Proposition 1. The persistence effect accounts for 84.15 percent of $\bar{\Phi} - \Phi^b$, while the concentration effect accounts for another 17.05 percent. The risk-premium component offsets these forces by 1.20 percent. Thus, present bias in the calibrated economy is driven primarily by persistence in trading, with a quantitatively meaningful contribution from ownership concentration and only a small opposing risk-premium effect.

Finally, Proposition 2 establishes that, holding $\tilde{\delta}$ fixed, capital in the no-commitment economy is increasing in $\tilde{\beta}$. Figure 1 presents the related full general-equilibrium comparative statics. We solve the model in general equilibrium for different values of the transaction-cost

³⁰In Subsection 6.4, we study how the results change when the idiosyncratic shocks are less persistent. This exercise is motivated by interpreting the shocks more broadly than labor income risk: they may also capture heterogeneity in investment opportunities, which is likely to be less persistent. Reducing persistence raises the liquidity premium and transaction costs relative to GDP toward their empirical counterpart while generating a corporate discount-rate wedge of a similar magnitude.

Table 4: Theoretical Mechanisms in the Calibrated Economy

	Value
<i>Panel A: Pricing condition in Lemma 2</i>	
Stock buyers choosing nonnegative bond positions, percent	98.40
<i>Panel B: Decomposition in Proposition 1</i>	
Persistence-effect contribution, percent	84.15
Concentration-effect contribution, percent	17.05
Risk-premium contribution, percent	−1.20
<i>Panel C: Regression summary of Figure 1</i>	
Semi-elasticity of log capital with respect to the corporate wedge	−0.00194

Note: Panel A reports the stationary-mass share of stock buyers whose chosen next-period bond position is nonnegative in the benchmark no-commitment economy. Panel B reports each component's share of $\bar{\Phi} - \Phi^b$ in the benchmark no-commitment economy. Panel C summarizes the same no-commitment general-equilibrium economies reported in Figure 1. It reports the coefficient from an OLS regression of log steady-state capital on the annualized corporate discount-rate wedge, measured in basis points, across the capital-wedge pairs generated by varying the transaction-cost scale.

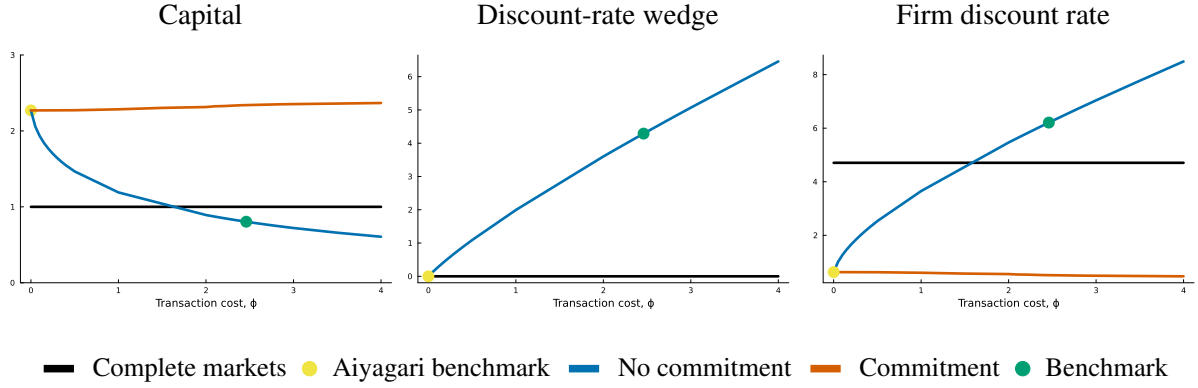
scale. Each value of the transaction-cost scale generates both a level of steady-state capital and a corporate discount-rate wedge, as reported in the first and second panels of the figure. Panel C summarizes these same no-commitment model outcomes in regression form by regressing log capital on the corporate discount-rate wedge. The estimated semi-elasticity implies that a 10-basis-point increase in the annualized corporate discount-rate wedge is associated with approximately 1.9 percent lower steady-state capital.

6.2 Transaction Costs and Investment in General Equilibrium

Relative to complete markets, the quantitative model adds three economic forces: precautionary savings, transaction costs, and lack of commitment. We disentangle them through a sequence of comparisons. Moving from complete markets to an incomplete-markets economy with $\phi = 0$ isolates precautionary savings. Varying ϕ with commitment captures the role of transaction costs. Comparing economies with and without commitment at a given ϕ identifies the additional effect of time inconsistency in firms' investment decisions.

Figure 1 presents the equilibrium outcomes for economies with different levels of the transaction cost parameter, ϕ . The black line corresponds to the complete-markets benchmark. In this economy, the firm's quarterly discount rate is $r = 1/\beta - 1$, independent of ϕ ; the third panel reports the corresponding annualized, continuously compounded rate, $4 \log(1 + r) = -4 \log \beta$. The yellow point depicts the frictionless incomplete-markets benchmark ($\phi = 0$), which is isomorphic to a one-asset Aiyagari economy under our calibration. In this case, the absence of trading frictions implies that the firm's problem is time-consistent. Precautionary savings lower the equilibrium interest rate and raise aggregate capital relative to the complete-markets benchmark.

Figure 1: Transaction Costs and Investment in General Equilibrium



Note: The figure reports equilibrium outcomes under different transaction costs, ϕ . Capital is normalized by its complete-markets level. The discount-rate wedge and firm discount rate are annualized using continuous compounding. The wedge is reported in percentage points and the firm discount rate in percent. The yellow marker denotes the incomplete-markets economy with $\phi = 0$, and the green marker denotes the calibrated benchmark, $\phi = 2.46$.

We first consider the joint effect of transaction costs and lack of commitment. The blue line in Figure 1 shows outcomes when markets are incomplete, transaction costs are positive ($\phi > 0$), and firms cannot commit to future policies. In this case, capital decreases monotonically with ϕ . For instance, when $\phi = 2.46$, the annualized corporate discount-rate wedge is 4.3 percentage points, and capital is about 20 percent lower than in the complete markets economy. Thus, the combination of trading frictions and lack of commitment overturns the typical over-accumulation of capital found in the Aiyagari framework.³¹

To isolate the additional effect of lack of commitment, we compare the blue and red lines at a given value of ϕ . The red line shows equilibrium outcomes when firms can commit. Under commitment, capital changes little with ϕ because the effective discount factor, $\tilde{\delta} = (1 - \Phi^b)/(1 + r)$, is nearly constant. This occurs because, as transaction costs rise, bonds become relatively more attractive than equity, which simultaneously increases the liquidity premium and lowers the interest rate. These two forces approximately offset, yielding only a small net effect on the stock of capital.

In contrast, without commitment, firms value next-period payouts using the effective discount factor $\tilde{\beta}\tilde{\delta} = \frac{1-\Phi}{1+r}$, where $\tilde{\beta} = \frac{1-\Phi}{1-\Phi^b}$. The third panel of Figure 1 reports the corresponding annualized, continuously compounded firm discount rate, $r_{\text{firm}}^{NC} = -4\log(\tilde{\beta}\tilde{\delta})$, which rises with transaction costs. A higher discount rate is equivalent to placing less weight on future payouts. Because equilibrium capital is increasing in $\tilde{\beta}\tilde{\delta}$, as shown in Proposition 2, this increase in the firm discount rate reduces the stock of capital. With commitment, the corresponding firm discount rate, $r_{\text{firm}}^C = -4\log\tilde{\delta}$, remains nearly constant. Thus, the absence of commitment con-

³¹The magnitude of the response of capital partly reflects the sensitivity of capital to the interest rate in neo-classical production models; see Winberry (2021) and Wroblewski (2025). In our model, general-equilibrium adjustment in the wage moderates this partial-equilibrium response.

verts trading frictions into a higher effective hurdle rate for investment. Section 7 returns to this comparison in a setting with aggregate shocks, where the wedge varies over time and shapes the transmission of those shocks to investment.

These aggregate results provide a mechanism consistent with concerns about corporate short-termism. Laurence Fink, CEO of BlackRock, warned that firms often “deliver immediate returns to shareholders, such as buybacks or dividend increases, while underinvesting in innovation, skilled workforces or essential capital expenditures” (Fink, 2015). In the model, the wedge arises only when firms cannot commit, and it reflects their present bias, which leads them to place less weight on future payouts than they would under commitment, and hence to invest less.

Appendix D studies a separate implication using an extension with two types of firms. The extension shows that both the commitment and no-commitment versions of the model can reproduce the cross-sectional pattern in Amihud and Levi (2023). The distinction is that, without commitment, higher transaction costs also generate a larger corporate discount-rate wedge and reduce investment; with commitment, they affect asset prices but do not create the same investment distortion.

6.3 Liquidity and the Corporate Discount Rate

In Appendix D we extend our baseline model to incorporate two types of firms and show that, without commitment, firms whose equity is less liquid have a higher corporate discount-rate wedge. We test this prediction by combining the firm-quarter corporate discount rates and costs of capital from Gormsen and Huber (2025) with firm-quarter relative spreads from CRSP.³²

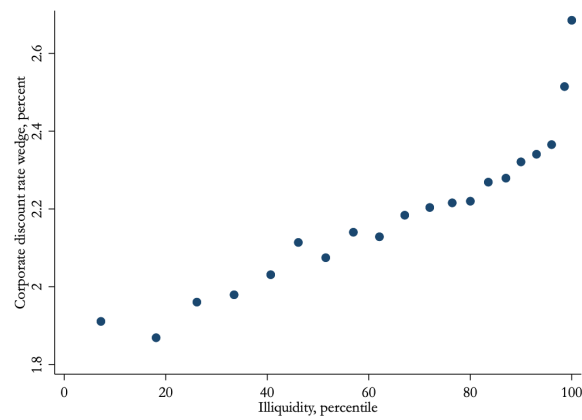
The relative spread provides a firm-level proxy for stock illiquidity.³³ We follow Naes et al. (2011) and consider daily stock data from the Center for Research in Security Prices (CRSP). For each day, we compute the relative spread as the quoted spread (the difference between the best ask and bid quotes) as a fraction of the midpoint price (the average of the best ask and bid quotes). We then aggregate the data to the firm-quarter level by averaging daily relative spreads within the quarter.

We start by showing that the corporate discount-rate wedge (discount rate net of cost of capital) and firms’ illiquidity are positively correlated. Figure 2 shows a binscatter of relative spreads and corporate discount-rate wedges: greater illiquidity is associated with higher discount-rate wedges, consistent with the no-commitment mechanism.

³²Here we use the publicly available raw firm-quarter wedge. Our preferred specifications include firm and time fixed effects and therefore identify the relationship between illiquidity and the wedge from within-firm variation over time. This focus on within-firm variation is consistent with the construction of the series used to calibrate the quantitative model.

³³We follow Amihud and Levi (2023) and use relative spread to measure illiquidity. In the model, the relative spread is $\tilde{\mathbb{E}} \left[\frac{\phi}{2} \Delta_t^- \mid \Delta_t^- > 0 \right]$, a cross-sectional expectation that reflects the expected transaction cost per share sold. Both the relative spread and the liquidity premium, Φ^b , are increasing functions of the expected marginal transaction cost faced by sellers, which is the key object in our model that captures illiquidity.

Figure 2: Illiquidity and Corporate Discount-Rate Wedge



Note: This figure is a binscatter. Firms are ranked into market-capitalization-weighted percentiles of relative spreads, and each dot reports the mean corporate discount-rate wedge within a bin of this percentile measure.

Next, Table 5 exploits within- and across-firm variation to show the strong relationship between illiquidity and firms' discount rates. The first column of Panel A regresses the corporate discount rate on relative spreads and includes time fixed effects. It uses cross-firm variation within quarters while absorbing aggregate variation across quarters. The second includes firm and time fixed effects and therefore uses within-firm variation over time, net of aggregate quarterly variation.³⁴ The table reports standardized beta coefficients, so the estimates are in standard-deviation units. We find that firms with higher relative spreads have higher discount rates, with coefficients that are statistically significant and economically large. Across firms, the standardized coefficient is about 0.58; adding firm fixed effects yields a coefficient of about 0.32.

The relationship in Panel A could reflect pass-through from illiquidity to the financial cost of capital rather than the wedge emphasized by the model. Panel B therefore uses the corporate discount-rate wedge as the dependent variable. The positive coefficients remain: the standardized coefficient is about 0.35 across firms and 0.28 when firm fixed effects are included. Columns 3 and 4 show that the results are robust to controlling for market capitalization. These findings alleviate potential concerns that our results are driven by variation in firm size or the cost of capital.

Overall, firm illiquidity is positively associated with the corporate discount-rate wedge: more illiquid firms apply discount rates that are higher relative to their cost of capital, consistent with the model. In the model, variation in this wedge can disconnect firms' investment decisions from movements in their cost of capital.

³⁴Firm fixed effects absorb all time-invariant firm characteristics. Therefore, to the extent that alternative determinants of the corporate discount-rate wedge, such as market power, are stable over our sample, they cannot explain the within-firm relationship between relative spreads and the wedge.

Table 5: Relative Spreads and Corporate Discount Rates

<i>Panel A: Corporate Discount Rate</i>				
Relative spread	0.582*** (0.030)	0.322*** (0.018)	0.568*** (0.031)	0.318*** (0.018)
Observations	27163	27158	27163	27158
R-squared	0.236	0.805	0.238	0.805
FE	Time	Firm, Time	Time	Firm, Time
Controls			Market Cap.	Market Cap.
<i>Panel B: Corporate Discount-Rate Wedge</i>				
Relative spread	0.349*** (0.024)	0.282*** (0.018)	0.353*** (0.024)	0.278*** (0.018)
Observations	27163	27158	27163	27158
R-squared	0.266	0.668	0.266	0.669
FE	Time	Firm, Time	Time	Firm, Time
Controls			Market Cap.	Market Cap.

Notes: The dataset is at the firm-quarter level and runs from 2002 to 2021. All specifications include time fixed effects; columns 2 and 4 add firm fixed effects. Columns 3 and 4 control for market capitalization. The table reports standardized beta coefficients. Standard errors (in parentheses) are clustered by firm. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.4 What Moves the Wedge

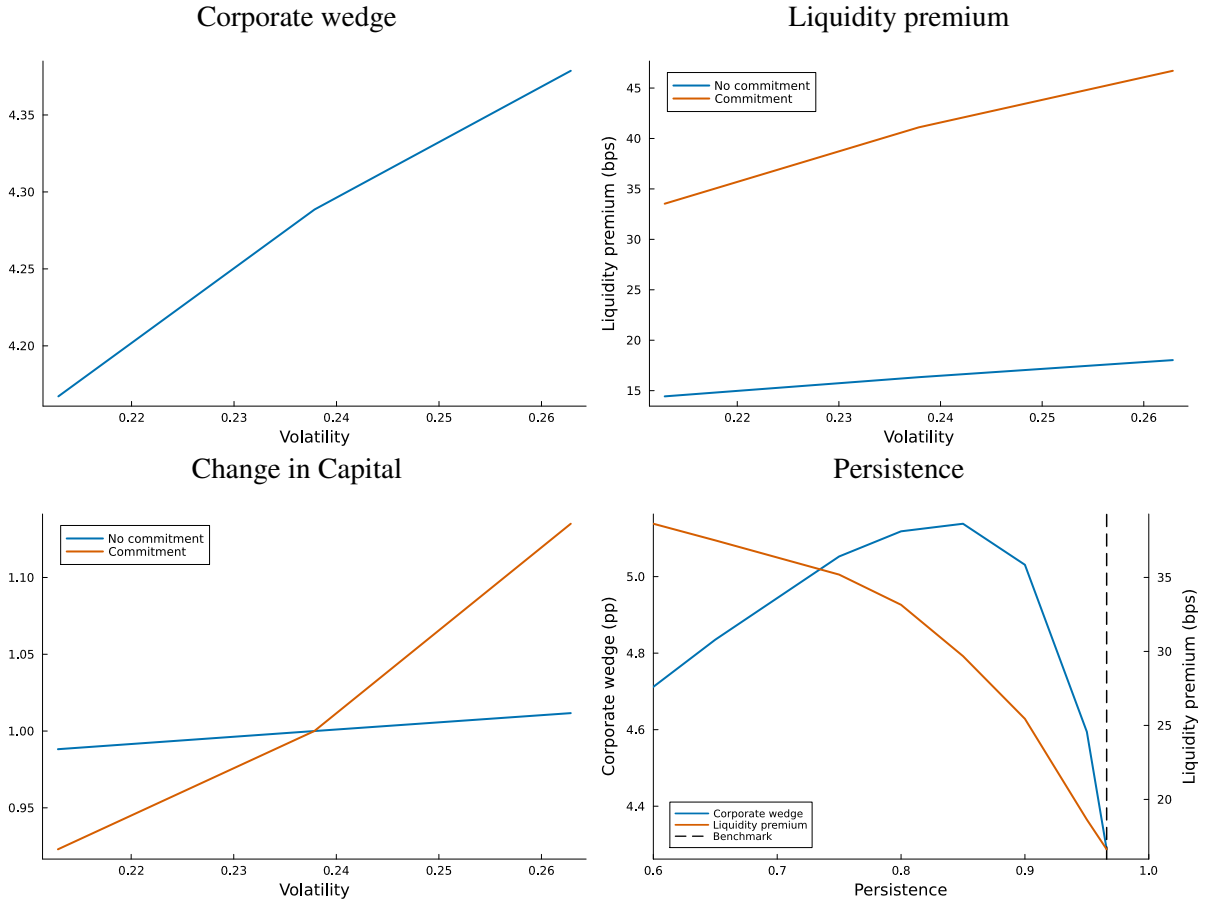
We now examine what moves the corporate discount-rate wedge and discuss its potential empirical implications. We organize the analysis around the demand for liquidity (idiosyncratic income risk and the persistence of income shocks) and its supply (government bonds).

Demand for Liquidity. We begin by examining the implications of rising uncertainty, which we interpret as an increase in the demand for liquidity. To do so, we increase σ_h , the innovation standard deviation of log labor income. Because mean labor income is renormalized, this change produces a mean-preserving increase in idiosyncratic income risk while holding persistence ρ_h fixed.

Higher uncertainty sets two forces in motion. It strengthens the precautionary savings motive and raises the demand for liquid assets, consistent with the mechanism in [Aiyagari \(1994\)](#). Without commitment, it also lowers $\tilde{\beta}$ by exacerbating the frequency and severity of reallocations between firm owners, which widens the corporate discount-rate wedge.

Figure 3 shows both forces at work. The first panel reports the corporate discount-rate wedge, which increases as uncertainty rises. The second reports the liquidity premium, which rises. The third displays capital with and without commitment: it rises in both economies, but by much less without commitment. While canonical heterogeneous-agent models predict that higher uncertainty raises capital through precautionary savings, our framework uncovers a

Figure 3: Income Risk and Persistence



Note: The first three panels report equilibrium outcomes as volatility, σ_h , varies. The first panel reports the corporate discount-rate wedge, κ in percentage points. The second reports the annualized liquidity premium, in basis points, with and without commitment. The third reports capital relative to the calibrated benchmark within each regime, $K(\sigma_h)/K(\sigma_h = 0.238)$. The fourth reports the annualized corporate discount-rate wedge, κ , in percentage points and the annualized liquidity premium corresponding to Φ^b , in basis points, as persistence, ρ_h , varies.

countervailing force strong enough to offset most of that channel.

These comparative statics provide a possible explanation for long-run trends in corporate discount rates. Gormsen and Huber (2025) document a secular increase in the corporate discount-rate wedge. One possibility is that trading frictions, captured by ϕ , have increased. However, Greenwood and Scharfstein (2013) show that transaction costs as a share of GDP, an empirical object closely linked to ϕ in our model, have not risen over time. By contrast, Guvenen (2009a) and Moffitt and Zhang (2018) document increases in idiosyncratic income risk. Braxton et al. (2026) reach the same conclusion in administrative tax data, and find that the rise in income risk is sharpest among high earners. Our comparative statics show that higher income risk widens the corporate discount-rate wedge even when ϕ remains constant. Rising income risk can therefore be a plausible explanation for the increase in the wedge.

Persistence Effect. Next we study the effects of changing the persistence of the income process. Specifically, we vary persistence ρ_h while adjusting σ_h to keep the unconditional variance of log labor income constant, so that only persistence changes. Although we model shocks as income shocks, this exercise is also informative about investment opportunities, which may be less persistent than labor income.

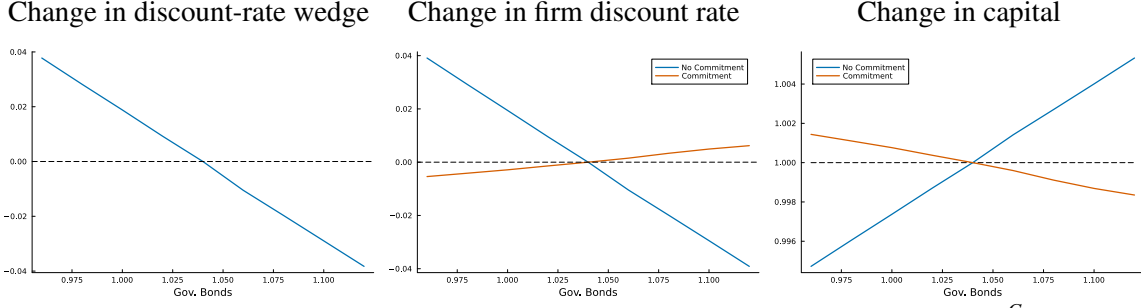
The fourth panel of Figure 3 reports the results. Lower persistence substantially raises the liquidity premium, from 17 basis points at the benchmark to close to its empirical counterpart of 35 basis points, while the corporate discount-rate wedge remains of a similar order of magnitude. The effect on the wedge is non-monotone, reflecting two competing forces. The persistence effect identified in Proposition 1 tends to widen the wedge, and corresponds to the rising portion of the wedge. Working against it, highly persistent economies are characterized by very little trade, with no trade at all in the $\rho_h = 1$ limit, which dampens both the wedge and the liquidity premium. Proposition 1 therefore identifies persistence in trading roles as a force toward a larger wedge, while changes in the persistence of income shocks also change the amount of trade and so have a non-monotone overall effect.

Supply of Liquidity. Now we study the effect of changing the supply of liquid assets, which in our model is the stock of government bonds, B^G , available to households. As B^G rises, the scarcity value of bonds falls, so bond prices decrease and the interest rate increases. Households rely less on equity for liquidity, and the liquidity premium falls.

Figure 4 shows that the effect on capital depends on the firm's ability to commit. On the one hand, with commitment, firms use the effective discount factor $\tilde{\delta} = (1 - \Phi^b)/(1 + r)$. With an increase in B^G , the rise in the interest rate more than offsets the fall in the liquidity premium, and so capital slightly falls. On the other hand, when firms cannot commit, the same increase in B^G also raises $\tilde{\beta}$, narrowing the corporate discount-rate wedge. This effect dominates, and capital rises.

The exercise varies government debt, but the corporate bonds of Section 5.5 supply liquidity through the same channel. They can be incorporated into general equilibrium by interpreting $\tilde{\phi}/(1 + r)$ as a per-unit issuance cost borne by the firm, with financial intermediaries holding the bonds and financing them by issuing claims to households that have the same payoff and liquidity properties as b_{t+1} . The household asset menu is then unchanged, and bond-market clearing depends on $B^G + B^{cb}$ rather than on B^G alone, where B^{cb} denotes corporate debt. Varying B^G therefore measures the liquid-asset-supply channel common to both, rather than corporate borrowing itself. Because Section 5.5 restricts attention to default-free bonds, the relevant empirical counterpart is AAA-rated corporate debt, which amounts to only about 0.3% of annual GDP (S&P Global Ratings, 2025). Incorporating it would therefore not materially affect our conclusions. The much larger stock of risky corporate debt would require modeling default and recovery, which is beyond the scope of the paper.

Figure 4: Government Bond Supply



Note: The figure reports equilibrium outcomes under different levels of government bonds, B^G . The first panel reports the change in the annualized corporate discount-rate wedge, in percentage points, relative to the no-commitment benchmark at $B^G = 1.04$; under commitment, this wedge is identically zero. The second panel reports the change in the annualized, continuously compounded firm discount rate, in percentage points, relative to each regime's benchmark at $B^G = 1.04$. The blue line corresponds to no commitment, with $r_{\text{firm}}^{NC} = -4 \log(\tilde{\beta} \tilde{\delta})$, and the red line corresponds to commitment, with $r_{\text{firm}}^C = -4 \log \tilde{\delta}$. The dashed horizontal lines in the first two panels mark zero changes. The third panel reports capital relative to the corresponding benchmark, $K(B^G)/K(B^G = 1.04)$.

Sensitivity Analysis. Appendix E.3 complements these exercises with local comparative statics around the benchmark. The results confirm the expected signs: a higher household discount factor raises the capital-output ratio, whereas a higher transaction cost increases the corporate discount-rate wedge and the liquidity premium while reducing capital. They also reinforce the role of household liquidity needs: greater income-shock volatility increases the liquidity premium and the wedge. The effect of income persistence is more nuanced: unlike the exercise in Subsection 6.4, which adjusts σ_h to hold the unconditional variance of income fixed, the perturbation here changes the unconditional variance as well as the amount of trade.

7 Time Inconsistency and the Transmission of Aggregate Shocks

The steady-state analysis establishes how trading frictions affect capital accumulation, but it does not reveal how the mechanism operates over the business cycle. Transitory aggregate shocks can simultaneously move the risk-free rate, the liquidity premium, and the corporate discount-rate wedge, thereby affecting firms' discount rates and investment. We study these dynamic responses to assess whether endogenous variation in the corporate discount-rate wedge is quantitatively important for shock transmission, and to separate it from movements in the liquidity premium, the channel emphasized in the asset-pricing literature on illiquidity.

More specifically, we study shocks to productivity, financial frictions, and income uncertainty and evaluate how they interact with firms' ability to commit. We consider standard *MIT shocks*: unanticipated transitory aggregate shocks with perfect foresight over subsequent aggregate paths, while idiosyncratic income risk remains stochastic. We compute first-order equilibrium responses using the sequence-space Jacobian method of Auclert et al. (2021). We compare the benchmark no-commitment economy ($\phi = 2.46$) with two alternatives: the same

calibration with commitment and a no-commitment economy with a higher transaction-cost scale ($\phi = 3$).³⁵ Appendix E.4 states the firm's dynamic problem.

The main takeaway is that firms' ability to commit is quantitatively more important for aggregate shock transmission than empirically plausible variation in the level of transaction costs: relative to the commitment economy, no commitment materially changes investment responses and reverses their sign following an uncertainty shock, whereas higher transaction costs within the no-commitment economy have only modest additional effects.

To interpret the impulse responses, we first decompose the rate at which firms discount future payouts. Let f denote the length of one model period in years, so that $f = 0.25$ corresponds to one quarter. Let $j = N$ denote the no-commitment economy and $j = C$ the commitment economy. Define the annualized, continuously compounded risk-free rate, liquidity premium, and corporate discount-rate wedge as

$$i_t^{\text{rf}} \equiv \frac{1}{f} \log(1 + r_t), \quad \ell_t \equiv -\frac{1}{f} \log(1 - \Phi_{t-1}^b), \quad \kappa_t \equiv -\frac{1}{f} \log \tilde{\beta}_t.$$

The lag in ℓ_t reflects that Φ_{t-1}^b is the expectation, formed at $t - 1$, of the transaction cost buyers face at t , so it is predetermined when a date- t shock arrives. We refer to $i_t^{\text{fin}} \equiv i_t^{\text{rf}} + \ell_t$ as the financial cost of capital. The discount rate of the firm without commitment is

$$i_t^{F,N} = -\frac{1}{f} \log(\tilde{\beta}_t \tilde{\delta}_t) = i_t^{\text{rf}} + \ell_t + \kappa_t = i_t^{\text{fin}} + \kappa_t,$$

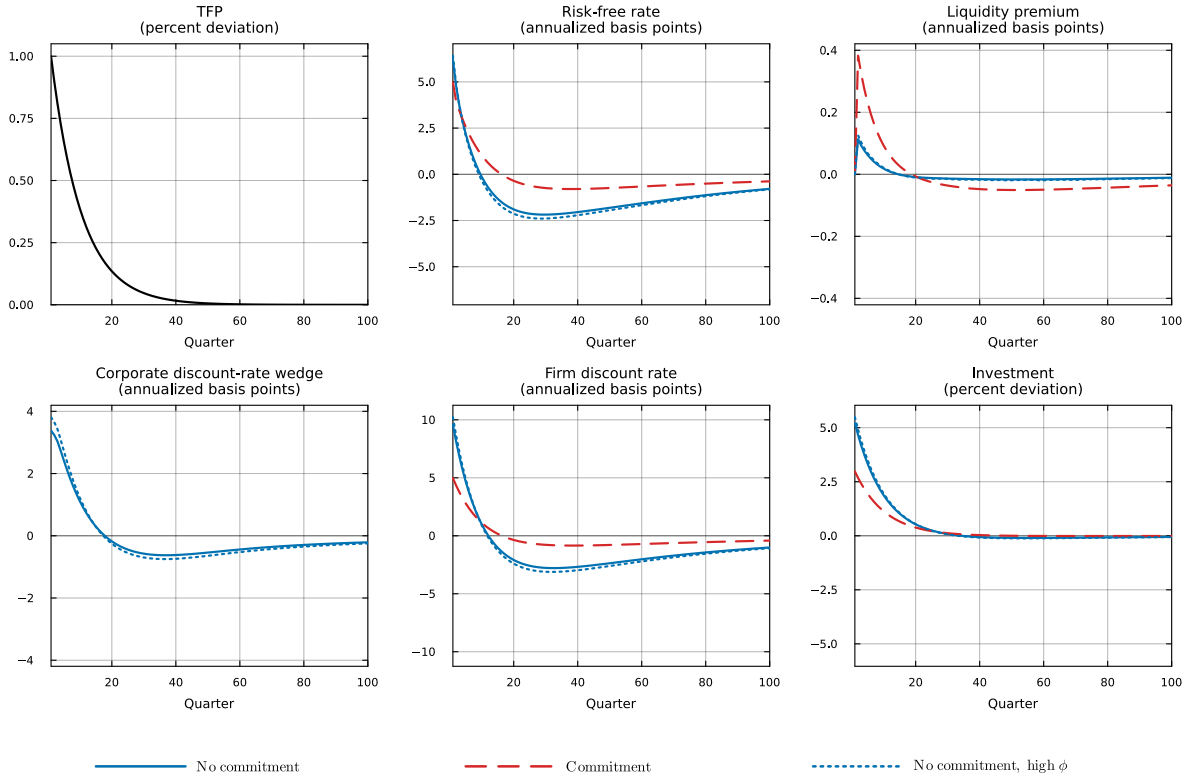
whereas under commitment it is

$$i_t^{F,C} = -\frac{1}{f} \log \tilde{\delta}_t = i_t^{\text{rf}} + \ell_t = i_t^{\text{fin}},$$

The corporate discount-rate wedge is therefore the additional component generated by lack of commitment.

This decomposition connects the model to the evidence in [Gormsen and Huber \(2025\)](#). They document that firms' discount rates move substantially less than one-for-one with the cost of capital in the short and medium run, generating time-varying discount-rate wedges. Under commitment, $i_t^{F,C} = i_t^{\text{fin}}$, implying one-for-one pass-through. Without commitment, $i_t^{F,N} = i_t^{\text{fin}} + \kappa_t$, so endogenous movements in κ_t can insulate the firm discount rate from movements in the financial cost of capital.

Figure 5: Impulse Responses to a TFP Shock



Note: The figure reports impulse responses to an unanticipated one-percent increase in TFP with persistence 0.9. The shock and investment are expressed as percent deviations from steady state. The risk-free rate, liquidity premium, corporate discount-rate wedge, and firm discount rate are annualized and expressed in basis points. The Liquidity premium panel reports $\ell_t = -f^{-1} \log(1 - \Phi_{t-1}^b)$, the predetermined liquidity component entering the commitment discount factor. The solid blue line represents the benchmark no-commitment economy ($\phi = 2.46$), the dashed red line the commitment economy with the same ϕ , and the dotted blue line the no-commitment economy with $\phi = 3$.

7.1 Productivity Shock

Figure 5 reports the response to an increase in productivity. In the benchmark economy without commitment, the risk-free rate rises by 6.2 basis points on impact and the corporate discount-rate wedge rises by 3.4 basis points. The liquidity premium is unchanged on impact because Φ_0^b is predetermined; it begins to rise in period 2. Together, these movements raise the firm discount rate by 9.6 basis points on impact.

In the economy with commitment, the corporate discount-rate wedge does not enter the firm's discount rate. The firm discount rate therefore consists only of the risk-free rate and the liquidity premium, which is predetermined on impact. As a result, the firm's discount rate rises by 5.0 basis points on impact, about half the increase in the benchmark no-commitment

³⁵Greenwood and Scharfstein (2013) report that fees and commissions from equity trading fell from 0.67 percent of GDP in 1997 to 0.53 percent in 2007, a change of about 20 percent over a decade. The increase in ϕ we consider is of the same order of magnitude.

economy.

The investment response reflects two opposing forces. Higher productivity raises the marginal product of capital and increases desired capital, whereas the increase in the firm discount rate raises the user cost of capital and reduces desired capital. Since $i_t^{F,j}$ is annualized, $\exp(fi_t^{F,j})$ is the corresponding gross rate over one quarter. The firm's desired capital can then be written as

$$k_{t+1}^j = \left[\frac{(1-\gamma)A_{t+1}H^\gamma}{\exp(fi_t^{F,j}) - (1-\delta)} \right]^{\frac{1}{\gamma}},$$

assuming constant returns to scale, $\psi = 1$. A first-order approximation around the steady state gives

$$\Delta \log k_{t+1}^j = \varepsilon_{k,A} \Delta \log A_{t+1} - \eta_{k,i}^j \Delta i_t^{F,j},$$

where Δ denotes a deviation from the steady state. The coefficients are

$$\varepsilon_{k,A} = \frac{1}{\gamma}, \quad \eta_{k,i}^j = \frac{f \exp(fi^{F,j,*})}{\gamma [\exp(fi^{F,j,*}) - (1-\delta)]}.$$

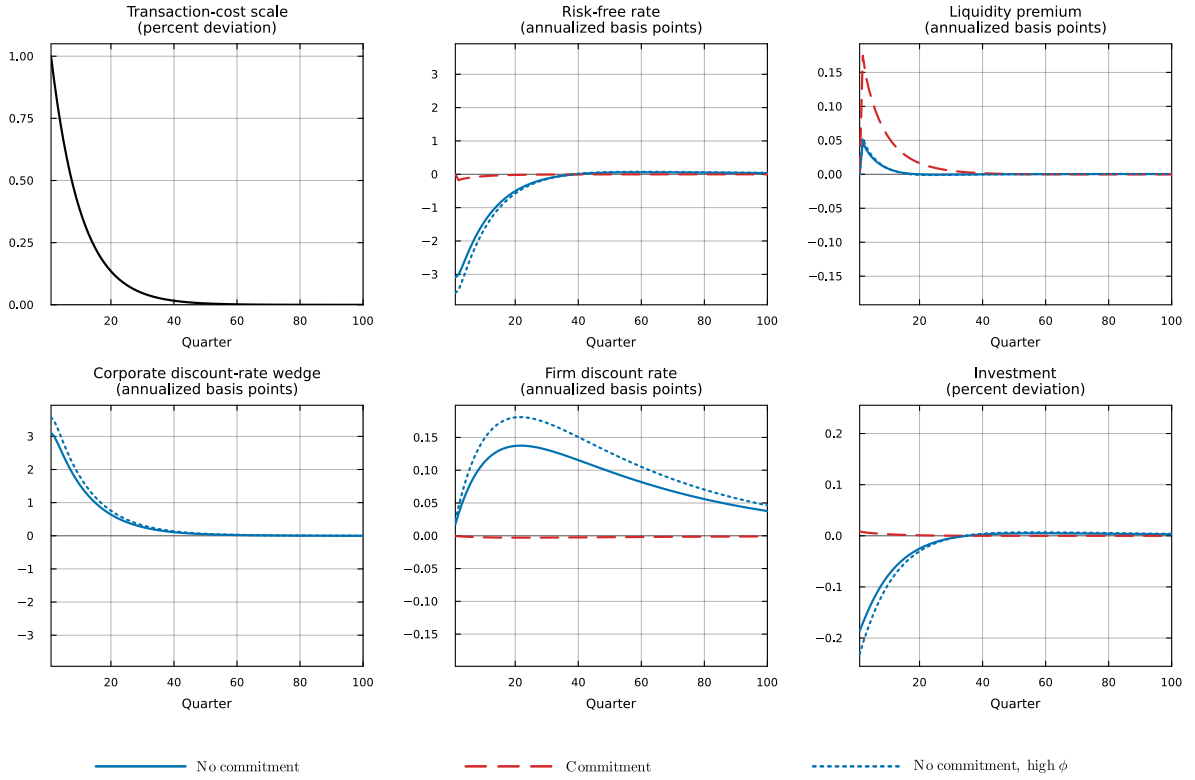
Thus, $\varepsilon_{k,A}$ is the elasticity of capital with respect to TFP, while $\eta_{k,i}^j$ is its semi-elasticity with respect to the annualized firm discount rate. The TFP elasticity is the same in both economies, and so the productivity channel generates the same proportional increase in desired capital.

The discount-rate semi-elasticity differs across the two economies because it is evaluated at each economy's steady-state firm discount rate, $i^{F,j,*}$. The semi-elasticity decreases with the steady-state firm discount rate. Because the commitment economy has a lower steady-state firm discount rate, its desired capital is more sensitive to a given change in that rate. Thus, although the firm discount rate increases by less under commitment, its negative effect on desired capital is slightly larger. Since the productivity channel is the same in both economies, the net increase in desired capital is larger without commitment. Because current capital is predetermined when the shock arrives, the impact response of investment equals the change in capital installed for the following period. Investment therefore increases by about 5.2 percent without commitment and by about 3.0 percent under commitment.

Finally, the figure also shows a no-commitment economy with a higher transaction-cost scale, $\phi = 3$. The responses are similar to those in the benchmark economy but modestly larger: on impact, the firm discount rate increases by 10.2 basis points and investment by about 5.5 percent.

Taken together, the TFP shock shows that firms' ability to commit is quantitatively more important than empirically plausible increases in transaction costs within the class of no-commitment economies. More generally, the investment response cannot be inferred from the risk-free rate alone: it depends both on how the shock changes the full firm discount rate, including the corporate discount-rate wedge, and on the semi-elasticity of desired capital to

Figure 6: Impulse Responses to a Financial-Friction Shock



Note: The figure reports impulse responses to an unanticipated one-percent increase in the transaction-cost scale ϕ with persistence 0.9. The shock and investment are expressed as percent deviations from steady state. The risk-free rate, liquidity premium, corporate discount-rate wedge, and firm discount rate are annualized and expressed in basis points. The Liquidity premium panel reports $\ell_t = -f^{-1} \log(1 - \Phi_{t-1}^b)$, the predetermined liquidity component entering the commitment discount factor. The solid blue line represents the benchmark no-commitment economy ($\phi = 2.46$), the dashed red line the commitment economy with the same ϕ , and the dotted blue line the no-commitment economy with $\phi = 3$.

that rate.

7.2 Financial-Friction Shock

We next examine the response to a one-percent increase in the transaction-cost scale ϕ , reported in Figure 6. In the benchmark no-commitment economy, the risk-free rate falls by 3.1 basis points on impact, while the predetermined liquidity component is unchanged, so the financial cost of capital falls by approximately 3.1 basis points. At the same time, the corporate discount-rate wedge rises by approximately the same amount, leaving the firm discount rate essentially unchanged. The resulting impact pass-through from the financial cost of capital to the firm discount rate is close to zero rather than one, providing a qualitative counterpart to the low short-run pass-through documented by [Gormsen and Huber \(2025\)](#). This disconnect arises even though the shock is fully observed and agents have perfect foresight. The liquidity premium begins rising in period 2, and the firm discount rate follows a small hump-shaped response,

peaking about twenty quarters out, as the corporate discount-rate wedge decays more slowly than the risk-free rate recovers.

Investment falls by approximately 0.19 percent on impact. Although the firm discount rate peaks later, investment falls most on impact because the existing capital stock is predetermined. The entire initial reduction in desired next-period capital must therefore occur through current investment.

The response is substantially different under commitment. The risk-free rate falls only slightly, while the liquidity premium is predetermined on impact, leaving the firm discount rate essentially unchanged because the corporate discount-rate wedge does not enter the firm's discount rate under commitment. As a consequence, investment also remains nearly unchanged under commitment.

The no-commitment economy with a higher transaction-cost scale displays the same mechanism as the benchmark, with modestly larger responses. The corporate discount-rate wedge rises more than in the benchmark economy, the firm discount rate follows a slightly more pronounced path, and the decline in investment is somewhat larger. Thus, higher initial trading frictions amplify the effects of the shock without changing its qualitative transmission.

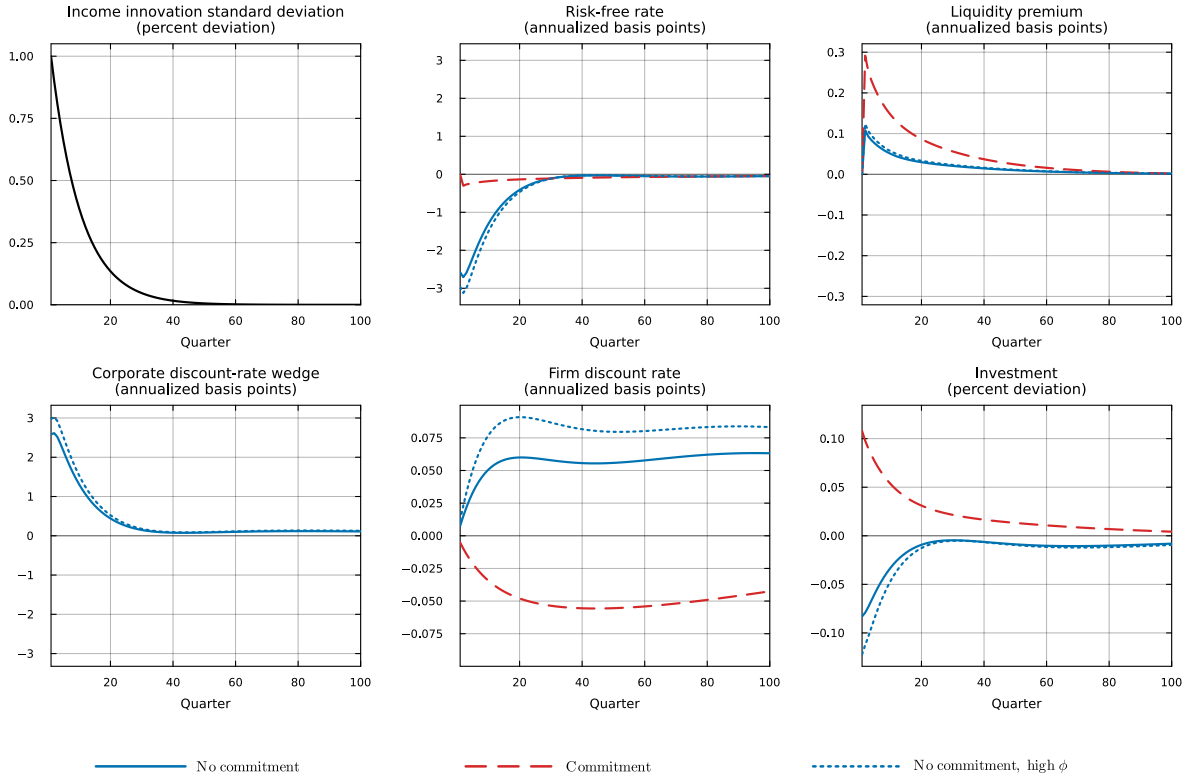
Taken together, these dynamic responses mirror the steady-state comparisons in Figure 1. First, at a given transaction-cost scale, commitment largely insulates investment from changes in trading frictions, whereas without commitment higher transaction costs raise the corporate discount-rate wedge and reduce investment. Second, within the class of no-commitment economies, a higher initial transaction-cost scale amplifies the investment response, consistent with the steady-state result that capital decreases with ϕ .

7.3 Income-Uncertainty Shock

We next examine the response to a one-percent increase in the innovation standard deviation of log income, σ_h , reported in Figure 7. Mean labor income remains unchanged, so the shock is a mean-preserving increase in idiosyncratic income risk. In the benchmark no-commitment economy, higher uncertainty strengthens precautionary demand for liquid assets, lowering the risk-free rate and raising the liquidity premium. At the same time, it widens the corporate discount-rate wedge by increasing disagreement between current and future owners. These components nearly offset on impact, leaving the firm discount rate slightly above its steady-state level. As the risk-free rate recovers, the firm discount rate rises further, even though the financial cost of capital fell on impact. The increase in the firm discount rate lowers desired capital. Existing capital is predetermined when the shock arrives, so investment falls most on impact. The no-commitment economy with a higher transaction-cost scale displays the same mechanism, with a somewhat larger increase in the corporate discount-rate wedge and a larger decline in investment.

The responses of the firm discount rate and investment have the opposite sign under com-

Figure 7: Impulse Responses to an Income-Uncertainty Shock



Note: The figure reports impulse responses to an unanticipated one-percent increase in the income innovation standard deviation σ_{η} with persistence 0.9. The shock and investment are expressed as percent deviations from steady state. The risk-free rate, liquidity premium, corporate discount-rate wedge, and firm discount rate are annualized and expressed in basis points. The Liquidity premium panel reports $\ell_t = -f^{-1} \log(1 - \Phi_{t-1}^b)$, the predetermined liquidity component entering the commitment discount factor. The solid blue line represents the benchmark no-commitment economy ($\phi = 2.46$), the dashed red line the commitment economy with the same ϕ , and the dotted blue line the no-commitment economy with $\phi = 3$.

mitment. The corporate discount-rate wedge does not enter the firm's discount rate, and the decline in the risk-free rate more than offsets the increase in the liquidity premium over the subsequent path. Thus, higher uncertainty raises the firm discount rate and lowers investment by 0.08 percent without commitment, but lowers the firm discount rate and raises investment by 0.11 percent under commitment.

These dynamic responses reflect the same competing forces as the comparative statics in Figure 3. In that figure the increase in uncertainty is permanent and capital rises under both regimes, though much more under commitment, because the corporate discount-rate wedge dampens the standard precautionary-saving channel. For the temporary shock considered here, the same countervailing force is strong enough to turn the no-commitment investment response negative.

The financial-friction and uncertainty shocks generate a qualitative counterpart to the low short-run pass-through from the cost of capital to firms' discount rates documented by [Gormsen](#)

and Huber (2025). In both cases, the financial cost of capital and the corporate discount-rate wedge move in opposite directions, leaving the firm discount rate largely insulated from movements in the cost of capital. This result is not mechanical. Following a productivity shock, the financial cost of capital and the corporate discount-rate wedge instead move in the same direction, so the firm discount rate responds more than one-for-one. The model therefore predicts that discount-rate pass-through depends on the source of aggregate fluctuations. The average pass-through measured in the data may consequently mask different responses to financial-friction, uncertainty, and productivity shocks.

8 Conclusion

This paper studies firms' investment decisions in heterogeneous-agent economies in which shareholders disagree about how to discount future cash flows. The main result is that when equity shares trade in frictional asset markets, the firm's problem is time-inconsistent, and the empirically relevant direction of inconsistency is present bias. Financial transaction costs cause buyers to value future firm investment more than owners (as if they were more patient) and sellers to prefer immediate dividend distribution (as if they were more impatient). While investment plans for the distant future are governed by asset prices, which reflect buyers' valuations, each period the current owners, sellers included, favor high contemporaneous dividends instead.

When firms are able to commit to future investments, transaction costs have large effects on asset prices but small consequences on real variables. However, without commitment, firms over-issue dividends and accumulate less capital than under commitment. The no-commitment case is consistent with recent empirical findings pertaining to the wedge between firms' discount rate and their cost of capital. Consistent with the mechanism, we find that firms whose equity is less liquid have larger wedges in the data.

The same mechanism operates over the business cycle. The wedge moves with the shock, so the risk-free rate is not a sufficient statistic for investment. Relative to the commitment economy, lack of commitment materially changes the investment responses to the shocks we consider, and even reverses their sign following an increase in income uncertainty.

While a transaction cost is a natural way to generate investor disagreement, it is not the only one. However, not all frictions generate a wedge: Section 5 shows that the firm's problem is time consistent at $\phi = 0$, despite the presence of borrowing constraints and incomplete markets. What is needed is a friction that generates a wedge between current owners' valuation of the ex-dividend price and the marginal buyer's. We show that a capital gains tax does exactly this.

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A Proofs

Proof of Lemma 1. In what follows, the dependence of each variable on (θ, b, h) is suppressed for brevity. The first-order condition for Δ_t^+ when household are buyers ($\eta_t = 0$) is given by $\mu_t = \lambda_t q_t$. The first-order condition for Δ_t^- when households are sellers is given by $\mu_t = \lambda_t q_t(1 - \phi \Delta_t^-) - \eta_t$. Expression (7) follows from plugging these conditions into the envelope condition $\tilde{q}_t = d_t + \frac{\mu_t + \eta_t}{\lambda_t}$. $\square$

Proof of Lemma 2. We first observe that no household both buys and sells, since reducing Δ_t^{j+} and Δ_t^{j-} by a common amount leaves θ_{t+1}^j unchanged and raises the household's resources. Hence a buyer ($\Delta_t^{j+} > 0$) has $\Delta_t^{j-} = 0$ and $\theta_{t+1}^j > \theta_t^j \geq 0$, while a household that is not a buyer has $\Delta_t^{j+} = 0$ and $\theta_{t+1}^j = \theta_t^j - \Delta_t^{j-}$. We begin with the Euler equation for bonds, equation (17). Bond market clearing gives $\int_{j \in [0,1]} b_{j,t+1} dj = B_{t+1}^G \geq 0 \geq \bar{b}$, while $b_{j,t+1} \geq \bar{b}$ for every j . Whenever $B_{t+1}^G > \bar{b}$, a positive mass of households has $b_{j,t+1} > \bar{b}$ and hence $v_t^j = 0$ by complementary slackness. In the remaining case, $B_{t+1}^G = \bar{b}$, we have $b_{j,t+1} = \bar{b}$ almost everywhere, so the bond is in zero net supply and market clearing does not pin down r_t : any rate with $1/(1+r_t) \geq \hat{M}_t$ is consistent with equilibrium, where $\hat{M}_t$ denotes the highest expected IMRS in the cross-section. We select $1+r_t = 1/\hat{M}_t$, the limit of the equilibrium rate as $B_{t+1}^G \downarrow \bar{b}$, under which the households attaining $\hat{M}_t$ have $v_t^j = 0$. Thus, in both cases, some households are unconstrained. Any household with $v_t^j = 0$ has the highest expected IMRS according to (17) and dual feasibility, $v_t^j \geq 0$; and furthermore, a household with the highest expected IMRS must satisfy $v_t^j = 0$.

Note that the expected IMRS equals the fraction in (19), up to the short-selling Lagrange multiplier η_t^j . Condition (19) is sufficient because $\eta_t^j = 0$ for buyers and

$$\frac{q_t(1 - \phi \Delta_t^{j-})}{d_{t+1} + (1 - \Phi_t^j)q_{t+1}} \geq \frac{q_t(1 - \phi \Delta_t^{j-}) - \eta_t^j}{d_{t+1} + (1 - \Phi_t^j)q_{t+1}}$$

for all other households. For the converse, suppose a borrowing-unconstrained buyer exists and consider any other household. If its short-selling constraint is slack, its expected IMRS precisely equals the fraction in (19), which the Euler equation for bonds bounds above by $1/(1+r_t)$, the value the unconstrained buyer attains. If instead the constraint binds, then $\Phi_t^j = 0$ because $\Delta_{t+1}^{j-} = 0$. But a buyer always has a higher expected IMRS than the fraction in (19) because

$$\frac{q_t}{d_{t+1} + (1 - \Phi_t^b)q_{t+1}} \geq \frac{q_t(1 - \phi \Delta_t^{j-})}{d_{t+1} + q_{t+1}}.$$

Note that the statement of the lemma requires amendment in the case of our extension from Appendix E.1: A stock buyer *who does not borrow* in the bond market exists if and only if (19)

holds. In this case, the Euler equation for bonds is

$$\frac{1}{1 + r_t + \xi_b(\mathcal{B}_{t+1}^{j-})} = \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} \right] + \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)(1 + r_t + \xi_b(\mathcal{B}_{t+1}^{j-}))} \frac{\partial \xi_b}{\partial b}(\mathcal{B}_{t+1}^{j-}) \mathcal{B}_{t+1}^{j-} \right] + v_t^j,$$

where $\xi_b(b)$ is defined in (43) and $\mathcal{B}_{t+1}^{j-}$ denotes the policy function for borrowing, $b_{j,t+1}^- \equiv \max\{-b_{j,t+1}, 0\}$ where $b_{j,t+1}$ denotes the net bond position. Similarly, saving is denoted $b_{j,t+1}^+ \equiv \max\{b_{j,t+1}, 0\}$. Like before, $\int_{j \in [0,1]} b_{j,t+1} dj = B_{t+1}^G \geq 0$. If a positive mass of households has $b_{j,t+1} > 0$, each of them satisfies $b_{j,t+1} > 0 \geq \bar{b}$ and hence $(v_t^j, \mathcal{B}_{t+1}^{j-}) = (0, 0)$. Otherwise $b_{j,t+1} \leq 0$ almost everywhere, which together with $B_{t+1}^G \geq 0$ forces $b_{j,t+1} = 0$ almost everywhere, so every household is a non-borrower; these households have $v_t^j = 0$ when $\bar{b} < 0$, and, when $\bar{b} = 0$, the selection of r_t above gives $v_t^j = 0$ for those attaining $\hat{M}_t$. In every case, a household with $(v_t^j, \mathcal{B}_{t+1}^{j-}) = (0, 0)$ exists. Any household with $(v_t^j, \mathcal{B}_{t+1}^{j-}) = (0, 0)$ has the highest expected IMRS according to the Euler equation for bonds; and furthermore, a household with the highest expected IMRS must satisfy $(v_t^j, \mathcal{B}_{t+1}^{j-}) = (0, 0)$.³⁶ $\square$

Proof of Proposition 1. Throughout, we suppress the dependence on (θ, b, h) whenever it does not lead to confusion. Because shares integrate to one, $\tilde{\mathbb{E}}_t[\theta^j] = 1$, and applying the definition of a covariance to (9) gives

$$\frac{\bar{\Phi}}{\phi} = \tilde{\mathbb{E}}_t[\theta^j \Delta_t^{j-}] = \tilde{\mathbb{E}}_t[\Delta_t^{j-}] + \widetilde{\text{cov}}_t(\theta^j, \Delta_t^{j-}). \quad (25)$$

We focus on equilibria with a buyer who is unconstrained by the borrowing limit in the bond market. For every such buyer the expected marginal cost Φ_t^j in (15) equals the common value Φ^b , so averaging (15) across the unconstrained buyers leaves Φ^b unchanged and yields

$$\frac{\Phi^b}{\phi} = \tilde{\mathbb{E}}_t \left[\mathbb{E}_t[\Delta_{t+1}^{j-}] \middle| \text{buyer} \right] + \tilde{\mathbb{E}}_t \left[\frac{\text{cov}_t(u'(\mathcal{C}_{t+1}^j), \Delta_{t+1}^{j-})}{\mathbb{E}_t[u'(\mathcal{C}_{t+1}^j)]} \middle| \text{buyer} \right]. \quad (26)$$

Subtracting (26) from (25) and multiplying through by ϕ delivers the decomposition of Proposition 1. The argument uses neither stationarity nor the stronger assumption that *all* buyers are unconstrained by the borrowing limit for bonds.

In the case of the extension from Appendix E.1, the statement of the proposition remains unchanged except for the following: Conditionings are on being a buyer *who does not borrow* today, that is, on belonging to the set of buyers that prices the stock. $\square$

Proof of Proposition 2. We prove the proposition for a more general firm problem with any

³⁶This follows from several useful properties of the function $\xi_b(b)$ from (43). First, $\xi_b(b) \geq 0$ for all b with $\xi_b(b) = 0 \iff b = 0$. Second, $\frac{\partial \xi_b}{\partial b}(b) \cdot b \geq 0$ for all b with $\frac{\partial \xi_b}{\partial b}(b) \cdot b = 0 \iff b = 0$.

twice-differentiable production function $y_t = f(l_t, k_t)$ that is weakly concave in both variables (jointly), and strictly concave in k_t . Profits are given by $\pi_t = y_t - w_t l_t$ so the intratemporal decision satisfies

$$f_1(l_t, k_t) = w_t \quad (27)$$

and the subscript denotes the partial derivative. Condition (27) implicitly defines the labor function, which we denote $l(k_t)$, and differentiating it gives its slope,

$$f_{11}l_1 + f_{12} = 0. \quad (28)$$

Let $\pi(k_t) \equiv f(l(k_t), k_t) - w l(k_t)$ denote per-period profits as a function of capital. Differentiating twice, and using (27) and (28) to cancel terms,

$$\pi_{11} = f_{12}l_1 + f_{22} = f_{22} - \frac{f_{12}^2}{f_{11}} \leq 0,$$

where the inequality follows from the Hessian condition equivalent to weak concavity of f . Hence $F(k, k') = \pi(k) + (1 - \delta)k - k'$ is weakly concave in k , so the firm's objective, a positively weighted sum of F across dates, is concave in the investment plan and its first-order condition (29) is sufficient. Next we solve the firm's intertemporal problem with no commitment. The first-order condition of the firm's problem is given by

$$F_2(k, k') + \tilde{\beta}\tilde{\delta}W'(k') = 0 \quad (29)$$

where

$$W'(k') \equiv F_1(k', g(k')) + g'(k') \left(F_2(k', g(k')) + \tilde{\delta}W'(g(k')) \right) \quad (30)$$

First we guess that $g'(k') = 0$. Plugging in the function $F(k, k') = f(l(k), k) - w l(k) + (1 - \delta)k - k'$:

$$\frac{1}{\tilde{\beta}\tilde{\delta}} = \pi'(k') + 1 - \delta$$

where $\pi'(k) = f_1(H, k)l_1(k) + f_2(H, k) - w l_1(k) = f_2(H, k)$ and the second equality follows from the intratemporal condition (27), and H is the unconditional expectation of the labor process. Because f is strictly concave in k , $\pi'(k)$ is a strictly decreasing function. This monotonicity uses only strict concavity in k ($f_{22} < 0$), evaluated at the market-clearing labor H ; it holds even under constant returns to scale, where the joint Hessian $\pi_{11} = f_{22} - f_{12}^2/f_{11}$ is singular and equilibrium capital is instead pinned down by the endogenous, market-clearing wage. This yields the steady-state level of capital $k^N = (\pi')^{-1} \left(\frac{1}{\tilde{\beta}\tilde{\delta}} - 1 + \delta \right)$ where π' has an inverse because π' was shown to be strictly decreasing. By the same reasoning, $(\pi')^{-1}$ must be strictly decreasing. Similar logic for the commitment case yields $k^C = (\pi')^{-1} \left(\frac{1}{\tilde{\delta}} - 1 + \delta \right)$ and so we have shown that $k^N < k^C$ if and only if the firm has present bias, $\tilde{\beta} < 1$. Now say $g'(k') \neq 0$, and one of two scenarios can occur. First, (29) and (30) could pin down an optimal value of k'

(notice that this value does not depend on the value of k). But since $g'(k') \neq 0$, $g(k')$ is not a constant function. By definition of MPE, $\zeta(k)$ is also not a constant function, which contradicts the optimal choice k' (which was irrespective of the value of k). Second, we cannot rule out a policy $g(k')$ where (29) and (30) are satisfied for any value of k' . Such a $g(k')$ would imply that any $\zeta(k)$ is optimal, and in particular one could select $\zeta(k) = g(k)$. Now consider a resource-exhausting finite-horizon version of the same problem.³⁷ Because there is zero investment in the final period, the firm objective in the preceding period would be

$$F(k, k') + \tilde{\beta} \tilde{\delta} F(k', 0)$$

and first-order conditions yield k^N . Using backward induction, the firm in each preceding period would also choose k^N and hence our claim is proved. Finally, from (23), $\log k^N = a \log \left[(1 - \gamma) \psi \tilde{\beta} \tilde{\delta} H^{\gamma\psi} / (1 - \tilde{\beta} \tilde{\delta} (1 - \delta)) \right]$ with $a = 1 / (1 - (1 - \gamma) \psi) > 0$, so holding $\tilde{\delta}$ fixed, $\partial \log k^N / \partial \tilde{\beta} = 1 / [\tilde{\beta} (1 - (1 - \gamma) \psi) (1 - \tilde{\beta} \tilde{\delta} (1 - \delta))] > 0$ under the interiority condition $1 - \tilde{\beta} \tilde{\delta} (1 - \delta) > 0$. At $\tilde{\beta} = 1$ this recovers (22), so $k^N = k^C$. $\square$

Proof of Lemma 3. Because the committing firms have mass zero, they leave all aggregate quantities, the wage w , the interest rate r , and the cross-sectional distribution Γ unchanged; the household problem and its stationary solution, including marginal utilities and trading policies, are those of the baseline economy. The two firm types issue equity that differs only in the firm's investment policy: both stocks carry the same transaction cost ϕ and are priced by the same households facing no aggregate risk. We assume the existence of a buyer who is unconstrained in the bond market, and we maintain that such a buyer trades both stocks interiorly. The budget constraint of the household becomes

$$c_t + q_t^C \Delta_t^{C+} + q_t^N \Delta_t^{N+} + b_{t+1} \\ \leq w_t h_t + d_t^C \theta_t^C + d_t^N \theta_t^N + \left(\Delta_t^{N-} - \frac{\phi}{2} (\Delta_t^{N-})^2 \right) q_t^N + \left(\Delta_t^{C-} - \frac{\phi}{2\epsilon} (\Delta_t^{C-})^2 \right) q_t^C + b_t (1 + r_{t-1}) - \mathcal{T}_t,$$

with extra constraints

$$\Delta_t^{C+} \geq 0, \quad \theta_t^C \geq \Delta_t^{C-} \geq 0,$$

and law of motion

$$\theta_{t+1}^C = \theta_t^C + \Delta_t^{C+} - \Delta_t^{C-}.$$

Clearing for the commitment firm is

$$\int_{j \in [0,1]} \theta_{j,t}^C dj = \epsilon.$$

Anticipating $\epsilon \rightarrow 0$, we leave labor market clearing conditions intact. The Euler equation for

³⁷Laibson (1997) uses the same equilibrium refinement.

stocks associated with the commitment firm is

$$\left(1 - \frac{\phi}{\varepsilon} \Delta_t^{C-}\right) q_t^C = \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1})}{u'(\mathcal{C}_t)} \left(d_{t+1}^C + \left(1 - \frac{\phi}{\varepsilon} \Delta_{t+1}^{C-}\right) q_{t+1}^C \right) \right] + \eta_t^C.$$

There is an equilibrium where $\Delta_t^{C-} = \varepsilon \Delta_t^{N-}$, $\Delta_t^{C+} = \varepsilon \Delta_t^{N+}$, $\theta_t^C = \varepsilon \theta_t^N$, $\eta_t^C/d_{t+1}^C = \eta_t^N/d_{t+1}^N$, and $q^C/d^C = q^N/d^N$; and variables without subscripts denote steady-state values. To see why, simply plug in these proposed values to recover analogous equations (Euler, clearing, law of motion, and constraints) associated with the no commitment firm. Importantly, as $\varepsilon \rightarrow 0$, the budget collapses to that of the baseline model; this ensures that all aggregate quantities remain unchanged. From the Euler equation it follows that Φ^b , and therefore $\tilde{\delta} = (1 - \Phi^b)/(1 + r)$, are common across firm types. That $\frac{q^i}{d^i} = \frac{1}{r + \Phi^b}$ for $i \in \{C, N\}$ follows from the Euler equation for unconstrained buyers.

With a zero mass of firms that can commit, the single wage in the economy w_t is entirely determined by firms that cannot commit. With decreasing returns technologies, $\alpha = \frac{(1-\gamma)\psi}{1-\gamma\psi} < 1$ and first-order conditions for the firm are

$$M(z\alpha k^{\alpha-1} + (1 - \delta)) = 1 \iff k = \left(\frac{z\alpha}{1/M - 1 + \delta} \right)^{\frac{1}{1-\alpha}},$$

where $M = \tilde{\delta}$ if the firms can commit, $M = \tilde{\beta}\tilde{\delta}$ if they cannot, and z is common across both firm types. Note that such a first-order condition is consistent with the same refinement of MPE used in Proposition 2. It is immediate from this expression above that k is increasing in M . $\square$

Proof of Proposition 3. Recall from Section 2 that $\pi_t = z_t k_t^\alpha$ with $\alpha = (1 - \gamma)\psi/(1 - \gamma\psi)$. Both firm types face the same wage, hence the same z , so the first-order condition in the proof of Lemma 3 gives $k^i = (z\alpha/(1/M^i - 1 + \delta))^{1/(1-\alpha)}$ for $i \in \{C, N\}$, with $M^C = \tilde{\delta}$ and $M^N = \tilde{\beta}\tilde{\delta}$. Using $\tilde{\delta} = 1/(1 + r^C)$ and $\tilde{\beta}\tilde{\delta} = 1/(1 + r^N)$, we have $1/M^C - 1 + \delta = r^C + \delta$ and $1/M^N - 1 + \delta = r^N + \delta$, so $\log(k^N/k^C) = \frac{1}{1-\alpha} [\log(r^C + \delta) - \log(r^N + \delta)]$. By Lemma 3, $\tilde{\delta}$ (hence r^C) is common across firms, while r^N depends on both $\tilde{\beta}$ and $\tilde{\delta}$. Differentiating with respect to $\mathcal{P}$, with $\partial r^C/\partial \mathcal{P} = -\tilde{\delta}^{-2} \partial \tilde{\delta}/\partial \mathcal{P}$ and $\partial r^N/\partial \mathcal{P} = -(\tilde{\beta}^2 \tilde{\delta})^{-1} \partial \tilde{\beta}/\partial \mathcal{P} - (\tilde{\beta} \tilde{\delta}^2)^{-1} \partial \tilde{\delta}/\partial \mathcal{P}$, gives (24) after dividing by $1/(1 - \alpha) > 0$. The coefficient on $\partial \tilde{\beta}/\partial \mathcal{P}$ is $(r^N + \delta)^{-1} (\tilde{\delta} \tilde{\beta}^2)^{-1} > 0$. The coefficient on $\partial \tilde{\delta}/\partial \mathcal{P}$ equals $(1 - \delta)(\tilde{\beta} - 1)/[\tilde{\delta}^2(r^N + \delta)\tilde{\beta}(r^C + \delta)]$, negative iff $\tilde{\beta} < 1$ since $r^N + \delta, r^C + \delta, \tilde{\beta}, \tilde{\delta} > 0$. $\square$

Proof of Proposition 4. The problem of the firm that can borrow up to a limit is given by

$$V^F(k_t, b_t^-, b_t^+) = \max_{\{k_{t+s}, b_{t+s}^-, b_{t+s}^+\}_{s \geq 1}} d_t + \tilde{\beta} \sum_{s=1}^{\infty} \tilde{\delta}^s d_{t+s}$$

subject to

$$d_t = F(k_t, k_{t+1}) + \frac{b_{t+1}^-(1 - \tilde{\phi})}{1 + r} - b_t^- - \frac{b_{t+1}^+}{1 + r} + b_t^+, \quad 0 \leq b_{t+1}^- \leq -\bar{b}^f, \quad 0 \leq b_{t+1}^+$$

The firm can borrow the amount b_{t+1}^- up to a limit $-\bar{b}^f$, where $\bar{b}^f$ is defined to be a non-positive number. The parameter $\tilde{\phi} \geq 0$ captures the illiquidity of corporate bonds. Firms can save the amount b_{t+1}^+ at the interest rate r . First-order conditions for capital do not change from Proposition 2. First-order conditions for bonds are

$$\begin{aligned} (b_{t+1}^-) : \frac{1 - \tilde{\phi}}{1 + r} - \tilde{\beta} \tilde{\delta} + \mu_t^1 - \mu_t^2 &= 0 \\ (b_{t+s}^-) : \frac{1 - \tilde{\phi}}{1 + r} - \tilde{\delta} + \mu_{t+s-1}^1 - \mu_{t+s-1}^2 &= 0 \\ (b_{t+1}^+) : -\frac{1}{1 + r} + \tilde{\beta} \tilde{\delta} + \mu_t^3 &= 0 \\ (b_{t+s}^+) : -\frac{1}{1 + r} + \tilde{\delta} + \mu_{t+s-1}^3 &= 0 \end{aligned}$$

where μ_t^1, μ_t^2 , and μ_t^3 are the Lagrange multipliers on the three inequality constraints, respectively. Because $\tilde{\beta} \tilde{\delta} = \frac{1 - \bar{\Phi}}{1 + r} < \frac{1}{1 + r}$ and $\tilde{\delta} = \frac{1 - \Phi^b}{1 + r} < \frac{1}{1 + r}$, we have that $b_{t+s}^+ = 0$ for all $s \geq 1$. In other words, no savings occurs. The firm's borrowing decision depends on the illiquidity parameter $\tilde{\phi}$. We consider the empirically relevant case of present bias, $\bar{\Phi} > \Phi^b$.

Case 1: $\tilde{\phi} > \bar{\Phi}$ and, hence, corporate bonds are extremely illiquid. Then $\tilde{\beta} \tilde{\delta} = \frac{1 - \bar{\Phi}}{1 + r} > \frac{1 - \tilde{\phi}}{1 + r}$ and $\tilde{\delta} = \frac{1 - \Phi^b}{1 + r} > \frac{1 - \tilde{\phi}}{1 + r}$ so $b_{t+s}^- = 0$ for all $s \geq 1$.

Case 2: $\tilde{\phi} < \Phi^b$ and, hence, corporate bonds are extremely liquid. Then $\tilde{\beta} \tilde{\delta} = \frac{1 - \bar{\Phi}}{1 + r} < \frac{1 - \tilde{\phi}}{1 + r}$ and $\tilde{\delta} = \frac{1 - \Phi^b}{1 + r} < \frac{1 - \tilde{\phi}}{1 + r}$ so $b_{t+s}^- = -\bar{b}^f$ for all $s \geq 1$.

Case 3a: $\Phi^b = \tilde{\phi} < \bar{\Phi}$ and, hence, corporate bonds are characterized by intermediate liquidity. Then $\tilde{\beta} \tilde{\delta} = \frac{1 - \bar{\Phi}}{1 + r} < \frac{1 - \tilde{\phi}}{1 + r}$ yet $\tilde{\delta} = \frac{1 - \Phi^b}{1 + r} = \frac{1 - \tilde{\phi}}{1 + r}$, so $b_{t+1}^- = -\bar{b}^f$ yet $0 \leq b_{t+s}^- \leq -\bar{b}^f$ for $s > 1$. This problem is time-inconsistent.

Case 3b: $\Phi^b < \tilde{\phi} < \bar{\Phi}$ and, again, corporate bonds are characterized by intermediate liquidity. Then $\tilde{\beta} \tilde{\delta} = \frac{1 - \bar{\Phi}}{1 + r} < \frac{1 - \tilde{\phi}}{1 + r}$ yet $\tilde{\delta} = \frac{1 - \Phi^b}{1 + r} > \frac{1 - \tilde{\phi}}{1 + r}$, so $b_{t+1}^- = -\bar{b}^f$ yet $b_{t+s}^- = 0$ for $s > 1$. This problem is time-inconsistent.

Case 3c: $\Phi^b < \tilde{\phi} = \bar{\Phi}$ and, again, corporate bonds are characterized by intermediate liquidity. Then $\tilde{\beta} \tilde{\delta} = \frac{1 - \bar{\Phi}}{1 + r} = \frac{1 - \tilde{\phi}}{1 + r}$ yet $\tilde{\delta} = \frac{1 - \Phi^b}{1 + r} > \frac{1 - \tilde{\phi}}{1 + r}$, so $0 \leq b_{t+1}^- \leq -\bar{b}^f$ yet $b_{t+s}^- = 0$ for $s > 1$. This problem is time-inconsistent.

In the time-inconsistent parameter range, we must consider a MPE when the firm cannot commit. Letting $g(b)$ denote the policy function, the first-order condition with respect to b_{t+1}^- is

$$\frac{1 - \tilde{\phi}}{1 + r} + \mu^1 - \mu^2 + \tilde{\beta} \tilde{\delta} W'(b') = 0$$

where $W'(b') \equiv -1 + g'(b') \left[\frac{1 - \tilde{\phi}}{1 + r} + \mu^{1'} - \mu^{2'} + \tilde{\delta} W'(g(b')) \right]$ and one of the solutions is found by setting $g'(b') = 0$. This collapses the first-order condition into those of Case 3 above. One can apply similar arguments to those in Proposition 2 to show that this is, in fact, the only equilibrium that is the limit of a finite-horizon version of the same problem. $\square$

B Firm's Objective

B.1 Generalized Weights

In this appendix, we generalize Assumption 1 to encompass three important ways of aggregating shareholder valuations: that of Grossman and Hart (1979), Dreze (1974), and DeMarzo (1993). Grossman and Hart (1979) is the familiar case, discussed in the main text, of the firm maximizing an ownership-weighted valuation. Dreze (1974) makes a similar assumption, except that the weighting is done by *the next period's* (that is, future) ownership. Finally, DeMarzo (1993) suggests a majority voting rule that works in the following way. A majority stable allocation is defined by the absence of an alternative allocation preferred by at least half of the shares. Due to the continuum of traders in our setting, for tractability, we modify this definition to the absence of any alternative allocation preferred by *more than* half of the shares.

Assumption 3. *The firm maximizes a weighted valuation given by:*

$$\int_{\theta, b, h} w(\theta, b, h) [d_t + (1 - \phi \Delta_t^-(\theta, b, h)) q_t] d\Gamma_t(\theta, b, h) = d_t + (1 - \bar{\Phi}_t) q_t, \quad (31)$$

where $\Gamma_t(\theta, b, h)$ denotes the cross-section distribution over the portfolio holdings and employment status, $d_t = F_t(k_t, k_{t+1})$, and

$$\bar{\Phi}_t \equiv \phi \int_{\theta, b, h} w(\theta, b, h) \Delta_t^-(\theta, b, h) d\Gamma_t(\theta, b, h). \quad (32)$$

Weights are assumed to be non-negative and to integrate to one, $\int_{\theta, b, h} w(\theta, b, h) d\Gamma_t(\theta, b, h) = 1$. We allow $w d\Gamma_t$ to be a probability measure rather than requiring w to be a density, which is needed for the median case below. Current shareholder weighting from Grossman and Hart (1979) is given by $w(\theta, b, h) = \theta$, and future shareholder weighting from Dreze (1974) is given by $w(\theta, b, h) = \Theta_{t+1}(\theta, b, h)$. For the case of DeMarzo (1993), consider the median m satisfy-

ing

$$\int_{\theta, b, h} 1_{\{\Delta_t^-(\theta, b, h) \leq m\}} \theta d\Gamma_t(\theta, b, h) = \frac{1}{2}$$

and let w place unit mass on the households with $\Delta_t^-(\theta, b, h) = m$, so that (32) gives $\bar{\Phi}_t = \phi m$.³⁸ Then the optimum of (31) satisfies majority stability. To see why, say there is an alternative allocation preferred by more than half of the shares, where shareholders rank allocations by their valuations, (7). In this group, there must exist two households such that $\Delta_t^-(\bar{\theta}, \bar{b}, \bar{h}) \geq m$ and $\Delta_t^-(\underline{\theta}, \underline{b}, \underline{h}) \leq m$. Then if the above-median household prefers a particular deviation (d_t^a, q_t^a) ,

$$d_t^a + (1 - \phi \Delta_t^-(\bar{\theta}, \bar{b}, \bar{h})) q_t^a > d_t + (1 - \phi \Delta_t^-(\bar{\theta}, \bar{b}, \bar{h})) q_t$$

where (d_t^a, q_t^a) satisfy constraints of the firm problem and $(d_t, q_t) \in \operatorname{argmax} d_t + (1 - \phi m) q_t$ subject to constraints of the firm problem, then the below-median household must not,

$$d_t^a + (1 - \phi \Delta_t^-(\underline{\theta}, \underline{b}, \underline{h})) q_t^a \leq d_t + (1 - \phi \Delta_t^-(\underline{\theta}, \underline{b}, \underline{h})) q_t$$

For if this were not the case, we could take an appropriate convex combination of the two inequalities above to contradict the initial optimality of (d_t, q_t) . This last inequality contradicts the assumption that both households prefer the deviation (d_t^a, q_t^a) .

Next we show that our theoretical results from the main text hold under this generalized setup. This next proposition generalizes Proposition 1, which decomposed the sources of time inconsistency.

Proposition 5. *The difference $\bar{\Phi} - \Phi^b$ is the sum of a persistence effect, a concentration effect, and a risk premium:*

$$\begin{aligned} \bar{\Phi} - \Phi^b = & \underbrace{\phi \left(\tilde{\mathbb{E}}_t \left[\Delta_t^{j-} \right] - \tilde{\mathbb{E}}_t \left[\mathbb{E}_t \left[\Delta_{t+1}^{j-} \right] \mid \text{buyer} \right] \right)}_{\text{persistence effect}} + \underbrace{\phi \widetilde{\text{cov}}_t \left(w_t^j, \Delta_t^{j-} \right)}_{\text{concentration effect}} - \\ & \underbrace{\phi \tilde{\mathbb{E}}_t \left[\frac{\text{cov}_t \left(u'(\mathcal{C}_{t+1}^j), \Delta_{t+1}^{j-} \right)}{\mathbb{E}_t \left[u'(\mathcal{C}_{t+1}^j) \right]} \mid \text{buyer} \right]}_{\text{risk premium}}, \end{aligned}$$

where, exactly as in Proposition 1, $\tilde{\mathbb{E}}_t$ and $\widetilde{\text{cov}}_t$ are the equal-weighted cross-sectional mean and covariance across agents under $d\Gamma$, $\mathbb{E}_t$ and cov_t are taken over an agent's own next shock $dF(h'|h)$, and the conditionings are on being a buyer who is unconstrained by the borrowing limit today.

Proof. We follow the same two steps as in the proof of Proposition 1, replacing the ownership

³⁸This weighting is a point mass rather than a density: quadratic transaction costs leave no inaction region, so Δ_t^- is continuously distributed above zero and has no atom at m , and $m > 0$ whenever buyers hold less than half of the shares.

weight θ by the general aggregation weight w_t . Splitting the expectation into a product of means and a covariance, and then using that the weights integrate to one, $\mathbb{E}_t [w_t^j] = 1$,

$$\frac{\bar{\Phi}}{\phi} = \mathbb{E}_t [w_t^j \Delta_t^{j-}] = \mathbb{E}_t [\Delta_t^{j-}] + \widetilde{\text{cov}}_t (w_t^j, \Delta_t^{j-}). \quad (33)$$

The buyer side is unchanged, since it does not depend on the firm's weights: (26) continues to hold. Subtracting (26) from (33) and multiplying by ϕ delivers Proposition 5. Like in the case of Proposition 1, the statement of Proposition 5 requires slight amendment in the case of the extension from Appendix E.1: the conditionings are on being a buyer *who does not borrow* today. $\square$

Just like in the special case of Grossman and Hart (1979) from the main text, Proposition 5 shows that time-inconsistency can only emerge in an economy with transaction costs, i.e., $\phi > 0$. Proposition 5 is the exact counterpart of Proposition 1 with the ownership weight θ replaced by the aggregation weight w_t , which is the sense in which the concentration effect of the main text is not special to Grossman and Hart (1979).

Recall that in the Dreze (1974) setup, $w_t = \Theta_{t+1}$. Focusing on the concentration effect where these weights appear, $\widetilde{\text{cov}}_t (\Theta_{t+1}^j, \Delta_t^{j-}) = \widetilde{\text{cov}}_t (\theta^j + \Delta_t^{j+} - \Delta_t^{j-}, \Delta_t^{j-}) \leq \widetilde{\text{cov}}_t (\theta^j, \Delta_t^{j-})$, where the equality uses the law of motion $\Theta_{t+1}^j = \theta^j + \Delta_t^{j+} - \Delta_t^{j-}$. For the inequality, we note two things: first, $\widetilde{\text{cov}}_t (\Delta_t^{j+}, \Delta_t^{j-}) \leq 0$, since no household both buys and sells (see the proof of Lemma 2) and both quantities are non-negative; and, second, $\widetilde{\text{var}}_t (\Delta_t^{j-}) \geq 0$. Altogether, evaluated at the same cross-section, the Dreze (1974) weighting delivers a smaller concentration effect than the Grossman and Hart (1979) weighting, a force towards future bias.

In the case of majority voting from DeMarzo (1993), instead of studying the decomposition in Proposition 5, we directly consider whether the median shareholder is a seller. In numerical simulations, we find that sixty to seventy percent of shares belong to sellers. That is, a large number of households sell small amounts, while a small number of households buy large amounts; this is partly due to the quadratic nature of transaction costs. The median shareholder is therefore a seller, so $\bar{\Phi} = \phi m > 0$. When buyers tend to stay buyers, Φ^b is small, so $\bar{\Phi} > \Phi^b$ and the median rule also delivers present bias.

The net effect of all three competing forces depends on parameter values. However, in our numerical simulations, we find negligible quantitative differences in the wedge and resulting capital levels between the three weighting schemes.

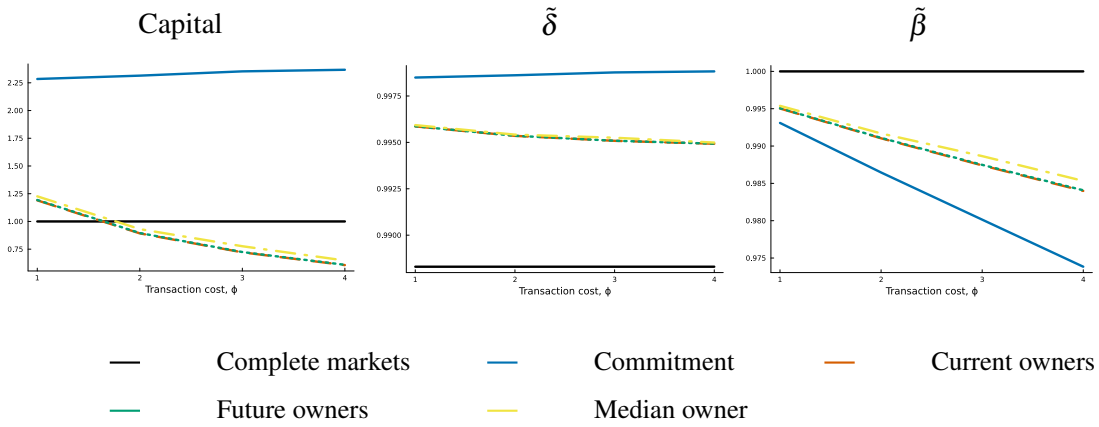
B.2 Quantitative Analysis

In this section, we quantitatively solve the general-equilibrium model under alternative assumptions about the weights in the firm's objective. For reference, we also report the complete-markets allocation. We then consider the model with commitment, whose steady state is inde-

pendent of the weighting rule because the steady-state investment condition depends on $\tilde{\delta}$ but not on $\tilde{\beta}$. Finally, we solve the model without commitment under three alternative weighting rules: (i) current owners (our benchmark specification, [Grossman and Hart, 1979](#)), (ii) future owners ([Dreze, 1974](#)), and (iii) the median owner ([DeMarzo, 1993](#)).

Figure 8 reports equilibrium capital and the effective discount factors $\tilde{\delta}$ and $\tilde{\beta}$ for different transaction costs, ϕ . Across all three no-commitment specifications, higher transaction costs lower $\tilde{\beta}$ and capital. Moreover, each specification generates present bias and substantially less capital than the commitment economy. Thus, the qualitative implications of trading frictions and lack of commitment are robust to the weighting rule used in the firm's objective.

Figure 8: Generalized Weights



Note: The panels report equilibrium capital, $\tilde{\delta}$, and $\tilde{\beta}$ under different transaction costs, ϕ . The black line denotes complete markets, the blue line denotes commitment, and the remaining lines denote no commitment under the current-owner, future-owner, and median-owner weighting rules. Although $\tilde{\beta}$ is reported for the commitment economy, it does not enter the firm's steady-state investment condition in that economy.

B.3 Grossman and Hart Derivation

In this section, we provide a foundation for an ownership-weighted objective of the form in (8). While derivations mirror those of [Grossman and Hart \(1979\)](#), our settings differ which makes the exercise worthwhile. An appropriate objective for the firm is one that has it choosing an investment plan $\{k_{t+s}\}_{s \geq 1}$ if there is no other plan $\{\hat{k}_{t+s}\}_{s \geq 1}$ which makes its shareholders better off. However, since the firm's ownership structure may change over time, it is not trivial to determine which shareholders should be considered.

Changes in a firm's investment plan will affect its shareholders in two ways. First, the dividends the firm pays will be altered. Higher investment in period t will lower the dividend in period t and increase it in $t + 1$. Second, the market price of the firm may change. While a single firm is atomistic with respect to the price of other assets (including the risk-free interest rate), its investment decisions do have an impact on its own share price. Formally, let $V_t^i(\{k_{t+s}\}_{s \geq 1})$ denote the maximized value of utility of an agent i (characterized by some state

variables (θ, b, h) as a function of the firm's investment plan, $\{k_{t+s}\}_{s \geq 1}$, and let $\mathcal{P}_{t,t+z}^i$ denote agent's i marginal propensity to pay in period t to change the capital in period $t+z$ by dk_{t+z} , for some $z \geq 1$. Then, by direct computation:

$$\begin{aligned} \mathcal{P}_{t,t+z}^i &\equiv \frac{1}{u'(\mathcal{C}_t^i)} \frac{\partial V_t^i(\{k_{t+s}\}_{s \geq 1})}{\partial k_{t+z}} \\ &= \mathbb{E}_t \left[\sum_{j=0}^{\infty} \left[\beta^j \frac{u'(\mathcal{C}_{t+j}^i)}{u'(\mathcal{C}_t^i)} \left(\theta_{t+j}^i \frac{\partial d_{t+j}}{\partial k_{t+z}} + \left(\Delta_{t+j}^{i,-} - \frac{\phi}{2} (\Delta_{t+j}^{i,-})^2 - \Delta_{t+j}^{i,+} \right) \left(\frac{\partial q_{t+j}}{\partial k_{t+z}} \right)^i \right) \right] \right]. \end{aligned} \quad (34)$$

Expression (34) includes two elasticities. The first describes how dividends are affected by changes in capital, and is given by $\frac{\partial d_{t+j}}{\partial k_{t+z}} = \frac{\partial F_{t+j}(k_{t+j}, k_{t+j+1})}{\partial k_{t+z}}$. We assume that all agents agree on this, as d_t is determined by technological factors. The second elasticity describes how the firm's share price is affected by changes in capital, and households need not agree on this elasticity.³⁹ That is, agents might disagree about how stock prices would change in the case of a change in the investment plan. Following Grossman and Hart (1979), we say that $\left(\frac{\partial q_{t+j}}{\partial k_{t+z}} \right)^i$ represents household i 's perception of the change in the market value of the firm caused by a change in its investment. The following lemma helps understand the role of the households' perceptions in expression (34).

Lemma 4. Let $M_{t+j,t+1+j}^i \equiv \beta \frac{u'(\mathcal{C}_{t+1+j}^i)}{u'(\mathcal{C}_{t+j}^i)}$ denote household i 's discount factor. Then, $\mathcal{P}_{t,t+z}^i$ can be written as

$$\begin{aligned} \mathcal{P}_{t,t+z}^i &= \theta_t^i \left(\frac{\partial d_t}{\partial k_{t+z}} + \left(1 - \frac{\phi}{2} \Delta_t^{i,-} \right) \left(\frac{\partial q_t}{\partial k_{t+z}} \right)^i \right) + \\ &\mathbb{E}_t \left[\sum_{j=0}^{\infty} \left(\prod_{u=0}^{j-1} M_{t+u,t+1+u}^i \right) \theta_{t+1+j}^i \left[M_{t+j,t+1+j}^i \left(\frac{\partial d_{t+1+j}}{\partial k_{t+z}} + \left(1 - \frac{\phi}{2} \Delta_{t+1+j}^{i,-} \right) \left(\frac{\partial q_{t+1+j}}{\partial k_{t+z}} \right)^i \right) \right. \right. \\ &\quad \left. \left. - \left(1 - \frac{\phi}{2} \Delta_{t+j}^{i,-} \right) \left(\frac{\partial q_{t+j}}{\partial k_{t+z}} \right)^i \right] \right]. \end{aligned} \quad (35)$$

Proof. Write $A_{t+j}^i \equiv \left(1 - \frac{\phi}{2} \Delta_{t+j}^{i,-} \right) \left(\frac{\partial q_{t+j}}{\partial k_{t+z}} \right)^i$, and note that $\prod_{u=0}^{j-1} M_{t+u,t+1+u}^i = \beta^j u'(\mathcal{C}_{t+j}^i) / u'(\mathcal{C}_t^i)$. Because no household both buys and sells (see the proof of Lemma 2), $\Delta_{t+j}^{i,+} \Delta_{t+j}^{i,-} = 0$, so the law of motion (3) gives

$$\Delta_{t+j}^{i,-} - \frac{\phi}{2} (\Delta_{t+j}^{i,-})^2 - \Delta_{t+j}^{i,+} = (\Delta_{t+j}^{i,-} - \Delta_{t+j}^{i,+}) \left(1 - \frac{\phi}{2} \Delta_{t+j}^{i,-} \right) = (\theta_{t+j}^i - \theta_{t+1+j}^i) \left(1 - \frac{\phi}{2} \Delta_{t+j}^{i,-} \right).$$

³⁹All shareholders must agree on the *level* of the firm's share price in equilibrium.

Substituting into (34) and splitting the sum,

$$\mathcal{P}_{t,t+z}^i = \mathbb{E}_t \left[\sum_{j=0}^{\infty} \beta^j \frac{u'(\mathcal{C}_{t+j}^i)}{u'(\mathcal{C}_t^i)} \theta_{t+j}^i \left(\frac{\partial d_{t+j}}{\partial k_{t+z}} + A_{t+j}^i \right) \right] - \mathbb{E}_t \left[\sum_{j=0}^{\infty} \beta^j \frac{u'(\mathcal{C}_{t+j}^i)}{u'(\mathcal{C}_t^i)} \theta_{t+1+j}^i A_{t+j}^i \right].$$

Re-indexing the first sum from j to $j+1$, separating its $j=0$ term, and using $\beta^{j+1} u'(\mathcal{C}_{t+1+j}^i) / u'(\mathcal{C}_t^i) = \left(\prod_{u=0}^{j-1} M_{t+u,t+1+u}^i \right) M_{t+j,t+1+j}^i$ delivers (35). $\square$

Lemma 4 shows that $\mathcal{P}_{t,t+z}^i$ can be decomposed into two main terms. The first term captures household i 's *current* wealth change due to the firm's investment change (note that changes in the household's choice variables have no first order effects because of the envelope theorem). It is given by the sum of the change in the dividend in period t plus the change in the firm's (ex-dividend) valuation. Crucially, the change in the firm's valuation takes into account that sellers do not receive the entire proceeds from selling stocks, as they have to pay the transaction cost. In particular, household i receives q_t per unit sold but it has to pay an average cost of $\frac{\phi}{2} (\Delta_t^{i,-}) q_t$ for that sale. Thus, $\left(1 - \frac{\phi}{2} (\Delta_t^{i,-}) \right) q_t$ represents household i 's net average valuation of the firm. Naturally, the effect of this change on the household's wealth is proportional to their initial holdings, θ_t^i .

The second term captures discounted gains (or losses) from holding θ_{t+1+j}^i amount of the stock in the future. Inside of this discounted sum, there are two terms: the change in household i 's perception of the average cost of the share in $t+j$, $\left(1 - \frac{\phi}{2} \Delta_{t+j}^{i,-} \right) \left(\frac{\partial q_{t+j}}{\partial k_{t+z}} \right)^i$, and the change in the perceived benefits of holding the stock for one period, $\mathbb{E}_{t+j} \left[M_{t+j,t+1+j}^i \left(\frac{\partial d_{t+1+j}}{\partial k_{t+z}} + \left(1 - \frac{\phi}{2} \Delta_{t+1+j}^{i,-} \right) \left(\frac{\partial q_{t+1+j}}{\partial k_{t+z}} \right)^i \right) \right]$. These terms depend on two household-specific objects, besides $\left(\frac{\partial q_{t+j}}{\partial k_{t+z}} \right)^i$: the household's own discount factor, $M_{t+j,t+1+j}^i$, and its own path of sales, $\Delta_{t+j}^{i,-}$, which determines the average transaction cost it pays. Nonzero transaction costs are a necessary condition for household disagreement in these idiosyncratic variables, as we show next.

Consider the benchmark with $\phi = 0$, so that there are no transaction costs associated with trading stocks, and fix a particular period, $t+j$. Then, the future gain (or loss) for household i in period $t+j$ simplifies to

$$\mathbb{E}_{t+j} \left[M_{t+j,t+1+j}^i \left(\frac{\partial d_{t+1+j}}{\partial k_{t+z}} + \left(\frac{\partial q_{t+1+j}}{\partial k_{t+z}} \right)^i \right) \right] - \left(\frac{\partial q_{t+j}}{\partial k_{t+z}} \right)^i.$$

Note that household i will only care about this difference if $\theta_{t+1+j}^i > 0$ (see equation 35). In that case its short-selling constraint is slack, so its Euler equation for the stock holds with equality. Because there is no aggregate risk, d_{t+1+j} and q_{t+1+j} are deterministic, and that Euler equation reads $q_{t+j} = \mathbb{E}_{t+j} \left[M_{t+j,t+1+j}^i \right] (d_{t+1+j} + q_{t+1+j})$. Hence $\mathbb{E}_{t+j} \left[M_{t+j,t+1+j}^i \right] = 1/(1+r_{t+j})$ takes the same value for every household that holds the stock: all stockholders

price the same riskless payoff.⁴⁰ Since $\frac{\partial d_{t+1+j}}{\partial k_{t+z}}$ is also deterministic, rational expectations then imply that perceptions solve the common recursion

$$\left(\frac{\partial q_{t+j}}{\partial k_{t+z}}\right)^i = \frac{\frac{\partial d_{t+1+j}}{\partial k_{t+z}} + \left(\frac{\partial q_{t+1+j}}{\partial k_{t+z}}\right)^i}{1 + r_{t+j}},$$

and so, absent bubbles, all stockholders share the same perception and we obtain

$$\mathcal{P}_{t,t+z}^i = \theta_t^i \left(\frac{\partial d_t}{\partial k_{t+z}} + \frac{\partial q_t}{\partial k_{t+z}} \right).$$

This analysis recovers the standard result that all households agree that the problem of the firm is to maximize the *current* value of the firm. This is the natural result when all agents agree on the appropriate discount factor for the firm's dividends. An immediate implication of this result is that households do not benefit from future changes in valuations. Any benefit arising from the change in investment plans is reflected in current valuations.

When $\phi > 0$, this fails to be the case, as transaction costs generate dispersion in the household idiosyncratic variables. In the spirit of [Grossman and Hart \(1979\)](#), we assume that households' perceptions are such that they will not benefit or lose from future capital gains and losses; asset prices will react to a change in investment in such a way that the future benefits of holding the stock are precisely offset by the cost. Note that each household considers their own idiosyncratic variables, $\frac{\phi}{2}\Delta_{t+j}^{i,-}$ and $M_{t+j,t+1+j}^i$, in lieu of those of the overall market when forming this perception. The belief is that, when a firm alters its production plan, utility will not change except to the extent that the household is an initial shareholder of the firm. Because of this property of the assumption, [Grossman and Hart \(1979\)](#) describe this as a “utility-taking” assumption. In their paper, they call the assumption *competitive perceptions*.

Assumption 4 (Competitive Perceptions). *For all households with $\theta_{t+j+1}^i > 0$,*

$$\left(1 - \frac{\phi}{2}\Delta_{t+j}^{i,-}\right) \left(\frac{\partial q_{t+j}}{\partial k_{t+z}}\right)^i = \mathbb{E}_{t+j} \left[M_{t+j,t+1+j}^i \left(\frac{\partial d_{t+1+j}}{\partial k_{t+z}} + \left(1 - \frac{\phi}{2}\Delta_{t+1+j}^{i,-}\right) \left(\frac{\partial q_{t+1+j}}{\partial k_{t+z}}\right)^i \right) \right],$$

for all $t, j, z \geq 0$.

Given Assumption 4, the households' marginal propensity to pay in period t for a change in investment in period $t + z$ simplifies to

$$\mathcal{P}_{t,t+z}^i = \theta_t^i \left(\frac{\partial d_t}{\partial k_{t+z}} + \left(1 - \frac{\phi}{2}\Delta_t^{i,-}\right) \left(\frac{\partial q_t}{\partial k_{t+z}}\right)^i \right).$$

To determine the problem of the firm, we follow [Dreze \(1974\)](#) and [Grossman and Hart \(1979\)](#),

⁴⁰This common value is the risk-free rate: by Lemma 2 some stockholder is not constrained by the borrowing limit for bonds, and its Euler equation for bonds gives $\mathbb{E}_{t+j}[M_{t+j,t+1+j}^i] = 1/(1 + r_{t+j})$.

and assume that the firm's manager is able to transfer current income from those who favor the change in investment plan ($\mathcal{P}_{t,t+z}^i > 0$) to those who oppose it ($\mathcal{P}_{t,t+z}^i < 0$). Therefore, the firm increases its investment up to the point where

$$\int_{\theta,b,h} \mathcal{P}_{t,t+z}(\theta,b,h) d\Gamma_t(\theta,b,h) = 0, \quad (36)$$

where $\Gamma_t(\theta,b,h)$ denotes the cross-section distribution over the portfolio holdings and employment status.⁴¹

To summarize, we can substitute Assumption 4 for Assumption 1 in the main text, then use Lemma 4 to arrive at the first-order condition (36) of an objective of the same form as (8), with the average transaction cost, $\frac{\phi}{2}\Delta_t^{i,-}$, in place of the marginal transaction cost, $\phi\Delta_t^{i,-}$. While each household uses its own competitive perceptions to calculate price elasticities, this still leaves an open question of firm perceptions. The analog of Assumption 2 is that the firm uses the unconstrained buyer's competitive perception from Assumption 4, which carries the same factor (of two). Both objects that enter the firm's problem, the wedge it internalizes and the discount factor it applies to future dividends, therefore carry it: the objective of this appendix is that of (8) with $\frac{\phi}{2}$ in place of ϕ in both $\bar{\Phi}_t$ and Φ_t^b . Since every term in Proposition 1 is proportional to ϕ , the same decomposition applies, with the same three components and the same signs, and $\tilde{\beta} = 1$ if $\phi = 0$.

C 3-Period Model

C.1 Stationary Model

Consider a stationary version of the 3-period model laid out in Section 3. Like before, there are only three time periods $t = 0, 1, 2$ and $c_t = 0$ for $t \geq 3$; however $q_2 \neq 0$. The reason households hold stock beyond period 2 is due to an exogenous “continuation value” we append to the household's objective. This trick allows us to induce a stationary 3-period equilibrium with constant asset prices, dividends, and wages. The household objective becomes

$$\sum_{t=0}^2 \beta^t u(c_t^j) + \beta^2 \mu^j \theta_3^j$$

The Euler equation in $t = 2$ becomes

$$q_2 u'(c_2^j)(1 - \phi \Delta_2^{j,-}) = \mu^j$$

⁴¹Under the analog of Assumption 2 discussed at the end of this appendix, the firm takes the average transaction cost and Φ_t^b as given, so (36) is the first-order condition of $\sum_{s \geq 0} \omega_s F(k_{t+s}, k_{t+s+1})$, a positively weighted sum of concave dividends, and is therefore necessary and sufficient for optimality.

Table 6: Stationary Equilibrium

Household type	c_0	c_1	c_2	Δ_0^-	Δ_1^-	Δ_2^-	θ_0
<i>low</i>	2.0998	2.1680	2.0998	0.1614	0.0000	0.1614	1.0807
<i>high</i>	2.1680	2.0998	2.1680	0.0000	0.1614	0.0000	0.9193

If μ^j and initial assets θ_0^j are selected appropriately, then the resulting household equilibrium will be stationary. In this stationary equilibrium, consumption will simply fluctuate (i.e. c^l, c^h, c^l), asset sales will also fluctuate (i.e. $\Delta^-, 0, \Delta^-$), and asset prices, dividends, and wages will be constant over time. Using this notation, the required μ^j are given by

$$\mu^l = \beta u'(c^h)(d + q), \text{ and } \mu^h = \beta u'(c^l)(d + (1 - \phi\Delta^-)q)$$

We now re-solve the numerical example in Section 3 with the same parameter values, except for the initial conditions, which are set to their stationary levels. The household equilibrium is summarized in Table 6, asset prices are given by $q = 5.4489$, dividends are given by $d = 0.2821$, capital is $k = 5.1459$ (note that $k_0 = k_3$ are also fixed at this value), and wages are given by $w = 0.9295$.⁴²

The problem of the firm is defined as before, which, in period 0, becomes

$$V_0^F(k_0) = \max_{k_1, k_2 \geq 0} d_0 + 0.9480d_1 + 0.9025d_2,$$

and, in period 1, becomes

$$V_1^F(k_1) = \max_{k_2 \geq 0} d_1 + 0.9480d_2.$$

Due to the stationary nature of our new equilibrium, the one-period-ahead discount rate is always the same, 0.9480, and the two-period-ahead discount rate is simply β^2 (that is, consumption at time t and time $t + 2$ agree). From its period 0 perspective, the firm discounts between time $t = 1$ and $t = 2$ at the rate 0.9520, which is the ratio between 0.9025 and 0.9480. When period 1 arrives, the firm discounts between time $t = 1$ and $t = 2$ at the rate 0.9480. Hence the problem of the firm is time-inconsistent and, like in Section 3, the direction of inconsistency is towards present bias.

C.2 Symmetric Transaction Costs

Consider a 3-period economy, as in Section 3, but with symmetric transaction costs. That is, the budget constraint (2) now has one additional term $\frac{\phi}{2}(\Delta_t^{j+})^2 q_t$, which denotes the transaction cost incurred for buying the stock. Noting that $q_2 = 0$ like in Section 3, the new budget

⁴²Unlike the other three period examples, these calculations assume that the firm re-optimizes in period one. The reason for this change is the resulting choice of capital would otherwise be non-stationary.

Table 7: Equilibrium with Symmetric Transaction Costs

Household type	c_0	c_1	c_2	Δ_0^-	Δ_1^-	θ_1	θ_2
<i>low</i>	1.1721	1.7099	2.0760	0.3142	0.0000	0.6858	1.1827
<i>high</i>	1.6012	1.8281	2.7092	0.0000	0.4969	1.3142	0.8173

constraint for $t = 0, 1, 2$ is

$$c_t^j + q_t \Delta_t^{j+} \leq w_t h_t^j + d_t \theta_t^j + \left(\Delta_t^{j-} - \frac{\phi}{2} (\Delta_t^{j-})^2 - \frac{\phi}{2} (\Delta_t^{j+})^2 \right) q_t$$

and the Euler equation of the household becomes

$$(1 + \phi \Delta_t^j) q_t = \beta \frac{u'(c_{t+1}^j)}{u'(c_t^j)} d_{t+1} + \beta \frac{u'(c_{t+1}^j)}{u'(c_t^j)} (1 + \phi \Delta_{t+1}^j) q_{t+1}, \quad t \in \{0, 1\}$$

where $\Delta_t^j \equiv (\Delta_t^{j+} - \Delta_t^{j-})$. Intuitively, buyers now have an even higher valuation of the stock at the margin than in the asymmetric setting because they must incur a transaction cost to buy. Calculations similar to those of Lemma 1 yield a firm objective equal to

$$\sum_{j \in \{l, h\}} \frac{\theta_t^j}{2} [d_t + (1 + \phi \Delta_t^j) q_t]$$

With this new setup, we solve a numerical example using the same parameter values as in Section 3. The household equilibrium is summarized in Table 7, asset prices are given by $q_0 = 1.1267$ and $q_1 = 0.9172$, dividends are given by $d_0 = 0.2607$, $d_1 = 0.6131$, and $d_2 = 1.2886$, capital is $k_1 = 1.1267$ and $k_2 = 0.9061$, and wages are given by $w_0 = 0.5686$, $w_1 = 0.5893$, and $w_2 = 0.5520$.

The problem of the firm in period 0 becomes

$$V_0^F(k_0) = \max_{k_1, k_2 \geq 0} d_0 + 0.7417 d_1 + 0.5215 d_2,$$

and the problem of the firm in period 1 becomes

$$V_1^F(k_1) = \max_{k_2 \geq 0} d_1 + 0.6895 d_2.$$

so that, from its period 0 perspective, the firm discounts between time $t = 1$ and $t = 2$ at the rate 0.7031, which is the ratio between 0.5215 and 0.7417. When period 1 arrives, the firm discounts between time $t = 1$ and $t = 2$ at the rate 0.6895. Hence the problem of the firm remains time-inconsistent, with a direction of inconsistency towards present bias. Transaction costs on buyers further restrict asset trading (beyond the case of asymmetric transaction costs), which creates larger differences in IMRS across agents; this exacerbates problems of time-

inconsistency.

C.3 Shareholder Composition Across Firms

Our analysis assumes the existence of a representative firm, where the composition of shareholders is the same across firms. In this appendix, we ask whether, in the tractable stationary three-period setting of Appendix C.1, there exist other equilibria in which this is not the case. It turns out that there are none: any persistent difference would create an arbitrage opportunity through cross-firm portfolio rebalancing. In steady state this forces price-dividend ratios and seller measures to equalize across firms, so the representative firm assumption is without loss in the equilibrium class we study.

The way we model heterogeneous firms is with the following household budget

$$c_t + \sum_{f=1}^F q_t^f \Delta_t^{f+} \leq w_t h_t + \sum_{f=1}^F d_t^f \theta_t^f + \sum_{f=1}^F \left(\Delta_t^{f-} - \frac{\phi}{2} F (\Delta_t^{f-})^2 \right) q_t^f,$$

with market clearing conditions

$$\int_{j \in [0,1]} \theta_{j,t}^f dj = \frac{1}{F}$$

for each f , and the continuum is the limiting case as F tends to infinity. For a discussion about the extra factor F in the transaction cost, see Lemma 3 and Appendix D. Transaction costs imply that each household is on the same side of the market for all firms. To see this, suppose a household sells $\Delta^{f'-} > 0$ shares of firm f' but none of firm f in period t , so that, by stationarity, it buys back f' in period $t+1$, when it sells $\Delta^{f-} \geq 0$ shares of f . Since the household discounts both firms' payoffs with the same marginal rate of substitution, taking ratios of its Euler equations for the two firms in period t , and then in period $t+1$, gives

$$\begin{aligned} 1 + \frac{d^{f'}}{q^{f'}} &\leq \left(1 - \phi F \Delta^{f'-}\right) \left(1 + \frac{d^f}{q^f} - \phi F \Delta^{f-}\right) < 1 + \frac{d^f}{q^f}, \\ 1 + \frac{d^f}{q^f} &\leq \left(1 - \phi F \Delta^{f-}\right) \left(1 + \frac{d^{f'}}{q^{f'}} - \phi F \Delta^{f'-}\right) < 1 + \frac{d^{f'}}{q^{f'}}, \end{aligned}$$

a contradiction. Recall the stationary Euler equations,

$$q^f u'(c^l)(1 - \phi F \Delta^{f-}) = \beta u'(c^h)(d^f + q^f), \text{ and } q^f u'(c^h) = \beta u'(c^l)(d^f + (1 - \phi F \Delta^{f-})q^f),$$

which can be rearranged to

$$\frac{q^f}{d^f} = \frac{\beta u'(c^l) + \beta^2 u'(c^h)}{u'(c^h)(1 - \beta^2)},$$

and

$$\Delta^{f-} = \frac{\beta u'(c^h) + \beta^2 u'(c^l)(1 + q^f/d^f) - u'(c^l)q^f/d^f}{(\beta^2 - 1)u'(c^l)\phi F q^f/d^f}.$$

The first rearranged equation shows that steady-state price-dividend ratios, q^f/d^f , must equate across firms. The second rearranged equation shows that the quantity sold for the low-type, Δ^{f-} , consequently equates across firms. By stationarity, the quantity bought for the high-type, Δ^{f+} , must then equate across firms.

Recall that, for each firm f , the initial allocation θ_0^j for $j \in \{l, h\}$ must be picked to ensure stationarity. Explicitly, this requirement can be written

$$\frac{1}{2}\theta_0^l + \frac{1}{2}\theta_0^h = \frac{1}{F}, \text{ and } \theta_0^l - \Delta^- = \theta_0^h,$$

where the first equality reflects market clearing and the second equality is the law of motion for stocks. While the superscript f was suppressed to reduce clutter, Δ^{f-} equating across firms (along with our two equalities above) implies that (θ_0^l, θ_0^h) equates across firms. Hence, the intertemporal firm problem agrees across all firms in every period, and its solution d^f must agree across all firms. With equated price-dividend ratios, q^f must agree across all firms.

C.4 Capital Gains Tax

In this section, we substitute the transaction cost for a capital gains tax and derive similar results to those of Section 3. At the same time, we introduce a dividend tax to show that dividend taxes alone do not generate time inconsistency. Because there are no capital gains in period $t = 0$, the budget constraint (2) remains unchanged but with $\phi = 0$ and $(1 - \tau_d)d_0\theta_0^j$ in lieu of $d_0\theta_0^j$, where τ_d denotes the dividend tax parameter. In period $t = 1$, the budget becomes

$$c_1^j + q_1\Delta_1^{j+} \leq w_1h_1^j + (1 - \tau_d)d_1\theta_1^j + q_1\Delta_1^{j-} - \frac{\tau_c}{2}(\Delta_1^{j-})^2(q_1 - q_0) - \mathcal{T}_t$$

where τ_c denotes the capital gains tax parameter. We make the tax quadratic in the quantity sold, as with the transaction cost in Section 3, for tractability. What matters for the result is that the tax falls on the proceeds of sellers and not on holders, so it separates current owners' valuation of the stock from the marginal buyer's. $q_2 = 0$ because there is no need to hold assets beyond $t = 2$; otherwise budgets remain unchanged from (2) for $t = 2$ aside from, again, $(1 - \tau_d)d_2\theta_2^j$ in lieu of $d_2\theta_2^j$. Like in Section 3, we assume that households face no income risk and cannot trade bonds; however, we now re-introduce the lump-sum transfer $\mathcal{T}_t$ to redistribute revenues from both taxes back to the households. Government debt is set to zero. The envelope condition yields (focusing on the new $t = 1$):

$$\frac{V_{\theta_1^j}^j}{\lambda_1^j} = (1 - \tau_d)d_1 + (1 - \tau_c\Delta_1^{j-})q_1 + \tau_c\Delta_1^{j-}q_0$$

Invoking Assumption 1, firms maximize

$$\sum_{j \in \{l, h\}} \frac{\theta_1^j}{2} \left[(1 - \tau_d) d_1 + (1 - \tau_c \Delta_1^{j-}) q_1 + \tau_c \Delta_1^{j-} q_0 \right] \quad (37)$$

The households' Euler equations for $t = 0$ are

$$q_0 = \beta \frac{u'(c_1^j)}{u'(c_0^j)} ([1 - \tau_d] d_1 + [1 - \tau_c \Delta_1^{j-}] q_1 + \tau_c \Delta_1^{j-} q_0) \quad (38)$$

and for $t = 1$ are

$$(1 - \tau_c \Delta_1^{j-}) q_1 + \tau_c \Delta_1^{j-} q_0 = \beta \frac{u'(c_2^j)}{u'(c_1^j)} (1 - \tau_d) d_2 \quad (39)$$

We focus on equilibria where the short-selling constraint does not bind. Introducing (38) and (39) into the firm objective in period $t = 0$ (which follows the same calculations as those in Section 3), we get that the firm's objective in period 0 is given by

$$V_0^F(k_0) = \max_{k_1, k_2 \geq 0} \sum_{j \in \{l, h\}} \frac{\theta_0^j}{2} \left[(1 - \tau_d) d_0 + \beta \frac{u'(c_1^j)}{u'(c_0^j)} (1 - \tau_d) d_1 + \beta^2 \frac{u'(c_2^j)}{u'(c_0^j)} (1 - \tau_d) d_2 \right], \quad (40)$$

where $d_t = F_t(k_t, k_{t+1})$ and the firm takes the households' consumption plans as given. The firm's objective in period 1 is given by (37). Substituting in (39), the firm problem at $t = 1$ becomes

$$V_1^F(k_1) = \max_{k_2 \geq 0} \sum_{j \in \{l, h\}} \frac{\theta_1^j}{2} \left[(1 - \tau_d) d_1 + \beta \frac{u'(c_2^j)}{u'(c_1^j)} (1 - \tau_d) d_2 \right] \quad (41)$$

The firm's problem is time-consistent if and only if the discounting between $t = 1$ and $t = 2$ in problems (40) and (41) coincide. First, we show that $\tau_c = 0$ leads to a time-consistent firm problem. Then, we illustrate the firm's time inconsistency when $\tau_c > 0$ with a numerical example.

While the common factor $(1 - \tau_d)$ in problems (40) and (41) does not affect the firm's incentives, we still need to show that discounting between periods one and two from the period-0 perspective and period-1 perspective agree when $\tau_c = 0$. Following the same reasoning as that in Section 3, the Euler equation with $\tau_c = 0$ becomes, for $t = 0, 1$,

$$\frac{q_t}{(1 - \tau_d) d_{t+1} + q_{t+1}} = \beta \frac{u'(c_{t+1}^j)}{u'(c_t^j)}.$$

This equation shows that the IMRS are equalized across agents in each period. We then com-

Table 8: Equilibrium in the Capital Gains Model

Household type	c_0	c_1	c_2	Δ_0^-	Δ_1^-	θ_1	θ_2
<i>low</i>	1.4292	2.7040	5.3926	0.4880	0.0000	0.5120	1.4561
<i>high</i>	1.7412	3.0608	6.8485	0.0000	0.9441	1.4880	0.5439

pare the discount factor between period one and two from the period-0 perspective (leftmost term) to the corresponding discount factor from the period-1 perspective (rightmost term) below:

$$\frac{\sum_{j \in \{l, h\}} \frac{\theta_0^j}{2} \beta^2 \frac{u'(c_2^j)}{u'(c_0^j)}}{\sum_{j \in \{l, h\}} \frac{\theta_0^j}{2} \beta \frac{u'(c_1^j)}{u'(c_0^j)}} = \frac{\frac{q_0}{(1-\tau_d)d_1+q_1} \frac{q_1}{(1-\tau_d)d_2+q_2}}{\frac{q_0}{(1-\tau_d)d_1+q_1}} = \frac{q_1}{(1-\tau_d)d_2+q_2} = \sum_{j \in \{l, h\}} \frac{\theta_1^j}{2} \beta \frac{u'(c_2^j)}{u'(c_1^j)}.$$

We conclude that the problem is time consistent when $\tau_c = 0$.

In the numerical example, we maintain the parameter values of Section 3 aside from three changes: we replace the transaction cost with a capital gains tax, $\tau_c = 0.5$, we introduce the dividend tax $\tau_d = 0.1$, and let TFP grow, $A = (1, 2, 3)$, with the production function generalized to $y_t = A_t \left(\ell_t^\gamma k_t^{1-\gamma} \right)^\psi$. The remaining parameters are $\beta = 0.95$, $u = \ln(c)$, $h_{low} = 1$, $h_{high} = 3$, $\gamma = 0.7$, $\psi = 1$, $k_0 = 1$, and $\delta = 0.1$. Initial asset holdings are $\theta_0^j = 1$ for $j \in \{l, h\}$. Note the need for increasing TFP: this ensures capital gains, not losses, in a three-period model. The household equilibrium is summarized in Table 8, asset prices are given by $q_0 = 0.8454$ and $q_1 = 1.0982$, dividends are given by $d_0 = 0.4481$, $d_1 = 0.6504$, and $d_2 = 2.5616$, capital is $k_1 = 0.9393$ and $k_2 = 1.1515$, and wages are given by $w_0 = 0.5686$, $w_1 = 1.1160$, and $w_2 = 1.7795$.

Dropping the common factor $(1 - \tau_d)$, the problem of the firm in period 0 becomes

$$V_0^F(k_0) = \max_{k_1, k_2 \geq 0} d_0 + 0.5213d_1 + 0.2343d_2,$$

and the problem of the firm in period 1 becomes

$$V_1^F(k_1) = \max_{k_2 \geq 0} d_1 + 0.4378d_2.$$

We compare the firm's problem in $t = 0$ and $t = 1$ along the equilibrium path. From its period 0 perspective, the firm's discount factor between $t = 1$ and $t = 2$ is 0.4495, which is the ratio between 0.2343 and 0.5213. When period 1 arrives, the firm's discount factor between $t = 1$ and $t = 2$ is 0.4378. Hence, the problem of the firm is time-inconsistent and, in this calibration, the direction of inconsistency is towards present bias.

D Heterogeneous Firms: Liquid and Illiquid

In this appendix we extend the model to have heterogeneous firms. We quantitatively analyze the aggregate implications and show that the model is consistent with the cross-sectional evidence.

D.1 Model

Consider an economy with both liquid and illiquid stocks. To make the comparison stark, we assume that liquid stocks trade in markets with zero transaction costs but are still subject to the same short-selling constraint. The proportion of liquid stocks is $(1 - \omega)$, and the proportion of illiquid stocks is ω , so that the total mass of stocks is still normalized to one. The special case of $\omega = 1$ corresponds to our main model.

With two types of firms in the economy (with potentially differing discount rates), we must restrict our analysis to decreasing returns technologies to retain an interior solution to both firms' problems. The household budget constraint (2) becomes

$$c_t + q_t \Delta_t^+ + b_{t+1} + \hat{q}_t \hat{\theta}_{t+1} \leq w_t h_t + d_t \theta_t + (\hat{d}_t + \hat{q}_t) \hat{\theta}_t + \left(\Delta_t^- - \frac{\phi}{2\omega} (\Delta_t^-)^2 \right) q_t + b_t (1 + r_{t-1}) - \mathcal{T}_t$$

where $\hat{q}_t$ denotes the price of the liquid stock, $\hat{\theta}_{t+1}$ denotes the amount of liquid stock purchased, and $\hat{d}_t$ denotes dividends paid by the liquid firm.⁴³ Beyond the standard constraints (3), (4), and (5), we also have the short-selling constraint for liquid stocks

$$\hat{\theta}_{t+1} \geq 0. \tag{42}$$

To simplify this problem, we note that liquid stocks and bonds are equivalent, and hence we can combine them into one state variable $W_{t+1} \equiv (\hat{d}_{t+1} + \hat{q}_{t+1}) \hat{\theta}_{t+1} + b_{t+1} (1 + r_t)$ so that the budget constraint can be rewritten

$$c_t + q_t \Delta_t^+ + \frac{W_{t+1}}{1 + r_t} \leq w_t h_t + d_t \theta_t + W_t + \left(\Delta_t^- - \frac{\phi}{2\omega} (\Delta_t^-)^2 \right) q_t - \mathcal{T}_t.$$

The short-selling constraint (42), along with the borrowing limit (5), implies that

$$W_{t+1} \geq \bar{b}(1 + r_t)$$

⁴³As in the counterfactual of Lemma 3, the transaction cost applies to the *fraction* of the illiquid firms' outstanding shares that is traded: selling Δ_t^- shares out of a float of ω costs $\frac{\phi}{2} (\Delta_t^- / \omega)^2 q_t \omega$. This reduces to the cost in (2) at $\omega = 1$, the main model.

Market clearing conditions for bonds, illiquid stocks, and labor become

$$\begin{aligned}\int_{j \in [0,1]} W_{j,t} dj &= (1 - \omega)(\hat{d}_t + \hat{q}_t) + B^G(1 + r_{t-1}), \\ \int_{j \in [0,1]} \theta_{j,t} dj &= \omega, \quad \int_{j \in [0,1]} h_{j,t} dj = \omega l_t + (1 - \omega) \hat{l}_t.\end{aligned}$$

In the first equation above, the equality follows from the fact that the proportion of liquid stocks is $(1 - \omega)$. In the third equation above, $\hat{l}_t$ denotes labor demanded by the liquid firm, which is defined by the intratemporal first-order condition, $\hat{l}_t = \psi \gamma \frac{\hat{y}_t}{w_t}$, where $\hat{y}_t = [(\hat{l}_t)^\gamma (\hat{k}_t)^{1-\gamma}]^\psi$.⁴⁴ Labor demanded by the illiquid firm, l_t , is defined by similar conditions, $l_t = \psi \gamma \frac{y_t}{w_t}$, where $y_t = [(l_t)^\gamma (k_t)^{1-\gamma}]^\psi$. Combining the expression for output with the optimality condition for labor, we get $l_t = (\gamma \psi)^{\frac{1}{1-\gamma\psi}} \frac{k_t^{\frac{(1-\gamma)\psi}{1-\gamma\psi}}}{w_t^{\frac{1}{1-\gamma\psi}}}$, and $\hat{l}_t = (\gamma \psi)^{\frac{1}{1-\gamma\psi}} \frac{\hat{k}_t^{\frac{(1-\gamma)\psi}{1-\gamma\psi}}}{w_t^{\frac{1}{1-\gamma\psi}}}$. Thus, market clearing in the labor market implies

$$\omega l_t + (1 - \omega) \hat{l}_t = H = \omega (\gamma \psi)^{\frac{1}{1-\gamma\psi}} \frac{k_t^{\frac{(1-\gamma)\psi}{1-\gamma\psi}}}{w_t^{\frac{1}{1-\gamma\psi}}} + (1 - \omega) (\gamma \psi)^{\frac{1}{1-\gamma\psi}} \frac{\hat{k}_t^{\frac{(1-\gamma)\psi}{1-\gamma\psi}}}{w_t^{\frac{1}{1-\gamma\psi}}}$$

or

$$w_t = \frac{\gamma \psi}{H^{1-\gamma\psi}} \left(\omega k_t^{\frac{(1-\gamma)\psi}{1-\gamma\psi}} + (1 - \omega) \hat{k}_t^{\frac{(1-\gamma)\psi}{1-\gamma\psi}} \right)^{1-\gamma\psi}.$$

Now, let's use the optimality conditions for capital. Focusing on a steady-state equilibrium, the analysis in Section 5 implies

$$M(z\alpha k^{\alpha-1} + (1 - \delta)) = 1$$

for the illiquid firms, and

$$\hat{M}(z\alpha \hat{k}^{\alpha-1} + (1 - \delta)) = 1$$

for the liquid firms, where $M = \tilde{\delta}$ if the firms can commit to an investment plan and $M = \tilde{\beta}\tilde{\delta}$ if they cannot, $\hat{M} = \frac{1}{1+r}$, and $z = (1 - \gamma\psi) \left(\frac{\gamma\psi}{w} \right)^{\frac{\gamma\psi}{1-\gamma\psi}} = \frac{(1-\gamma\psi)H^{\gamma\psi}}{\left(\omega k^{\frac{(1-\gamma)\psi}{1-\gamma\psi}} + (1-\omega)\hat{k}^{\frac{(1-\gamma)\psi}{1-\gamma\psi}} \right)^{\gamma\psi}}$. That is, the

liquid stock has the same return as the bond, so it discounts the dividends it generates at the risk-free rate $\frac{1}{1+r}$, with standard exponential discounting. In contrast, the discount factor of illiquid

⁴⁴In our extension from Appendix E.1, the budget is

$$c_t + q_t \Delta_t^+ + \frac{W_{t+1}}{1+r_t} - b_{t+1}^- \leq w_t h_t + d_t \theta_t + W_t + \left(\Delta_t^- - \frac{\phi}{2\omega} (\Delta_t^-)^2 \right) q_t - b_t^- [1 + r_{t-1} + \xi_b(b_t^-)] - \mathcal{T}_t$$

where $W_{t+1} \equiv (\hat{d}_{t+1} + \hat{q}_{t+1})\hat{\theta}_{t+1} + b_{t+1}^+(1 + r_t) \geq 0$, and the market clearing condition for bonds becomes

$$\int_{j \in [0,1]} W_{j,t} dj = (1 - \omega)(\hat{d}_t + \hat{q}_t) + \left(B^G + \int_{j \in [0,1]} b_{j,t}^- dj \right) (1 + r_{t-1}).$$

firms is $\frac{1-\Phi^b}{1+r} < \frac{1}{1+r}$ with commitment, and $\frac{1-\bar{\Phi}}{1+r} < \frac{1}{1+r}$ without commitment. Moreover, the two types also differ in their corporate discount-rate wedges. Because sellers of liquid equity pay no transaction cost, $\bar{\Phi} = \Phi^b = 0$ for liquid firms and their wedge is zero; for illiquid firms the wedge is zero under commitment and equal to $-\log \tilde{\beta} > 0$ without it.

Combining the two optimality conditions, we get $\hat{k} = \left(\frac{\frac{1}{\hat{M}} - (1-\delta)}{\frac{1}{\bar{M}} - (1-\delta)} \right)^{\frac{1}{1-\alpha}} k$. Since, $M < \hat{M}$, then $\hat{k} > k$. To solve for k , we plug the expressions for z and $\alpha = \frac{(1-\gamma)\psi}{1-\gamma\psi}$ into the optimality condition for illiquid firms,

$$k = \left[\frac{1}{\left(\omega + (1-\omega) \left(\frac{\frac{1}{\hat{M}} - (1-\delta)}{\frac{1}{\bar{M}} - (1-\delta)} \right)^{\frac{(1-\gamma)\psi}{1-\gamma\psi}} \right)^{\gamma\psi} \frac{(1-\gamma)\psi M}{1-M(1-\delta)} H^{\gamma\psi}} \right]^{\frac{1}{1-(1-\gamma)\psi}}.$$

Thus, firms with illiquid stock always have less capital than firms with liquid stock. This result is consistent with [Amihud and Levi \(2023\)](#), who use cross-sectional data for US public firms to compare liquid and illiquid firms and conclude that illiquid firms invest less than liquid ones.⁴⁵ Next, we quantitatively analyze this extension.

D.2 Quantitative Results

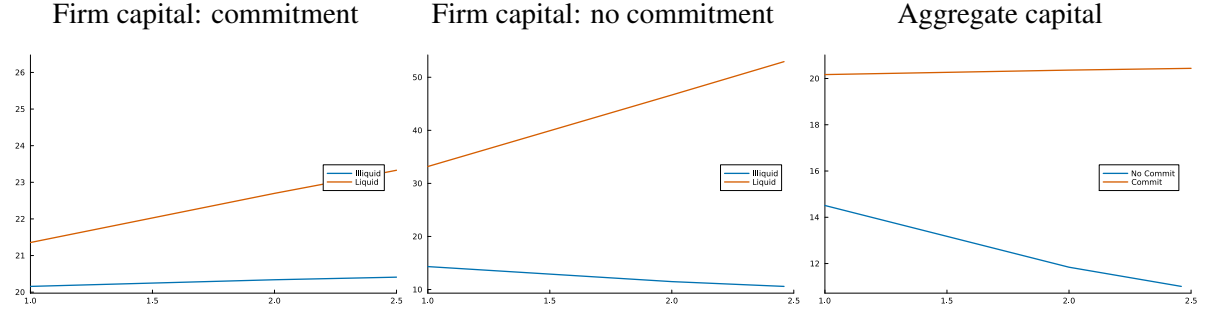
We now provide a quantitative evaluation of the model with two types of firms. Relative to the benchmark calibration, we make the following changes. First, as discussed in the previous subsection, an interior equilibrium with both firm types requires decreasing returns to scale; we therefore set $\psi = 0.9$. Second, we set government debt to zero, $B^G = 0$, so that liquid equity replaces government bonds as the source of the net supply of liquid assets. We choose $\omega = 0.99$ so that, at the no-commitment benchmark with $\phi = 2.46$, the supply of liquid-firm claims is approximately equal to the benchmark supply of government debt. This calibration implies liquid-firm claims equal to 28.2 percent of annual GDP, compared with government debt equal to 26 percent in the benchmark economy. Except for ϕ , which we vary in Figure 9, all remaining parameters are held at their benchmark values.

The exercise varies the transaction cost on illiquid equity, ϕ , while liquid equity continues to trade without transaction costs. The first two panels of Figure 9 confirm the model's cross-sectional implications. With commitment, firms with illiquid equity hold less capital than firms with liquid equity. This difference is substantially larger when illiquid firms cannot commit and widens as transaction costs increase. These results reproduce the negative cross-sectional relationship between stock illiquidity and investment documented by [Amihud and Levi \(2023\)](#).

⁴⁵ [Amihud and Levi \(2023\)](#) compare investment *rates* while we compare levels of installed capital. Our model generates the same investment rate for both types of firms in the steady state, namely, the depreciation rate δ . However, the two rates differ along a transition, which we do not study here.

This cross-sectional relationship, however, does not determine the aggregate effect of liquidity on investment. The third panel reports aggregate capital. With commitment, higher transaction costs lower the risk-free rate and raise capital at both liquid and illiquid firms, causing aggregate capital to increase slightly. Without commitment, higher transaction costs sharply reduce capital at illiquid firms, as the increase in present bias dominates the other general-equilibrium effects. Capital at liquid firms instead rises substantially. Because illiquid firms account for 99% of firms in the economy, their decline dominates and aggregate capital falls. Thus, the same negative cross-sectional relationship documented by [Amihud and Levi \(2023\)](#) is consistent with opposite aggregate responses to a liquidity dry-up depending on firms' ability to commit.

Figure 9: Liquidity and Investment in the Cross-Section and the Aggregate



Note: The panels vary the transaction cost on illiquid equity, ϕ , while liquid equity trades without transaction costs. The first two panels report capital at liquid and illiquid firms under commitment and no commitment, respectively. The third panel reports aggregate capital, $K = \omega k^I + (1 - \omega)k^L$, under the two regimes.

E Robustness and Extensions

E.1 Borrowing Wedge

In this appendix, we extend our theoretical analysis to the case where there is a positive wedge between the borrowing and saving rate for bonds, like in [Kaplan et al. \(2018\)](#). More specifically, household saving, denoted b_t^+ , yields a return r_{t-1} while household debt, denoted b_t^- , is subject to interest $r_{t-1} + \xi_b(b_t^-)$. While the wedge in [Kaplan et al. \(2018\)](#) is constant for all negative asset positions, we smooth the wedge around zero debt to avoid a kink in the household's bond-payoff schedule using the following function

$$\xi_b(b_t^-) = \bar{\xi}_b \left[1 - \frac{s_b}{b_t^-} \left(1 - \exp \left\{ -\frac{b_t^-}{s_b} \right\} \right) \right], \quad (43)$$

where $\bar{\xi}_b > 0$ is the upper bound for the borrowing wedge and $s_b > 0$ is a parameter that controls how quickly the wedge approaches this upper bound. The budget constraint becomes

$$\begin{aligned} & c_t + q_t \Delta_t^+ + b_{t+1}^+ - b_{t+1}^- \\ & \leq w_t h_t + d_t \theta_t + \left(\Delta_t^- - \frac{\phi}{2} (\Delta_t^-)^2 \right) q_t + b_t^+ (1 + r_{t-1}) - b_t^- [1 + r_{t-1} + \xi_b(b_t^-)] - \mathcal{T}_t, \end{aligned} \quad (44)$$

with additional constraints

$$b_{t+1}^+ \geq 0, \quad -\bar{b} \geq b_{t+1}^- \geq 0. \quad (45)$$

We assume that $\bar{b} \leq 0$ and that, for every $t \geq 0$, $\bar{b} > -\sum_{\tau=t+1}^{\infty} \left(\prod_{s=t+1}^{\tau} \frac{1}{1+r_{s-1}+\bar{\xi}_b} \right) (w_{\tau} h_1 - \mathcal{T}_{\tau})$. Thus, the problem of a household is to choose processes $\{c_t, b_{t+1}^+, b_{t+1}^-, \theta_{t+1}, \Delta_t^+, \Delta_t^-\}_{t=0}^{\infty}$ in order to maximize (1) subject to (44), (3), (4), and (45) for every $t \geq 0$, and given an initial portfolio (θ_0, b_0^+, b_0^-) and prices, dividends, and taxes $\{w_t, q_t, r_t, d_t, \mathcal{T}_t\}_{t=0}^{\infty}$. When defining value functions and cross-section distributions, we maintain the notation (θ, b, h) with the understanding that $b \equiv b^+ - b^-$ is the net bond position. The same comment applies to the market clearing condition for bonds, which remains unchanged with $b_{j,t} \equiv b_{j,t}^+ - b_{j,t}^-$.

The optimality condition for bonds becomes

$$\begin{aligned} & \frac{1}{1 + r_t + \xi_b(\mathcal{B}_{t+1}^{j-})} \\ & = \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} \right] + \mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)(1 + r_t + \xi_b(\mathcal{B}_{t+1}^{j-}))} \frac{\partial \xi_b(\mathcal{B}_{t+1}^{j-})}{\partial b} \mathcal{B}_{t+1}^{j-} \right] + v_t^j, \end{aligned}$$

where v_t^j is the Lagrange multiplier associated with the borrowing constraint in (45), and $\mathcal{B}_{t+1}^{j-}$ denotes the policy function for borrowing. For agents who do not borrow, $v_t^j = \mathcal{B}_{t+1}^{j-} = \xi_b(\mathcal{B}_{t+1}^{j-}) = 0$, and hence

$$\mathbb{E}_t \left[\beta \frac{u'(\mathcal{C}_{t+1}^j)}{u'(\mathcal{C}_t^j)} \right] = \frac{1}{1 + r_t} \implies q_t = \frac{d_{t+1}}{1 + r_t} + \frac{(1 - \Phi_t^b) q_{t+1}}{1 + r_t},$$

so Φ_t^b must be the same for all buyers who do not borrow, who we denote with the superscript b . Note that our proofs are written to accommodate this setup with a borrowing wedge when the difference from the baseline setting is substantial; see Appendix A for details. Our quantitative exercises from Sections 6 and 7 also include a positive wedge; see 6.1 for details about the calibration of parameters $\bar{\xi}_b$ and s_b .

E.2 Borrowing and Short-Selling

In the upcoming exercises, we hold all parameters other than the borrowing and short-selling limits at their benchmark values from Section 6, rather than recalibrating the model. The com-

parisons therefore isolate the effects of the alternative constraints, one by one, on the equilibrium.

In the benchmark economy, households may borrow subject to the constraint $b_{t+1} \geq \bar{b}$ with $\bar{b} = -0.66$. To evaluate whether private borrowing is important for our results, we instead impose $\bar{b} = 0$ in our first quantitative exercise. This restriction prohibits private debt but does not eliminate the bond market: households may continue to hold positive bond positions and save at the risk-free rate. All other household and firm problems and all market-clearing conditions remain unchanged.

The benchmark model imposes the equity constraint $\theta_{t+1} \geq 0$. In our second quantitative investigation, we consider an alternative economy in which households may take bounded short positions, $\theta_{t+1} \geq \bar{\theta}$ and we set $\bar{\theta} = -0.66$. Because average equity ownership is normalized to one, the limit allows an individual household to short approximately two thirds of the average household equity position. The value of this limit is chosen as a simple, economically meaningful bound for the robustness exercise.

Allowing negative equity positions requires specifying how short positions enter the firm's objective. Directly using θ as an ownership weight would assign negative weight to short sellers, which would translate to a preference for firm value minimization (for those agents). Instead, we assume that short positions receive zero firm weight, while positive positions continue to receive weights proportional to their equity holdings. Formally, the firm's objective (8) becomes

$$\int_{\theta,b,h} \max\{\theta, 0\} [d_t + (1 - \phi \Delta_t^-(\theta, b, h)) q_t] d\Gamma_t(\theta, b, h),$$

which is, up to multiplication by positive constants, equal to $d_t + (1 - \bar{\Phi}_t^{\text{short}}) q_t$ where

$$\bar{\Phi}_t^{\text{short}} = \phi \int_{\theta,b,h} w_t^{\text{short}}(\theta, b, h) \Delta_t^-(\theta, b, h) d\Gamma_t(\theta, b, h),$$

and

$$w_t^{\text{short}}(\theta, b, h) = \frac{\max\{\theta, 0\}}{\int_{\tilde{\theta}, \tilde{b}, \tilde{h}} \max\{\tilde{\theta}, 0\} d\Gamma_t(\tilde{\theta}, \tilde{b}, \tilde{h})}. \quad (46)$$

This exercise jointly changes households' feasible portfolios and specifies how the firm aggregates the valuations of investors with long and short positions.

Table 9 reports the results. The no-private-borrowing economy is close to the benchmark across all reported moments. The corporate discount-rate wedge declines slightly, from 4.29 to 4.16 percent, while the capital-to-annual-GDP ratio increases from 2.99 to 3.04. The corporate discount-rate wedge also remains positive when bounded short-selling is allowed, increasing from 4.29 to 5.14 percent. Thus, the mechanism generating the wedge does not depend on ruling out negative equity positions. Consistent with the larger wedge, the capital-to-annual-GDP ratio falls to 2.68. The ratio of consumption to income variance also rises, from 0.32 to

Table 9: Alternative Portfolio Constraints

Moment	Benchmark	No borrowing	Short-selling
Corporate discount-rate wedge, annualized percent	4.29	4.16	5.14
Capital to annual GDP	2.99	3.04	2.68
Liquidity premium, annualized bps	16.63	15.17	17.76
Transaction costs to annual GDP, percent	0.14	0.14	0.17
Variance log consumption / variance log income	0.32	0.31	0.48

Notes: The no-private-borrowing specification imposes $b_{t+1} \geq 0$ but retains positive bond saving. The bounded-short-selling specification imposes $\theta_{t+1} \geq -0.66$ and uses the normalized nonnegative firm weights in (46). All remaining parameters are held at their benchmark values; neither alternative model is recalibrated.

0.48.

E.3 Sensitivity Analysis

In this Appendix, we report local sensitivity analysis around the benchmark for the main household, income-process, and production parameters. For each parameter, we perturb its value one at a time on both sides of the benchmark, re-solve the general equilibrium, and calculate a two-sided finite-difference derivative, $\Delta m / \Delta \mathcal{P}$, for each outcome m . Table 10 reports the results.

The exercise delivers three main conclusions. First, it confirms the local identification of the two internally calibrated parameters. A higher household discount factor, β , raises the capital-output ratio, whereas a higher transaction cost, ϕ , increases the corporate discount-rate wedge and the liquidity premium while reducing capital. These responses discipline β through the capital-output target and ϕ through the corporate-wedge target.

Second, the results reinforce the role of household liquidity needs in the mechanism. Greater labor-income-shock volatility increases the liquidity premium and the corporate discount-rate wedge. Across the parameter perturbations, the share of buyers with nonnegative liquid-asset positions remains close to its benchmark value of 98 percent. Thus, the existence of an unconstrained buyer, investigated in Lemma 2, is robust locally.⁴⁶ To corroborate our analysis of the concentration effect from Proposition 1, here we investigate an alternative concentration metric. Following Gul et al. (2010), Rubin (2007), and Parigi and Pelizzon (2008), who measure ownership concentration as the share held by the top shareholders, we define

$$\mathcal{O} = \int_{\theta, b, h} 1_{\{\theta \geq \theta^*\}} \theta d\Gamma(\theta, b, h), \text{ where } 0.8 = \Gamma(\theta^*, b_{high}, h_{high}),$$

the share of equity held by the top 20% of shareholders. Higher income volatility σ_h raises precautionary demand for liquid assets, generating more heterogeneity among firm ownership. This increases ownership concentration, and at the same time amplifies disagreement between buyers and sellers (increasing the corporate discount-rate wedge). Table 10 shows that, while

⁴⁶The existence of a buyer who does not borrow is sufficient, but not necessary, for our results in the main text. However, the existence of a buyer who does not borrow is both necessary and sufficient for our results in the extension from Appendix E.1.

Table 10: Sensitivity Analysis

Parameter	Capital output ratio	Corporate wedge (percent)	Liquidity premium (bps)	Liquid rate (percent)	$\tilde{\delta}$	$\tilde{\beta}$	Ownership concentration	Share of buyers not borrowing
Benchmark value	2.9947	4.2873	16.6277	1.7725	0.9952	0.9893	0.3956	0.9840
Household discount factor	118.9338	-232.1129	70.5817	-216.5782	0.5271	0.5741	-3.8188	1.4087
Transaction cost	-0.4180	1.4440	10.3878	0.0144	-0.0003	-0.0036	-0.0026	0.0018
CRRRA Coefficient	0.5656	-1.0921	-9.9161	-0.9391	0.0025	0.0027	-0.0355	0.0098
Labor share	-9.6884	7.5669	50.2924	-4.8638	0.0106	-0.0187	0.1489	-0.0479
Depreciation Rate	-110.3807	52.2935	234.3680	-37.0676	0.0848	-0.1293	0.9952	-0.3568
Volatility of employment shocks	0.9489	4.1263	71.9755	-8.5374	0.0191	-0.0102	0.0250	0.0283
Persistence of employment shocks	14.6909	-16.3658	13.1095	-37.7433	0.0919	0.0405	1.1953	-1.0723

Note: The first row reports benchmark levels; the remaining rows report two-sided finite-difference derivatives of each moment with respect to its corresponding parameter. For each parameter (as indicated by the row), the model is solved under parameter perturbations and the derivative is calculated as the change in the moment divided by the change in the parameter, $\Delta m / \Delta \mathcal{P}$. Because parameters are measured in different units, derivative magnitudes are not directly comparable across rows.

concentration and the corporate wedge tend to co-move like in the case of income volatility perturbations, the sign of the relationship depends on the parameter being perturbed.

Third, the effect of income persistence is more nuanced. Holding innovation volatility fixed, greater persistence raises unconditional income risk and the liquidity premium, but locally lowers the corporate discount-rate wedge because it reduces trade and weakens disagreement between current and future owners. This experiment differs from Figure 3, where we adjust σ_h to hold the unconditional variance of income constant. Overall, the local sensitivity analysis supports the main mechanism while clarifying that the effect of persistence depends on what is held fixed.

E.4 Aggregate Shocks

This appendix extends the firm's problem to time-varying aggregate paths. We write the problem explicitly for an aggregate TFP path; the financial-friction and income-uncertainty shocks studied in Section 7 affect the firm through the induced equilibrium paths of the risk-free rate, the liquidity premium, and the corporate discount-rate wedge. Each shock is unanticipated when it arrives in period 1. From that point onward, agents have perfect foresight over aggregates, while households continue to face idiosyncratic employment risk. For the TFP experiment,

$$A_t = A^* [1 + \varepsilon_A \rho_A^{t-1}], \quad t \geq 1.$$

Capital k_1 and the initial cross-sectional distribution are predetermined. Production is

$$y_t = A_t \left(H_t^\gamma k_t^{1-\gamma} \right)^\psi.$$

The intratemporal labor problem is unchanged. The stock-pricing equation of a buyer who does not borrow in the bond market is

$$q_t = \frac{d_{t+1}}{1+r_t} + \frac{1-\Phi_t^b}{1+r_t} q_{t+1}, \quad (47)$$

and current owners value the firm at $d_t + (1 - \bar{\Phi}_t)q_t$. Iterating equation (47) forward gives

$$V_t^F = \max_{\{k_{t+s}\}_{s \geq 1}} \left\{ d_t + \frac{1 - \bar{\Phi}_t}{1 + r_t} \left[d_{t+1} + \sum_{s=2}^{\infty} \left(\prod_{z=1}^{s-1} \frac{1 - \Phi_{t+z-1}^b}{1 + r_{t+z}} \right) d_{t+s} \right] \right\}.$$

Define the time-varying discount factors

$$\tilde{\delta}_t \equiv \frac{1 - \Phi_{t-1}^b}{1 + r_t}, \quad \tilde{\beta}_t \equiv \frac{1 - \bar{\Phi}_t}{1 - \Phi_{t-1}^b}. \quad (48)$$

Thus,

$$\tilde{\beta}_t \tilde{\delta}_t = \frac{1 - \bar{\Phi}_t}{1 + r_t}.$$

Note the distinction between Φ_{t-1}^b and $\bar{\Phi}_t$ in (48). Like in the main text, one distinction arises from the persistence, concentration, and risk premium effects from Proposition 1. Unlike the main text, here there is also a distinction in *timing*. While $\bar{\Phi}_t$ captures shareholder disagreement at time t , Φ_t^b captures the expected marginal transaction cost for buyers at time $(t + 1)$. Hence, Φ_{t-1}^b and $\bar{\Phi}_t$ effectively compare transaction costs within the same period.

Because the initial shock arrives after k_1 has been installed, the relevant policy formulas describe capital installed from period 2 onward. With commitment, the firm chooses a state-contingent rule for next period's capital before the surprise shock is realized. Along the realized perfect-foresight path, this rule gives

$$k_{t+1}^C = \left[\frac{(1 - \gamma)\psi \tilde{\delta}_t}{1 - \tilde{\delta}_t(1 - \delta)} A_{t+1} H^{\gamma\psi} \right]^{\frac{1}{1 - (1 - \gamma)\psi}}. \quad (49)$$

In particular, because the period-1 shock is unanticipated, Φ_0^b remains at its steady-state value. Hence, $\tilde{\delta}_1$ can respond on impact through r_1 , but not through the liquidity component. The impact response of Φ_1^b first enters the commitment discount factor in period 2.

Without commitment, the Markov-perfect policy is

$$k_{t+1}^N = \left[\frac{(1 - \gamma)\psi \tilde{\beta}_t \tilde{\delta}_t}{1 - \tilde{\beta}_t \tilde{\delta}_t(1 - \delta)} A_{t+1} H^{\gamma\psi} \right]^{\frac{1}{1 - (1 - \gamma)\psi}}. \quad (50)$$

Equations (49) and (50) are the time-varying counterparts of the stationary formulas in Proposition 2. Commitment removes $\tilde{\beta}_t$, while the no-commitment firm responds to both market discounting and shareholder disagreement.